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Statistics

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Chapter 1

Introduction

In this class, we will study statistical methods to analyse data. The goal of these techniques is to infer from some observations to an underlying population. Since there is usually randomness in the data, one has to decide whether some structure in the data might be really present in the population or whether it is only caused by random fluctuations.

Let us briefly discuss three statistical problems, which we will consider in this class.

1) In order to study whether males in the northern part of Germany are taller than males in Switzerland, the heights of 145 students at University of Bern and 26 students at University of Lübeck were compared (see [1, p. 158]). The average height of the students in Bern was about 5 cm less than that of the students in Lübeck. Can we conclude that males in northern Germany are taller than males in Switzerland?

2) In a famous genetics experiment (see e.g. [4, Example 11.13]), Gregor Mendel classified 556 peas according to their shape and colour:

	yellow	green
round	315	108
wrinkled	101	32

According to a theory by Mendel the frequencies of the different combinations must be 9/16, 3/16, 3/16 and 1/16. Do the experimental data support this theory?

3) In a physical problem, two quantities x and y satisfy the linear relation $y = \beta x$, but the exact value for β is unknown. In order to determine β, for several values $x_1, \ldots, x_n$ for x the corresponding values $y_1, \ldots, y_m$ for y are measured, but the measurements are corrupted by some noise, i.e.,

$$y_i = \beta x_i + \varepsilon_i, \quad i \in \{1, \ldots, n\},$$

with i.i.d. random variables $\varepsilon_1, \ldots, \varepsilon_n$. How can one estimate β?

Since data is collected and interpreted in a huge variety of fields, statistics is essential to many sciences. Although the studied topics and research questions might be often completely different, the employed statistical methods are often the same.

In order to understand and develop statistical methods, a solid background in probability theory is necessary. Thus, we will recall and newly introduce several concepts and results from probability theory throughout this class.

Chapter 2

Preliminaries

The focus of this chapter is on the theoretical foundations that we will need for this class. Mostly, we recall and extend concepts that are known from the class on Stochastics.

2.1 Probability spaces and random elements

First we remember the definition of a σ-algebra (see e.g. [7, Definition 2.45]):

Definition 2.1 (σ-algebra)**.** Let Ω be a non-empty set. A family $\mathscr{F}$ of subsets of Ω is called a σ-algebra on Ω if the following properties are satisfied:

(i) $\Omega \in \mathscr{F}$

(ii) $A \in \mathscr{F} \quad \Longrightarrow \quad A^c \in \mathscr{F}$

(iii) $A_1, A_2, \ldots \in \mathscr{F} \quad \Longrightarrow \quad \cup_{i=1}^{\infty} A_i \in \mathscr{F}$

We provide several examples for σ-algebras:

Example 2.2.

(i) Obviously, $\{\emptyset, \Omega\}$ and the power set $\mathcal{P}(\Omega)$ of Ω are σ-algebras.

(ii) For each family $\mathscr{A}$ of subsets of Ω there exists a smallest σ-algebra that contains all sets from $\mathscr{A}$. We say that this σ-algebra is generated by $\mathscr{A}$ and denote it by $\sigma(\mathscr{A})$.

(iii) The Borel σ-algebra $\mathcal{B}(\mathbb{R})$ on $\mathbb{R}$ is the σ-algebra generated by $\{(a, b] : a, b \in \mathbb{R}, a \leq b\}$.

(iv) The Borel σ-algebra $\mathcal{B}(\mathbb{R}^m)$ on $\mathbb{R}^m$ is generated by $\{\times_{i=1}^{m}(a_i, b_i] : a_i, b_i \in \mathbb{R}, a_i \leq b_i, i \in \{1, \ldots, m\}\}$.

Now we are prepared for the definition of a probability space:

Definition 2.3 (Probability space)**.** A probability space $(\Omega, \mathscr{F}, \mathbf{P})$ consists of a non-empty set Ω, a σ-algebra $\mathscr{F}$ on Ω and a function $\mathbf{P} : \mathscr{F} \to [0, 1]$ such that

(i) $\mathbf{P}(\Omega) = 1$

(ii) for all disjoint $A_1, A_2, \ldots \in \mathscr{F}$ (i.e., $A_i \cap A_j = \emptyset$ for $i \neq j$),

$$\mathbf{P}(\cup_{k=1}^{\infty} A_k) = \sum_{k=1}^{\infty} \mathbf{P}(A_k). \tag{2.1}$$

The set Ω is denoted as sample space and $\mathbf{P}$ is called probability or probability measure.

A further object we need are measurable spaces.

Definition 2.4 (Measurable space)**.** A pair $(\Omega, \mathscr{F})$ of a non-empty set Ω and a σ-algebra $\mathscr{F}$ on Ω is called a measurable space.

Note that every probability space is also a measurable space. An example for a measurable space is $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Definition 2.5. Let $(\Omega_1, \mathscr{F}_1)$ and $(\Omega_2, \mathscr{F}_2)$ be measurable spaces. A map $T : \Omega_1 \to \Omega_2$ is called measurable if $T^{-1}(A) = \{\omega \in \Omega_1 : T(\omega) \in A\} \in \mathscr{F}_1$ for all $A \in \mathscr{F}_2$.

Note that we equip $\mathbb{R}$ and $\mathbb{R}^m$ always with their Borel σ-algebras $\mathcal{B}(\mathbb{R})$ and $\mathcal{B}(\mathbb{R}^d)$ if not stated otherwise. Thus, we mean measurablity with respect to these σ-algebras when dealing with functions to $\mathbb{R}$ and $\mathbb{R}^m$. In this class, we usually do not check measurability since all functions we deal with are measurable.

Example 2.6. Let $(\Omega, \mathscr{F}, \mathbf{P})$ be a probability space, let $(M, \mathscr{M})$ be a measurable space and let $X : \Omega \to M$ be measurable. Then,

$$\mathbf{P}_X : \mathscr{M} \ni A \mapsto \mathbf{P}_X(A) := \mathbf{P}(X \in A) := \mathbf{P}(X^{-1}(A))$$

is well defined and is a probability (measure) on $(M, \mathscr{M})$, i.e., $(M, \mathscr{M}, \mathbf{P}_X)$ is a probability space.

Definition 2.7. Let $(\Omega, \mathscr{F}, \mathbf{P})$ be a probability space and let $(M, \mathscr{M})$ be a measurable space. A measurable map $X : \Omega \to M$ is called a random element on M and $\mathbf{P}_X$ from Example 2.6 is called the distribution of X. A measurable map $X : \Omega \to \mathbb{R}$ is denoted as random variable and a measurable map $\mathbf{X} : \Omega \to \mathbb{R}^m$ as random vector.

For $m = 1$ a random vector is a random variable. We usually denote the components of a random $\mathbf{X}$ in $\mathbb{R}^m$ by $\mathbf{X}_1, \ldots, \mathbf{X}_m$. The components of a random vector are random variables. If one writes several random variables as a vector, one obtains a random vector.

Definition 2.8. The distribution function of a random vector $\mathbf{X} = (\mathbf{X}_1, \ldots, \mathbf{X}_m)$ in $\mathbb{R}^m$ is given by

$$F_{\mathbf{X}} : \mathbb{R}^m \ni \mathbf{x} = (x_1, \ldots, x_m) \mapsto \mathbf{P}(\mathbf{X}_1 \leq x_1, \ldots, \mathbf{X}_m \leq x_m).$$

For $m = 1$ the previous definition coincides with that of the distribution function of a random variable (see e.g. [7, Definition 3.13]). Note that the distribution of a random vector is completely determined by its distribution function.

A very important concept in probability theory is independence.

Definition 2.9. Let $(\Omega, \mathscr{F}, \mathbf{P})$ be a probability space, let $(M_1, \mathscr{M}_1), \ldots, (M_n, \mathscr{M}_n)$, $n \in \mathbb{N}$, be measurable spaces and let X_i be a random element on M_i for $i \in \{1, \ldots, n\}$. One says that $X_1, \ldots, X_n$ are independent if

$$\mathbf{P}(X_1 \in A_1, \ldots, X_n \in A_n) = \prod_{i=1}^{n} \mathbf{P}(X_i \in A_i)$$

for all $A_1 \in \mathscr{M}_1, \ldots, A_n \in \mathscr{M}_n$. An infinite collection of random elements is called independent if each finite subset of them is independent.

We illustrate independence of random variables by the following example:

Example 2.10. For independent random variables $Y_1, \ldots, Y_m$, $m \in \mathbb{N}$, define the random vector $\mathbf{X} = (Y_1, \ldots, Y_m)$. Then the distribution function of $\mathbf{X}$ is given by

$$F_{\mathbf{X}}(x_1, \ldots, x_m) = \mathbf{P}(Y_1 \leq x_1, \ldots, Y_m \leq x_m) = \prod_{i=1}^{m} \mathbf{P}(Y_i \leq x_i) = \prod_{i=1}^{m} F_{Y_i}(x_i)$$

for $x_1, \ldots, x_m \in \mathbb{R}$, where independence was used for the second equality.

Recall that $\mathbf{E}$ denotes expectation and that the covariance $\mathbf{Cov}(Y, Z)$ of two random variables Y and Z with $\mathbf{E}[|Y|], \mathbf{E}[|Z|] < \infty$ is given by

$$\mathbf{Cov}(Y, Z) = \mathbf{E}[(Y - \mathbf{E}[Y])(Z - \mathbf{E}[Z])] = \mathbf{E}[YZ] - \mathbf{E}[Y]\mathbf{E}[Z].$$

For an m-dimensional random vector $\mathbf{X} = (\mathbf{X}_1, \ldots, \mathbf{X}_m)$ we define its expectation

$$\mathbf{E}[\mathbf{X}] = (\mathbf{E}[\mathbf{X}_1], \ldots, \mathbf{E}[\mathbf{X}_m])$$

if $\mathbf{E}[|\mathbf{X}_1|], \ldots, \mathbf{E}[|\mathbf{X}_m|] < \infty$ and its covariance matrix

$$\mathbf{Cov}(\mathbf{X}) = \big(\mathbf{Cov}(\mathbf{X}_i, \mathbf{X}_j)\big)_{i,j \in \{1,\ldots,m\}}$$

if $\mathbf{E}[\mathbf{X}_1^2], \ldots, \mathbf{E}[\mathbf{X}_m^2] < \infty$, i.e., the expectation of a random vector is obtained by taking the expectation component-wise and the covariance matrix contains all pair-wise covariances of the components.

Lemma 2.11. Let $\mathbf{X}$ be an m-dimensional random vector, let $\Theta \in \mathbb{R}^{n \times m}$ and let $\mathbf{a} \in \mathbb{R}^n$.

(i) If $\mathbf{E}[|\mathbf{X}_1|], \ldots, \mathbf{E}[|\mathbf{X}_m|] < \infty$,

$$\mathbf{E}[\Theta\mathbf{X} + \mathbf{a}] = \Theta\mathbf{E}[\mathbf{X}] + \mathbf{a}.$$

(ii) If $\mathbf{E}[\mathbf{X}_1^2], \ldots, \mathbf{E}[\mathbf{X}_m^2] < \infty$,

$$\mathbf{Cov}(\Theta\mathbf{X} + \mathbf{a}) = \Theta\mathbf{Cov}(\mathbf{X})\Theta^T.$$

Proof. See Exercise class. □

2.2 Absolutely continuous random vectors

A random variable X has a density f_X if its distribution function F_X can be written as

$$F_X(x) = \int_{-\infty}^{x} f_X(y)\,\mathrm{d}y$$

for $x \in \mathbb{R}$. On the other hand, any function $f : \mathbb{R} \to \mathbb{R}$ is the density of a distribution if and only if

$$f(x) \geq 0 \quad \text{for all } x \in \mathbb{R} \quad \text{and} \quad \int_{-\infty}^{\infty} f(y)\,\mathrm{d}y = 1.$$

Thus, a common way to construct the distribution of a random variable is to provide a density. Our next goal is to extend the concept of a density to random vectors, which requires multidimensional integration.

Intermezzo on multidimensional integration

For a measurable function $f : \mathbb{R}^m \to [0, \infty)$, we define

$$\int_{\mathbb{R}^m} f(\mathbf{x})\,\mathrm{d}\mathbf{x}$$

as an element of $[0, \infty) \cup \{\infty\}$ similarly as the expectation of a random variable by replacing the probability measure by the Lebesgue measure λ_m on $\mathbb{R}^m$ (one can think of λ_m as the volume, see e.g. [7, Example 2.54]):

(i) For $f = \mathbf{1}_A$ with $A \in \mathcal{B}(\mathbb{R}^m)$,

$$\int_{\mathbb{R}^m} f(\mathbf{x})\,\mathrm{d}\mathbf{x} = \lambda_m(A).$$

(ii) For $f = \sum_{i=1}^{k} c_i \mathbf{1}_{A_i}$ with $k \in \mathbb{N}$, $c_1, \ldots, c_k \in [0, \infty)$ and $A_1, \ldots, A_k \in \mathcal{B}(\mathbb{R}^m)$ (i.e., f is simple),

$$\int_{\mathbb{R}^m} f(\mathbf{x})\,\mathrm{d}\mathbf{x} = \sum_{i=1}^{k} c_i \lambda_m(A_i).$$

(iii) For $f : \mathbb{R}^m \to [0, \infty)$ and simple functions $(f_n)_{n \in \mathbb{N}}$ converging monotonously to f,

$$\int_{\mathbb{R}^m} f(\mathbf{x})\,\mathrm{d}\mathbf{x} = \lim_{n \to \infty} \int_{\mathbb{R}^m} f_n(\mathbf{x})\,\mathrm{d}\mathbf{x}.$$

As in the univariate case, one can think of the integral as the $(m+1)$-dimensional volume below the function. For a measurable function $f : \mathbb{R}^m \to \mathbb{R}$ one defines

$$\int_{\mathbb{R}^m} f(\mathbf{x})\,\mathrm{d}\mathbf{x} = \int_{\mathbb{R}^m} f_+(\mathbf{x})\,\mathrm{d}\mathbf{x} - \int_{\mathbb{R}^m} f_-(\mathbf{x})\,\mathrm{d}\mathbf{x}$$

with $f_+ = \max\{f, 0\}$ and $f_- = \max\{-f, 0\}$ if

$$\int_{\mathbb{R}^m} f_+(\mathbf{x})\,\mathrm{d}\mathbf{x} < \infty \quad \text{or} \quad \int_{\mathbb{R}^m} f_-(\mathbf{x})\,\mathrm{d}\mathbf{x} < \infty.$$

If both integrals are finite, we call f integrable. For a set $A \in \mathcal{B}(\mathbb{R}^d)$ we use the notation

$$\int_A f(\mathbf{x})\,\mathrm{d}\mathbf{x} = \int_{\mathbb{R}^m} \mathbf{1}_A(\mathbf{x}) f(\mathbf{x})\,\mathrm{d}\mathbf{x}.$$

An important theorem from measure theory, Fubini's theorem, yields that, for a measurable $f : \mathbb{R}^m \to [0, \infty)$,

$$\int_{\mathbb{R}^m} f(\mathbf{x})\,\mathrm{d}\mathbf{x} = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x_1, \dots, x_m)\,\mathrm{d}x_1 \dots \mathrm{d}x_m,$$

i.e., one can compute multidimensional integral by successively computing one-dimensional integrals. Here, the order of the integrations is not relevant, i.e.,

$$\int_{\mathbb{R}^m} f(\mathbf{x})\,\mathrm{d}\mathbf{x} = \int_{-\infty}^{\infty} \dots \int_{-\infty}^{\infty} f(x_1, \dots, x_m)\,\mathrm{d}x_{\sigma(1)} \dots \mathrm{d}x_{\sigma(m)}$$

for all permutations σ of $\{1, \dots, m\}$. This result can be extended to integrable $f : \mathbb{R}^m \to \mathbb{R}$ as soon as any of the integrals over $|f|$ is finite.

Example 2.12. Our goal is to compute

$$I = \int_\Delta 1\,\mathrm{d}\mathbf{x}$$

for $\Delta = \{(x_1, x_2) \in [0, 1]^2 : x_1 + x_2 \le 1\}$. We have

$$\begin{aligned} I &= \int_{\mathbb{R}^2} \mathbf{1}\{x_1, x_2 \in [0, 1], x_1 + x_2 \le 1\}\,\mathrm{d}(x_1, x_2) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathbf{1}\{x_1, x_2 \in [0, 1], x_1 + x_2 \le 1\}\,\mathrm{d}x_1\,\mathrm{d}x_2 \\ &= \int_0^1 \int_0^1 \mathbf{1}\{x_1 + x_2 \le 1\}\,\mathrm{d}x_1\,\mathrm{d}x_2 = \int_0^1 \int_0^1 \mathbf{1}\{x_1 \le 1 - x_2\}\,\mathrm{d}x_1\,\mathrm{d}x_2. \end{aligned}$$

For $x_2 \in [0, 1]$ we obtain

$$\int_0^1 \mathbf{1}\{x_1 \le 1 - x_2\}\,\mathrm{d}x_1 = \int_0^{1-x_2} 1\,\mathrm{d}x_1 = 1 - x_2$$

and, thus,

$$I = \int_0^1 1 - x_2\,\mathrm{d}x_2 = \left[x_2 - \frac{x_2^2}{2}\right]_0^1 = \frac{1}{2}.$$

Example 2.13. We have

$$I = \int_{[0,1]^2} xy \,\mathrm{d}(x,y) = \int_0^1 \int_0^1 xy \,\mathrm{d}x \,\mathrm{d}y.$$

Since we can take constants out of integrals, we obtain

$$\int_0^1 \int_0^1 xy \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 y \int_0^1 x \,\mathrm{d}x \,\mathrm{d}y = \int_0^1 x \,\mathrm{d}x \int_0^1 y \,\mathrm{d}y = \left[\frac{x^2}{2}\right]_0^1 \left[\frac{y^2}{2}\right]_0^1 = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4},$$

where we treated y as a constant in the first step and $\int_0^1 x \,\mathrm{d}x$ as a constant in the second step.

We will also use the following special case of the multivariate integration by substitution rule. For an integrable function $f : \mathbb{R}^m \to \mathbb{R}$, an invertible matrix $\Theta \in \mathbb{R}^{m \times m}$, $\boldsymbol{\mu} \in \mathbb{R}^m$ and $A \in \mathcal{B}(\mathbb{R}^m)$ one has

$$\int_{\Theta^{-1}(A-\boldsymbol{\mu})} f(\mathbf{x}) \,\mathrm{d}\mathbf{x} = \frac{1}{\det(\Theta)} \int_A f(\Theta^{-1}(\mathbf{y} - \boldsymbol{\mu})) \,\mathrm{d}\mathbf{y}, \tag{2.2}$$

where Θ^{-1} is the inverse of Θ, $\det(\Theta)$ is the determinant of Θ and $\Theta^{-1}(A - \boldsymbol{\mu}) = \{\Theta^{-1}(\mathbf{x} - \boldsymbol{\mu}) : \mathbf{x} \in A\}$.

After having introduced multidimensional integrals, we can define densities as in the univariate case.

Definition 2.14. A measurable function $f : \mathbb{R}^m \to \mathbb{R}$ is called a density if

$$f(\mathbf{x}) \geq 0 \quad \text{for all} \quad \mathbf{x} \in \mathbb{R}^m \quad \text{and} \quad \int_{\mathbb{R}^m} f(\mathbf{x}) \,\mathrm{d}\mathbf{x} = 1.$$

Example 2.15. It follows from Example 2.12 that

$$f : \mathbb{R}^2 \ni (x_1, x_2) \mapsto 2 \cdot \mathbf{1}_\Delta(x_1, x_2)$$

is a density since it is non-negative and its integral equals one.

Theorem 2.16. Let $f : \mathbb{R}^m \to [0, \infty)$ be a density. Then, there exists a probability space $(\Omega, \mathscr{F}, \mathbf{P})$ with a random vector $\mathbf{X}$ in $\mathbb{R}^m$ such that

$$\mathbf{P}(\mathbf{X} \in A) = \int_A f(\mathbf{x}) \,\mathrm{d}\mathbf{x} \tag{2.3}$$

for $A \in \mathcal{B}(\mathbb{R}^m)$ and

$$F_{\mathbf{X}}(x_1, \ldots, x_m) = \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_m} f(y_1, \ldots, y_m) \,\mathrm{d}y_m \ldots \mathrm{d}y_1 \tag{2.4}$$

for $x_1, \ldots, x_m \in \mathbb{R}$.

Proof. Note that (2.4) follows immediately from (2.3) for $A = (-\infty, x_1] \times \ldots \times (-\infty, x_m]$. The proof of (2.3) requires measure theory and is, thus, out of the scope of this class. □

Remark 2.17. If a function $f : \mathbb{R}^m \to \mathbb{R}$ satisfies (2.4) for an m-dimensional random vector $\mathbf{X}$, then f must be a density and (2.3) is satisfied as well. In this case we call f the density of the random vector $\mathbf{X}$. If a random vector possesses a density, the density is unique. In fact, this holds only up to modifications on sets with Lebesgue measure zero, but we will ignore this issue in the sequel.

The distribution of a random vector is the joint distribution of its components. Recall that the distributions of the components are called marginal distributions.

Corollary 2.18. Let $\mathbf{X} = (\mathbf{X}_1, \ldots, \mathbf{X}_m)$ be a random vector in $\mathbb{R}^m$ with density f. Then, for $i \in \{1, \ldots, m\}$, $\mathbf{X}_i$ has the density

$$f_{\mathbf{X_i}}(y) = \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} f(x_1, \ldots, x_{i-1}, y, x_{i+1}, \ldots, x_m) \, \mathrm{d}x_m \ldots \mathrm{d}x_{i+1} \, \mathrm{d}x_{i-1} \ldots \mathrm{d}x_1$$

for $y \in \mathbb{R}$ and

$$\begin{aligned} F_{\mathbf{X}_i}(y) = \int_{-\infty}^{y} \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} & f(x_1, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_m) \\ & \times \mathrm{d}x_m \ldots \mathrm{d}x_{i+1} \, \mathrm{d}x_{i-1} \ldots \mathrm{d}x_1 \, \mathrm{d}x_i \end{aligned}$$

for $y \in \mathbb{R}$.

Proof. It follows from Theorem 2.16 with $A = \mathbb{R}^{i-1} \times (-\infty, y] \times \mathbb{R}^{m-i}$ that, for $y \in \mathbb{R}$,

$$\begin{aligned} F_{\mathbf{X}_i}(y) = \mathbf{P}(\mathbf{X}_i \leq y) = \mathbf{P}(\mathbf{X} \in A) &= \int_A f(\mathbf{x}) \, \mathrm{d}\mathbf{x} \\ &= \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} \int_{-\infty}^{y} \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} f(x_1, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_m) \\ & \qquad \times \mathrm{d}x_m \ldots \mathrm{d}x_{i+1} \, \mathrm{d}x_i \, \mathrm{d}x_{i-1} \ldots \mathrm{d}x_1. \end{aligned}$$

Since, by Fubini's theorem, this can be rewritten as

$$\begin{aligned} F_{\mathbf{X}_i}(y) = \int_{-\infty}^{y} \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \ldots \int_{-\infty}^{\infty} & f(x_1, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_m) \\ & \times \mathrm{d}x_m \ldots \mathrm{d}x_{i+1} \, \mathrm{d}x_{i-1} \ldots \mathrm{d}x_1 \, \mathrm{d}x_i, \end{aligned}$$

$\mathbf{X}_i$ has the density from the assertion. □

Example 2.19. Let $\mathbf{X}$ be a random vector in $\mathbb{R}^2$ with the density f from Example 2.15, i.e., we can think of $\mathbf{X}$ as uniformly distributed on Δ. For $x_1 \notin [0, 1]$ we have

$$\int_{-\infty}^{\infty} f(x_1, x_2) \, \mathrm{d}x_2 = 0,$$

while for $x_1 \in [0,1]$, we obtain

$$\int_{-\infty}^{\infty} f(x_1, x_2)\,\mathrm{d}x_2 = 2\int_0^1 \mathbf{1}\{x_1 + x_2 \le 1\}\,\mathrm{d}x_2 = 2 - 2x_1.$$

Thus, $\mathbf{X}_1$ has the density

$$f_{\mathbf{X}_1}(x_1) = \begin{cases} 2 - 2x_1, & x_1 \in [0,1], \\ 0, & x_1 \notin [0,1], \end{cases}$$

and the distribution function

$$F_{\mathbf{X}_1}(x_1) = \begin{cases} 0, & x_1 < 0, \\ 2x_1 - x_1^2, & x_1 \in [0,1], \\ 1, & x_1 > 1. \end{cases}$$

By symmetry, $\mathbf{X}_2$ has the same marginal distribution. Since

$$\mathbf{P}(\mathbf{X}_1 \in (1/2,1], \mathbf{X}_2 \in (1/2,1]) = \mathbf{P}(\mathbf{X} \in (1/2,1]^2) = \int_{(1/2,1]^2} f(\mathbf{x})\,\mathrm{d}\mathbf{x} = 0,$$

but

$$\mathbf{P}(\mathbf{X}_1 \in (1/2,1])\mathbf{P}(\mathbf{X}_2 \in (1/2,1]) = \mathbf{P}(\mathbf{X}_1 \in (1/2,1])^2 = (1 - F_{\mathbf{X}_1}(1/2))^2 = \left(1 - \frac{3}{4}\right)^2 = \frac{1}{16},$$

the random variables $\mathbf{X}_1$ and $\mathbf{X}_2$ are not independent.

Theorem 2.20. Let $X_1, \dots, X_m$, $m \in \mathbb{N}$, be random variables with densities $f_{X_1}, \dots, f_{X_m}$ and let $\mathbf{X} = (X_1, \dots, X_m)$. Then, $X_1, \dots, X_m$ are independent if and only if the random vector $\mathbf{X}$ has the density

$$f_{\mathbf{X}}(x_1, \dots, x_m) = \prod_{i=1}^m f_{X_i}(x_i) \tag{2.5}$$

for $x_1, \dots, x_m \in \mathbb{R}$.

Proof. If $X_1, \dots, X_m$ are independent, we obtain

$$\begin{aligned} F_{\mathbf{X}}(x_1, \dots, x_m) &= \mathbf{P}(X_1 \le x_1, \dots, X_m \le x_m) = \prod_{i=1}^m \mathbf{P}(X_i \le x_i) = \prod_{i=1}^m \int_{-\infty}^{x_i} f_{X_i}(y_i)\mathrm{d}y_i \\ &= \int_{-\infty}^{x_1} \cdots \int_{-\infty}^{x_m} \prod_{i=1}^m f_{X_i}(y_i)\,\mathrm{d}y_m \dots \mathrm{d}y_1 \end{aligned}$$

for $x_1, \dots, x_m \in \mathbb{R}$, whence the right-hand side of (2.5) is a density of $f_{\mathbf{X}}$. On the other hand, if (2.5) is satisfied, it follows from Theorem 2.16 that

$$\begin{aligned}\mathbf{P}(X_1 \in A_1, \dots, X_m \in A_m) = \mathbf{P}((X_1, \dots, X_m) \in \times_{i=1}^m A_i) &= \int_{\times_{i=1}^m A_i} f_{\mathbf{X}}(\mathbf{x}) \, \mathrm{d}\mathbf{x} \\ &= \int_{A_1} \dots \int_{A_m} f_{\mathbf{X}}(x_1, \dots, x_m) \, \mathrm{d}x_m \dots \mathrm{d}x_1 \\ &= \int_{A_1} \dots \int_{A_m} \prod_{i=1}^m f_{X_i}(x_i) \, \mathrm{d}x_m \dots \mathrm{d}x_1 = \prod_{i=1}^m \int_{A_i} f_{X_i}(x_i) \, \mathrm{d}x_i \\ &= \prod_{i=1}^m \mathbf{P}(X_i \in A_i)\end{aligned}$$

for $A_1, \dots, A_m \in \mathcal{B}(\mathbb{R}^m)$. Thus, $X_1, \dots, X_m$ are independent. □

Example 2.21 (Multivariate normal distribution)**.** For $m \in \mathbb{N}$ let $N_1, \dots, N_m \sim \mathrm{Normal}(0, 1)$ be independent and define $\mathbf{N} = (N_1, \dots, N_m)$. By Theorem 2.20, $\mathbf{N}$ has the density

$$\varphi_m(x_1, \dots, x_m) = \prod_{i=1}^m \frac{1}{\sqrt{2\pi}} e^{-\frac{x_i^2}{2}} = \frac{1}{\sqrt{2\pi}^m} e^{-\frac{1}{2}\sum_{i=1}^m x_i^2}.$$

The distribution of $\mathbf{N}$ is called the m-dimensional standard normal distribution.

For $\boldsymbol{\mu} \in \mathbb{R}^m$ and a positive semi-definite matrix $\Sigma \in \mathbb{R}^{m \times m}$, we say that a random vector $\mathbf{Y}$ follows a multivariate normal distribution with parameters $\boldsymbol{\mu}$ and Σ if it has the same distribution as $\Sigma^{\frac{1}{2}}\boldsymbol{N} + \boldsymbol{\mu}$, where $\Sigma^{\frac{1}{2}}$ is the unique positive semi-definite matrix such that $\Sigma^{\frac{1}{2}}\Sigma^{\frac{1}{2}} = \Sigma$ (note that $\Sigma^{1/2}$ is symmetric as a positive semi-definite matrix). We write $\mathbf{Y} \sim \mathrm{Normal}(\mu, \Sigma)$, this is in contrast to the notation in the class on Stochastics (see [7]), where in the one-dimensional case the second parameter was the standard deviation and not the variance. By Lemma 2.11, we have

$$\mathbf{E}[Y] = \mathbf{E}[\Sigma^{\frac{1}{2}}\boldsymbol{N} + \boldsymbol{\mu}] = \boldsymbol{\mu} \quad \text{and} \quad \mathbf{Cov}(Y) = \mathbf{Cov}(\Sigma^{\frac{1}{2}}\boldsymbol{N} + \boldsymbol{\mu}) = \Sigma^{\frac{1}{2}}\Sigma^{\frac{1}{2}} = \Sigma,$$

i.e., the parameters μ and Σ are the expectation and the covariance matrix.

Next we assume that Σ is even positive definite and, thus, invertible. For $A \in \mathcal{B}(\mathbb{R}^m)$, we obtain

$$\mathbf{P}(\mathbf{Y} \in A) = \mathbf{P}(\Sigma^{\frac{1}{2}}\mathbf{N} + \boldsymbol{\mu} \in A) = \mathbf{P}(\mathbf{N} \in (\Sigma^{\frac{1}{2}})^{-1}(A - \boldsymbol{\mu})) = \int_{(\Sigma^{\frac{1}{2}})^{-1}(A-\boldsymbol{\mu})} \varphi_m(\mathbf{x}) \, \mathrm{d}\mathbf{x}.$$

Now it follows from (2.2) that

$$\begin{aligned}\mathbf{P}(\mathbf{Y} \in A) &= \frac{1}{\det(\Sigma^{\frac{1}{2}})} \int_A \varphi_m((\Sigma^{\frac{1}{2}})^{-1}(\mathbf{x} - \boldsymbol{\mu})) \, \mathrm{d}\mathbf{x} \\ &= \frac{1}{(\sqrt{2\pi})^m \sqrt{\det(\Sigma)}} \int_A e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T (\Sigma^{-\frac{1}{2}})^2 (\mathbf{x}-\boldsymbol{\mu})} \, \mathrm{d}\mathbf{x} \\ &= \frac{1}{(\sqrt{2\pi})^m \sqrt{\det(\Sigma)}} \int_A e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})} \, \mathrm{d}\mathbf{x},\end{aligned}$$

whence $\mathbf{Y}$ has the density

$$f_{\mathbf{Y}}(\mathbf{x}) = \frac{1}{(\sqrt{2\pi})^m \sqrt{\det(\Sigma)}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \Sigma^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$

for $\mathbf{x} \in \mathbb{R}^m$. For $m = 1$ this is the well-known formula from the one-dimensional case.

As in the univariate case (see e.g. [7, Theorem 3.40]), densities can be used to compute expectations. We do not prove the following result:

Theorem 2.22. Let $\mathbf{X}$ be a random vector in $\mathbb{R}^m$ with density f and let $h : \mathbb{R}^m \to \mathbb{R}$ be measurable. Then,

$$\mathbf{E}[h(\mathbf{X})] = \int_{\mathbb{R}^m} h(\mathbf{x}) f(\mathbf{x}) \, \mathrm{d}\mathbf{x}$$

if

$$\int_{\mathbb{R}^m} |h(\mathbf{x})| f(\mathbf{x}) \, \mathrm{d}\mathbf{x} < \infty.$$

Let $\langle \cdot, \cdot \rangle$ denote the scalar product in $\mathbb{R}^m$, i.e.,

$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{m} \mathbf{x}_i \mathbf{y}_i = \mathbf{x}^T \mathbf{y}$$

for $\mathbf{x} = (\mathbf{x}_1, \ldots, \mathbf{x}_m), \mathbf{y} = (\mathbf{y}_1, \ldots, \mathbf{y}_m) \in \mathbb{R}^m$. We write $\stackrel{d}{=}$ to indicate that two random elements follow the same distribution. The next result, which we will not prove, says that the distribution of a random vector is uniquely determined by the distributions of its one-dimensional projections.

Theorem 2.23. Let $\mathbf{X}$ and $\mathbf{Y}$ be random vectors in $\mathbb{R}^m$. If, for all $\mathbf{a} \in \mathbb{R}^m$, $\langle \mathbf{a}, \mathbf{X} \rangle \stackrel{d}{=} \langle \mathbf{a}, \mathbf{Y} \rangle$, then $\mathbf{X} \stackrel{d}{=} \mathbf{Y}$.

Next we consider transformations of the multivariate normal distribution.

Theorem 2.24. Let $\mathbf{X} = (\mathbf{X}_1, \ldots, \mathbf{X}_m)$ be a random vector in $\mathbb{R}^m$ such that $\mathbf{X} \sim \mathrm{Normal}(\boldsymbol{\mu}, \Sigma)$ with $\boldsymbol{\mu} \in \mathbb{R}^m$ and $\Sigma \in \mathbb{R}^{m \times m}$ positive semi-definite.

(i) For $\mathbf{a} \in \mathbb{R}^m$, $\langle \mathbf{a}, \mathbf{X} \rangle \sim \mathrm{Normal}(\mathbf{a}^T \boldsymbol{\mu}, \mathbf{a}^T \Sigma \mathbf{a})$.

(ii) For $A \in \mathbb{R}^{n \times m}$, $A\mathbf{X} \sim \mathrm{Normal}(A\boldsymbol{\mu}, A\Sigma A^T)$.

(iii) If Σ is a diagonal matrix (i.e., $\mathbf{Cov}(\mathbf{X}_i, \mathbf{X}_j) = 0$ for $i, j \in \{1, \ldots, m\}$ with $i \neq j$), then $\mathbf{X}_1, \ldots, \mathbf{X}_m$ are independent.

(iv) For $1 \leq i_1 < \ldots < i_\ell \leq m$ with $\mathbf{Cov}(\mathbf{X}_{i_j}, \mathbf{X}_{i_k}) = 0$ for $j, k \in \{1, \ldots, \ell\}$ with $j \neq k$, $\mathbf{X}_{i_1}, \ldots, \mathbf{X}_{i_\ell}$ are independent.

Proof. Let $\mathbf{N}$ be an m-dimensional standard Gaussian random vector. Then we have

$$\langle \mathbf{a}, \mathbf{X} \rangle \stackrel{d}{=} \langle \mathbf{a}, \Sigma^{1/2}\mathbf{N} + \boldsymbol{\mu} \rangle = \mathbf{a}^T \Sigma^{1/2}\mathbf{N} + \mathbf{a}^T \boldsymbol{\mu} = (\mathbf{a}^T \Sigma^{1/2})\mathbf{N} + \mathbf{a}^T \boldsymbol{\mu}.$$

Since it is a linear combination of i.i.d. standard normal random variables, $(\mathbf{a}^T \Sigma^{1/2})\mathbf{N}$ follows a normal distribution with mean zero and variance

$$(\mathbf{a}^T \Sigma^{1/2})(\mathbf{a}^T \Sigma^{1/2})^T = \mathbf{a}^T \Sigma^{1/2} (\Sigma^{1/2})^T \mathbf{a} = \mathbf{a}^T \Sigma \mathbf{a},$$

which proves (i).

Let $\mathbf{Y} \sim \mathrm{Normal}(A\boldsymbol{\mu}, A\Sigma A^T)$. For $\mathbf{a} \in \mathbb{R}^n$ it follows from (i) that

$$\langle \mathbf{a}, \mathbf{Y} \rangle \sim \mathrm{Normal}(\mathbf{a}^T A\boldsymbol{\mu}, \mathbf{a}^T A\Sigma A^T \mathbf{a})$$

and

$$\begin{aligned}
\langle \mathbf{a}, A\mathbf{X} \rangle &\stackrel{d}{=} \langle \mathbf{a}, A(\Sigma^{1/2}\mathbf{N} + \boldsymbol{\mu}) \rangle \\
&= \mathbf{a}^T A(\Sigma^{1/2}\mathbf{N} + \boldsymbol{\mu}) = ((\Sigma^{1/2})^T A^T \mathbf{a})^T \mathbf{N} + \mathbf{a}^T A\boldsymbol{\mu} = \langle (\Sigma^{1/2})^T A^T \mathbf{a}, \mathbf{N} \rangle + \mathbf{a}^T A\boldsymbol{\mu} \\
&\sim \mathrm{Normal}(\mathbf{a}^T A\boldsymbol{\mu}, \mathbf{a}^T A\Sigma^{1/2}(\Sigma^{1/2})^T A^T \mathbf{a}) = \mathrm{Normal}(\mathbf{a}^T A\boldsymbol{\mu}, \mathbf{a}^T A\Sigma A^T \mathbf{a}).
\end{aligned}$$

Thus, by Theorem 2.23, we have $A\mathbf{X} \stackrel{d}{=} \mathbf{Y}$, which yields (ii).

Assume that Σ is a diagonal matrix. Let $d_1, \ldots, d_m$ be the diagonal elements of Σ, which are variances and, thus, non-negative. Since $\mathbf{X} \stackrel{d}{=} \Sigma^{1/2}\mathbf{N} + \boldsymbol{\mu} = (\sqrt{d_1}\mathbf{N}_1 + \boldsymbol{\mu}_1, \ldots, \sqrt{d_m}\mathbf{N}_m + \boldsymbol{\mu}_m)$, we obtain

$$\begin{aligned}
F_{\mathbf{X}}(x_1, \ldots, x_m) &= \mathbf{P}(\mathbf{X}_1 \le x_1, \ldots, \mathbf{X}_m \le x_m) \\
&= \mathbf{P}(\sqrt{d_1}\mathbf{N}_1 + \boldsymbol{\mu}_1 \le x_1, \ldots, \sqrt{d_m}\mathbf{N}_m + \boldsymbol{\mu}_m \le x_m) \\
&= \prod_{i=1}^m \mathbf{P}(\sqrt{d_i}\mathbf{N}_i + \boldsymbol{\mu}_i \le x_i) = \prod_{i=1}^m \mathbf{P}(\mathbf{X}_i \le x_i) = \prod_{i=1}^m F_{\mathbf{X}_i}(x_i)
\end{aligned}$$

for $x_1, \ldots, x_m \in \mathbb{R}$, whence $\mathbf{X}_1, \ldots, \mathbf{X}_m$ are independent (see [7, Theorem 3.27]). This proves (iii).

Let A be the matrix whose row vectors are the i_1-th,..., i_ℓ-th unit vectors. By (ii), we have $(\mathbf{X}_{i_1}, \ldots, \mathbf{X}_{i_\ell}) = A\mathbf{X} \sim \mathrm{Normal}(A\boldsymbol{\mu}, A\Sigma A^T)$. Since $A\Sigma A^T$ is a diagonal matrix, (iv) follows from (iii). □

2.3 Stochastic convergence

Throughout this subsection, we consider different types of convergence for random variables and random vectors. First we recall convergence in probability and the weak law of large numbers.

Definition 2.25 (Convergence in probability)**.** Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables and let X be a random variable. We say that the sequence $(X_n)_{n\in\mathbb{N}}$ converges in probability to X if

$$\lim_{n\to\infty} \mathbf{P}(|X_n - X| \geq \varepsilon) = 0$$

for all $\varepsilon > 0$. We write $X_n \xrightarrow{\mathbf{P}} X$ as $n \to \infty$ for convergence in probability.

Theorem 2.26 (Weak law of large numbers)**.** Let $(X_i)_{i\in\mathbb{N}}$ be a sequence of i.i.d. random variables such that $\mathbf{E}[|X_1|] < \infty$ and define $S_n = \sum_{i=1}^n X_i$ for $n \in \mathbb{N}$. Then,

$$\frac{S_n}{n} \xrightarrow{\mathbf{P}} \mathbf{E}[X_1] \quad \text{as} \quad n \to \infty. \tag{2.6}$$

We will not prove this statement. Under the stronger assumption that the second moments are finite, a proof with the Chebyshev inequality is given in [7, Proof of Theorem 4.22].

In the following we always denote the underlying probability space by $(\Omega, \mathscr{F}, \mathbf{P})$ if not further specified.

Definition 2.27 ($\mathbf{P}$-a.s., $\mathbf{P}$-a.e.)**.** We say that an event $A \in \mathscr{F}$ holds $\mathbf{P}$-almost surely (we write $\mathbf{P}$-a.s.) if $\mathbf{P}(A) = 1$. An equivalent formulation is that A is satisfied for $\mathbf{P}$-almost every (we write $\mathbf{P}$-a.e.) $\omega \in \Omega$.

As an example, we have the following lemma.

Lemma 2.28. Let X be a random variable such that $\mathbf{E}[X^2] < \infty$. If $\mathbf{Var}(X) = 0$, then $X = \mathbf{E}[X]$ $\mathbf{P}$-a.s.

Proof. For each $u > 0$, we have by the Chebyshev inequality (see e.g. [7, Theorem 4.19]) that

$$\mathbf{P}(|X - \mathbf{E}[X]| \geq u) \leq \frac{\mathbf{Var}(X)}{u^2} = 0.$$

Thus, we obtain

$$\begin{aligned}\mathbf{P}(X \neq \mathbf{E}[X]) = \mathbf{P}(|X - \mathbf{E}[X]| > 0) &= \mathbf{P}\Big(\bigcup_{n=1}^{\infty} \{|X - \mathbf{E}[X]| \geq 1/n\}\Big) \\ &= \lim_{n\to\infty} \mathbf{P}(|X - \mathbf{E}[X]| \geq 1/n) = 0,\end{aligned}$$

where we used [7, Lemma 2.51] in the second last step. □

A further notion of convergence for random variables is almost sure convergence.

Definition 2.29 (Almost sure convergence)**.** We say that a sequence of random variables $(X_n)_{n\in\mathbb{N}}$ converges almost surely to a random variable X (we write $X_n \xrightarrow{a.s.} X$ as $n \to \infty$) if

$$\mathbf{P}(\lim_{n\to\infty} X_n = X) = \mathbf{P}(\{\omega \in \Omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\}) = 1,$$

i.e., $\lim_{n\to\infty} X_n = X$ $\mathbf{P}$-a.s. or equivalently $\lim_{n\to\infty} X_n(\omega) = X(\omega)$ for $\mathbf{P}$-a.e. $\omega \in \Omega$.

Note that almost sure convergence implies convergence in probability.

Lemma 2.30. Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables and let X be a random variable. If $X_n \xrightarrow{a.s.} X$ as $n \to \infty$, then $X_n \xrightarrow{\mathbf{P}} X$ as $n \to \infty$.

Proof. Let $\varepsilon > 0$. For $n \in \mathbb{N}$ we have that

$$\mathbf{P}(|X_n - X| \geq \varepsilon) \leq \mathbf{P}(\bigcup_{m\geq n} \{|X_m - X| \geq \varepsilon\}).$$

From [7, Lemma 2.51] and the definition of almost sure convergence it follows that

$$\begin{aligned}\lim_{n\to\infty} \mathbf{P}(\bigcup_{m\geq n} |X_m - X| \geq \varepsilon) &= \mathbf{P}(\bigcap_{n\in\mathbb{N}} \bigcup_{m\geq n} \{|X_m - X| \geq \varepsilon\}) \\ &\leq \mathbf{P}(\{\omega \in \Omega : (X_n(\omega))_{n\in\mathbb{N}} \text{ does not converge to } X(\omega)\}) = 0.\end{aligned}$$

This implies

$$\lim_{n\to\infty} \mathbf{P}(|X_n - X| \geq \varepsilon) = 0$$

and, thus, proves $X_n \xrightarrow{\mathbf{P}} X$ as $n \to \infty$. □

The next example shows that almost sure convergence is not implied by convergence in probability. Thus, we can think of almost sure convergence as a stronger type of convergence than convergence in probability.

Example 2.31. We consider the probability space $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbf{P})$ with $\mathbf{P}$ the uniform distribution on $[0, 1]$. For $n \in \mathbb{N}$ with $n = 2^k + j$, $k \in \mathbb{N}_0$ and $j \in \{0, 1, \ldots, 2^k - 1\}$, we define

$$X_n(\omega) = \begin{cases} 1, & \omega \in [j2^{-k}, (j+1)2^{-k}), \\ 0, & \text{otherwise}, \end{cases}$$

for $\omega \in \mathbb{R}$. This means that the $(X_n)_{n\in\mathbb{N}}$ are indicator functions of intervals which move from the left to the right of $[0, 1]$. After reaching the right side, the length is halved and the intervals start again on the left side. We have

$$\mathbf{P}(|X_n - 0| > 0) = \mathbf{P}([j2^{-k}, (j+1)2^{-k})) = 2^{-k},$$

whence $X_n \xrightarrow{\mathbf{P}} 0$ as $n \to \infty$. For each $\omega \in [0, 1)$, the sequence $(X_n(\omega))_{n\in\mathbb{N}}$ contains infinitely many ones and zeros, whence it does not converge. Thus, we do not have $X_n \xrightarrow{a.s.} 0$ as $n \to \infty$.

Our most important result for almost sure convergence is the strong law of large numbers, which we will not prove.

Theorem 2.32 (Strong law of large numbers)**.** Let $(X_i)_{i\in\mathbb{N}}$ be a sequence of i.i.d. random variables such that $\mathbf{E}[|X_1|] < \infty$ and define $S_n = \sum_{i=1}^n X_i$ for $n \in \mathbb{N}$. Then,

$$\frac{S_n}{n} \xrightarrow{a.s.} \mathbf{E}[X_1] \quad \text{as} \quad n \to \infty.$$

The strong law of large numbers deals with the same situation as the weak law of large numbers. By Lemma 2.30, the weak law of large numbers (see Theorem 2.26) is a consequence of the strong law of large numbers, which explains the terms weak and strong.

Example 2.33. Let $(X_i)_{i\in\mathbb{N}}$ be a sequence of i.i.d. random variables. In order to estimate the probability $\mathbf{P}(X_1 \le u)$ for some $u \in \mathbb{R}$ from the first n random variables, it is natural to consider the estimator

$$T_n = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \le u\},$$

which is the fraction of the first n random variables whose value is at most u. From the strong law of large numbers it follows that

$$T_n = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}\{X_i \le u\} \xrightarrow{a.s.} \mathbf{E}[\mathbf{1}\{X_1 \le u\}] = \mathbf{P}(X_1 \le u) \quad \text{as} \quad n \to \infty,$$

i.e., the estimators $(T_n)_{n\in\mathbb{N}}$ converge almost surely to the desired quantity.

A further type of convergence for random variables is L^p-convergence.

Definition 2.34. For $p > 0$ let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables and let X be a random variable such that $\mathbf{E}[|X_n|^p] < \infty$ for all $n \in \mathbb{N}$ and $\mathbf{E}[|X|^p] < \infty$. We say that $(X_n)_{n\in\mathbb{N}}$ converges in L^p to X if $\mathbf{E}[|X_n - X|^p] \to 0$ as $n \to \infty$ (we write $X_n \xrightarrow{L^p} X$ as $n \to \infty$).

Note that L^p-convergence implies convergence in probability.

Lemma 2.35. For a fixed $p > 0$ let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables and let X be a random variable such that $X_n \xrightarrow{L^p} X$ as $n \to \infty$. Then, $X_n \xrightarrow{\mathbf{P}} X$ as $n \to \infty$.

Proof. Exercise. □

The concepts of convergence in probability, almost sure convergence and L^p-convergence can be easily extended to random vectors and random vectors converge in these senses if and only if their components converge.

In the following, we study convergence in distribution of random vectors. Recall that a function $h : \mathbb{R}^m \to \mathbb{R}^\ell$ is continuous in $\mathbf{x} \in \mathbb{R}^m$ if

$$\lim_{\mathbf{y}\to\mathbf{x}} h(\mathbf{y}) = h(\mathbf{x}).$$

Definition 2.36 (Convergence in distribution)**.** Let $(\mathbf{X}_n)_{n\in\mathbb{N}}$ be a sequence of random vectors in $\mathbb{R}^m$ and let $\mathbf{X}$ be a random vector in $\mathbb{R}^m$. We say that $(\mathbf{X}_n)_{n\in\mathbb{N}}$ converges in distribution to $\mathbf{X}$ if

$$\lim_{n\to\infty} F_{\mathbf{X}_n}(\mathbf{t}) = F_{\mathbf{X}}(\mathbf{t})$$

for all $\mathbf{t} \in \mathbb{R}^m$ such that $\mathbf{t}$ is a continuity point of $F_{\mathbf{X}}$. For convergence in distribution we write $\mathbf{X}_n \xrightarrow{d} \mathbf{X}$ as $n \to \infty$.

For $m = 1$ in the previous definition, one obtains convergence in distribution of random variables. For random variables convergence in probability implies convergence in distribution and (this can be also extended to random vectors).

Lemma 2.37. Let $(X_n)_{n\in\mathbb{N}}$ be a sequence of random variables and let X be a random variable. If $X_n \xrightarrow{\mathbf{P}} X$ as $n \to \infty$, then $X_n \xrightarrow{d} X$ as $n \to \infty$.

Proof. Let u be a continuity point of F_X and let $\varepsilon > 0$. For $n \in \mathbb{N}$ we have

$$\mathbf{P}(X_n \leq u) \leq \mathbf{P}(X \leq u + \varepsilon) + \mathbf{P}(|X_n - X| \geq \varepsilon)$$

and

$$\mathbf{P}(X \leq u - \varepsilon) \leq \mathbf{P}(X_n \leq u) + \mathbf{P}(|X_n - X| \geq \varepsilon).$$

The convergence in probability implies

$$\lim_{n\to\infty} \mathbf{P}(|X_n - X| \geq \varepsilon) = 0,$$

whence $n \to \infty$ leads to

$$\mathbf{P}(X \leq u - \varepsilon) \leq \liminf_{n\to\infty} \mathbf{P}(X_n \leq u) \leq \limsup_{n\to\infty} \mathbf{P}(X_n \leq u) \leq \mathbf{P}(X \leq u + \varepsilon).$$

This can be rewritten as

$$F_X(u - \varepsilon) \leq \liminf_{n\to\infty} F_{X_n}(u) \leq \limsup_{n\to\infty} F_{X_n}(u) \leq F_X(u + \varepsilon).$$

Since u is a continuity point of F_X, we have $\lim_{\varepsilon\to 0} F_X(u-\varepsilon) = \lim_{\varepsilon\to 0} F_X(u+\varepsilon) = F_X(u)$ and $\varepsilon \to 0$ yields

$$\lim_{n\to\infty} F_{X_n}(u) = F_X(u),$$

which completes the proof. □

The most prominent example for convergence in distribution of random variables is the central limit theorem (see e.g. [7, Theorem 4.30]).

Theorem 2.38 (Central limit theorem)**.** Let $(X_i)_{i\in\mathbb{N}}$ be a sequence of i.i.d. random variables such that $\mathbf{E}[X_1^2] < \infty$ and $\mathbf{Var}[X_1] > 0$, define $S_n = \sum_{i=1}^n X_i$ for $n \in \mathbb{N}$ and let $N \sim \text{Normal}(0, 1)$. Then,

$$\lim_{n\to\infty} \mathbf{P}\left(\frac{S_n - n\mathbf{E}[X_1]}{\sqrt{\mathbf{Var}[X_1]}\sqrt{n}} \leq t\right) = \Phi(t)$$

for $t \in \mathbb{R}$ and

$$\frac{S_n - n\mathbf{E}[X_1]}{\sqrt{\mathbf{Var}[X_1]}\sqrt{n}} \xrightarrow{d} N \quad \text{as} \quad n \to \infty.$$

Next we define the characteristic function of a random vector. Recall that $\mathbf{i}$ stands for the imaginary unit, while $\langle \cdot, \cdot \rangle$ denotes the scalar product in $\mathbb{R}^m$. By the expectation of a complex valued random variable we mean the corresponding expectations of the real and the imaginary part.

Definition 2.39 (Characteristic function)**.** The characteristic function $\varphi_{\mathbf{X}} : \mathbb{R}^m \to \mathbb{C}$ of a random vector $\mathbf{X}$ in $\mathbb{R}^m$ is given by

$$\varphi_{\mathbf{X}}(\mathbf{t}) = \mathbf{E}[e^{\mathbf{i}\langle \mathbf{t}, \mathbf{X} \rangle}]$$

for $\mathbf{t} \in \mathbb{R}^m$.

Since the absolute values of the real and the imaginary part of $e^{\mathbf{i}a} = \cos(a) + \mathbf{i}\sin(a)$ are bounded by one for all $a \in \mathbb{R}$, the characteristic function is well defined for all random vectors and all arguments. This is an advantage compared to moment generating functions, where the imaginary unit is not present in the exponent.

The following theorem, which we will not prove, provides different characterisations of convergence in distribution.

Theorem 2.40. Let $(\mathbf{X}_n)_{n\in\mathbb{N}}$ be a sequence of random vectors in $\mathbb{R}^m$ and let $\mathbf{X}$ be a random vector in $\mathbb{R}^m$. Then the following statements are equivalent:

(i) $\mathbf{X}_n \xrightarrow{d} \mathbf{X} \quad \text{as} \quad n \to \infty$

(ii) For all $\mathbf{t} \in \mathbb{R}^m$,

$$\lim_{n\to\infty} \varphi_{\mathbf{X}_n}(\mathbf{t}) = \varphi_{\mathbf{X}}(\mathbf{t}).$$

(iii) For all continuous bounded functions $h : \mathbb{R}^m \to \mathbb{R}$,

$$\lim_{n\to\infty} \mathbb{E}[h(\mathbf{X}_n)] = \mathbb{E}[h(\mathbf{X})].$$

The next result characterises convergence in distribution of random vectors via convergence in distribution of one-dimensional projections.

Theorem 2.41 (Cramér-Wold device)**.** Let $(\mathbf{X}_n)_{n\in\mathbb{N}}$ be a sequence of random vectors in $\mathbb{R}^m$ and let $\mathbf{X}$ be a random vector in $\mathbb{R}^m$. Then, the following statements are equivalent:

(i) $\mathbf{X}_n \xrightarrow{d} \mathbf{X} \quad \text{as} \quad n \to \infty$

(ii) For all $\mathbf{a} \in \mathbb{R}^m$,

$$\langle \mathbf{a}, \mathbf{X}_n \rangle \xrightarrow{d} \langle \mathbf{a}, \mathbf{X} \rangle \quad \text{as} \quad n \to \infty.$$

Proof. First we show that (i) implies (ii). We fix $\mathbf{a} \in \mathbb{R}^m$. For $s \in \mathbb{R}$, it follows from Theorem 2.40 that

$$\begin{aligned}\lim_{n\to\infty} \varphi_{\langle \mathbf{a}, \mathbf{X}_n \rangle}(s) &= \lim_{n\to\infty} \mathbf{E}[e^{\mathbf{i}s\langle \mathbf{a}, \mathbf{X}_n \rangle}] = \lim_{n\to\infty} \mathbf{E}[e^{\mathbf{i}\langle s\mathbf{a}, \mathbf{X}_n \rangle}] = \lim_{n\to\infty} \varphi_{\mathbf{X}_n}(s\mathbf{a}) \\ &= \varphi_{\mathbf{X}}(s\mathbf{a}) = \mathbf{E}[e^{\mathbf{i}\langle s\mathbf{a}, \mathbf{X} \rangle}] = \mathbf{E}[e^{\mathbf{i}s\langle \mathbf{a}, \mathbf{X} \rangle}] = \varphi_{\langle \mathbf{a}, \mathbf{X} \rangle}(s).\end{aligned}$$

Thus, again by Theorem 2.40, $\langle \mathbf{a}, \mathbf{X}_n \rangle \xrightarrow{d} \langle \mathbf{a}, \mathbf{X} \rangle$ as $n \to \infty$. For the other direction let $\mathbf{t} \in \mathbb{R}^m$. From $\langle \mathbf{t}, \mathbf{X}_n \rangle \xrightarrow{d} \langle \mathbf{t}, \mathbf{X} \rangle$ as $n \to \infty$ and Theorem 2.40 we obtain

$$\lim_{n\to\infty} \varphi_{\mathbf{X}_n}(\mathbf{t}) = \lim_{n\to\infty} \mathbf{E}[e^{\mathbf{i}\langle \mathbf{t}, \mathbf{X}_n \rangle}] = \lim_{n\to\infty} \varphi_{\langle \mathbf{t}, \mathbf{X}_n \rangle}(1) = \varphi_{\langle \mathbf{t}, \mathbf{X} \rangle}(1) = \mathbf{E}[e^{\mathbf{i}\langle \mathbf{t}, \mathbf{X} \rangle}] = \varphi_{\mathbf{X}}(\mathbf{t})$$

and, thus, $\mathbf{X}_n \xrightarrow{d} \mathbf{X}$ as $n \to \infty$. □

By applying the Cramér-Wold device, we can extend the central limit theorem to the multivariate case.

Theorem 2.42 (Multivariate central limit theorem)**.** Let $(\mathbf{X}_j)_{j\in\mathbb{N}}$ be a sequence of i.i.d. random vectors in $\mathbb{R}^m$ with $\mathbf{E}[\|\mathbf{X}_1\|^2] < \infty$ and covariance matrix $\Sigma \in \mathbb{R}^{m\times m}$. Define $\mathbf{S}_n = \sum_{j=1}^n \mathbf{X}_j$ for $n \in \mathbb{N}$ and let $\mathbf{N} \sim \mathrm{Normal}(\mathbf{0}, \Sigma)$. Then,

$$\frac{1}{\sqrt{n}}\big(\mathbf{S}_n - n\mathbf{E}[\mathbf{X}_1]\big) \xrightarrow{d} \mathbf{N} \quad \text{as} \quad n \to \infty.$$

Proof. Fix $\mathbf{a} \in \mathbb{R}^m$. First assume that $\mathbf{a}^T\Sigma\mathbf{a} > 0$. By Lemma 2.11, we have $\mathbf{E}[\langle \mathbf{a}, \mathbf{X}_1 \rangle] = \langle \mathbf{a}, \mathbf{E}[\mathbf{X}_1] \rangle$ and $\mathbf{Var}(\langle \mathbf{a}, \mathbf{X}_1 \rangle) = \mathbf{a}^T\Sigma\mathbf{a}$. For $N \sim \mathrm{Normal}(0,1)$ it follows from Theorem 2.38 that

$$\frac{1}{\sqrt{\mathbf{a}^T\Sigma\mathbf{a}}}\langle \mathbf{a}, \frac{1}{\sqrt{n}}\big(\mathbf{S}_n - n\mathbf{E}[\mathbf{X}_1]\big)\rangle = \frac{1}{\sqrt{n}\sqrt{\mathbf{Var}(\langle \mathbf{a}, \mathbf{X}_1 \rangle)}}\Big(\sum_{j=1}^n \langle \mathbf{a}, \mathbf{X}_j \rangle - n\mathbf{E}[\langle \mathbf{a}, \mathbf{X}_1 \rangle]\Big) \xrightarrow{d} N$$

as $n \to \infty$. Since $\sqrt{\mathbf{a}^T\Sigma\mathbf{a}}N$ has the same distribution as $\langle \mathbf{a}, \mathbf{N} \rangle$, this is equivalent to

$$\langle \mathbf{a}, \frac{1}{\sqrt{n}}\big(\mathbf{S}_n - n\mathbf{E}[\mathbf{X}_1]\big)\rangle \xrightarrow{d} \langle \mathbf{a}, \mathbf{N} \rangle \quad \text{as} \quad n \to \infty. \tag{2.7}$$

In case that $\mathbf{a}^T\Sigma\mathbf{a} = 0$, we have $\mathbf{Var}[\langle \mathbf{a}, \mathbf{X}_1 \rangle] = \mathbf{a}^T\Sigma\mathbf{a} = 0$ and, thus, by Lemma 2.28, $\langle \mathbf{a}, \mathbf{X}_1 \rangle = \mathbf{E}[\langle \mathbf{a}, \mathbf{X}_1 \rangle]$ $\mathbf{P}$-a.s. and $\langle \mathbf{a}, \mathbf{N} \rangle = 0$ $\mathbf{P}$-a.s., whence (2.7) is satisfied in this situation as well. Now Theorem 2.41 completes the proof. □

The next result tells us that convergence in distribution carries over to continuous transformations.

Theorem 2.43 (Continuous mapping theorem)**.** Let $(\mathbf{X}_n)_{n\in\mathbb{N}}$ be a sequence of random vectors in $\mathbb{R}^m$ and let $\mathbf{X}$ be a random vector in $\mathbb{R}^m$ such that $\mathbf{X}_n \xrightarrow{d} \mathbf{X}$ as $n \to \infty$. Let $g : \mathbb{R}^m \to \mathbb{R}^\ell$ be a continuous function. Then,

$$g(\mathbf{X}_n) \xrightarrow{d} g(\mathbf{X}) \quad \text{as} \quad n \to \infty.$$

Proof. Exercise. □

Note that the statement of the continuous mapping theorem is also true for convergence in probability or for almost sure convergence, i.e., if $\xrightarrow{d}$ is everywhere replaced by $\xrightarrow{P}$ or if $\xrightarrow{d}$ is everywhere replaced by $\xrightarrow{a.s.}$. This does not hold for L^p-convergence.

2.4 Empirical distribution functions and the Glivenko-Cantelli theorem

In this subsection, we introduce the empirical distribution function and study its convergence.

Definition 2.44 (Empirical distribution function)**.** For $n \in \mathbb{N}$ and random variables $X_1, \dots, X_n$ the empirical distribution function F_n is given by

$$F_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{X_i \leq x\}$$

for $x \in \mathbb{R}$.

For a given $x \in \mathbb{R}$, the value of the empirical distribution function is the ratio of the $X_1, \dots, X_n$ that are at most x. Since it depends on the random variables $X_1, \dots, X_n$, F_n is a random function. For given realisations of $X_1, \dots, X_n$, F_n is even a distribution function. Thus, F_n is a random element in the space of distribution functions.

In statistics, the underlying distribution of a random sample is often unknown and the idea is to approximate it by the empirical distribution function. The following result tells us that the empirical distribution function converges to the underlying distribution function in the i.i.d. case.

Theorem 2.45 (Glivenko-Cantelli theorem)**.** Let $(X_i)_{i\in\mathbb{N}}$ be a sequence of i.i.d. random variables with distribution function F and let F_n be the empirical distribution function with respect to $X_1, \dots, X_n$ for $n \in \mathbb{N}$. Then,

$$\sup_{x\in\mathbb{R}} |F_n(x) - F(x)| \xrightarrow{a.s.} 0 \quad \text{as} \quad n \to \infty.$$

Proof. In this proof, we only treat the situation in detail that F is continuous, which we assume from now on. For a function $g : \mathbb{R} \to \mathbb{R}$ we define $\|g\|_\infty := \sup_{x\in\mathbb{R}} |g(x)|$.

Let $\varepsilon > 0$ be given. By the assumption that F is continuous there exist $m \in \mathbb{N}$ and $x_1, \dots, x_m \in \mathbb{R}^m$ such that

$$F(x_{j+1}) - F(x_j) \leq \varepsilon \quad \text{for} \quad j \in \{1, \dots, m-1\}$$

as well as $F(x_1) \leq \varepsilon$ and $1 - F(x_m) \leq \varepsilon$. By the monotonicity of F_n and F, we have for $x \in [x_j, x_{j+1}]$ with $j \in \{1, \dots, m-1\}$,

$$F_n(x) - F(x) \leq F_n(x_{j+1}) - F(x_j) = F_n(x_{j+1}) - F(x_{j+1}) + F(x_{j+1}) - F(x_j)$$

and

$$F_n(x) - F(x) \geq F_n(x_j) - F(x_{j+1}) = F_n(x_j) - F(x_j) + F(x_j) - F(x_{j+1}).$$

This implies that

$$|F_n(x) - F(x)| \leq \max\{|F_n(x_{j+1}) - F(x_{j+1})|, |F_n(x_j) - F(x_j)|\} + \varepsilon.$$

Similarly, we obtain, for $x \leq x_1$,

$$|F_n(x) - F(x)| \leq \max\{F_n(x), F(x)\} \leq \max\{F_n(x_1), F(x_1)\} \leq |F_n(x_1) - F(x_1)| + \varepsilon$$

and for $x \geq x_m$,

$$\begin{aligned} |F_n(x) - F(x)| &= |1 - F(x) - (1 - F_n(x))| \\ &\leq \max\{1 - F(x), 1 - F_n(x)\} \leq \max\{1 - F(x_m), 1 - F_n(x_m)\} \\ &\leq |1 - F(x_m)| + |F(x_m) - F_n(x_m)| \leq |F_n(x_m) - F(x_m)| + \varepsilon. \end{aligned}$$

Combining the estimates leads to

$$\|F_n - F\|_\infty = \sup_{x \in \mathbb{R}} |F_n(x) - F(x)| \leq \max_{j \in \{1,\dots,m\}} |F_n(x_j) - F(x_j)| + \varepsilon.$$

By the strong law of large numbers we have that, for each $j \in \{1, \dots, m\}$,

$$F_n(x_j) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{X_i \leq x_j\} \xrightarrow{a.s.} \mathbf{E}[\mathbf{1}\{X_1 \leq x_j\}] = \mathbf{P}(X_1 \leq x_j) = F(x_j) \quad \text{as} \quad n \to \infty.$$

This implies that

$$\max_{j \in \{1,\dots,m\}} |F_n(x_j) - F(x_j)| \xrightarrow{a.s.} 0 \quad \text{as} \quad n \to \infty.$$

Thus we have shown that

$$\mathbf{P}\big(\limsup_{n \to \infty} \|F_n - F\|_\infty \leq \varepsilon\big) = 1.$$

Since this holds for any $\varepsilon > 0$, we obtain

$$\mathbf{P}\big(\lim_{n \to \infty} \|F_n - F\|_\infty = 0\big) = 1,$$

which concludes the proof. □

2.5 Statistical models

The typical situation in statistics is that some data or observations are given that are realisations of an unknown probability distribution. Our aim is to infer from the observations to the distribution. In order to describe this situation formally and to specify the possible underlying distributions, we introduce statistical models.

Definition 2.46 (Statistical model)**.** A statistical model $(\mathcal{X}, \mathscr{F}_\mathcal{X}, (\mathbf{P}_\vartheta)_{\vartheta \in \Theta})$ consists of a sample space $\mathcal{X}$, a σ-algebra $\mathscr{F}_\mathcal{X}$ on $\mathcal{X}$ and a family of probability measures $(\mathbf{P}_\vartheta)_{\vartheta \in \Theta}$ on $\mathcal{X}$ with an index set Θ.

The sample space provides the set of all possible observations, which we denote as samples in the sequel. Note that $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, \mathbf{P}_\vartheta)$ is a probability space for each fixed $\vartheta \in \Theta$, whence a statistical model is a family of probability spaces. In statistics usually a sample $x \in \mathcal{X}$ is given and it is assumed that x is a realisation of a distribution $\mathbf{P}_{\vartheta_0}$ for some fixed $\vartheta_0 \in \Theta$, which is unknown. The goal of the statistical procedures studied in the following is to learn from the sample something about the unknown ϑ_0.

Throughout this class we usually consider the situation that we have n observations, i.e., that our sample x is of the form $x = (x_1, \dots, x_n)$, where $x_1, \dots, x_n$ all belong to a set A. In this case, we have the sample space $\mathcal{X} = A^n$. We call n the sample size.

Typically, one assumes that $x_1, \dots, x_n$ are independent realisations of the same underlying distribution. This means that $x_1, \dots, x_n$ are realisations of independent random elements $X_1, \dots, X_n \sim \widetilde{\mathbf{P}}_\vartheta$ with $\vartheta \in \Theta$. Here, $\widetilde{\mathbf{P}}_\vartheta$ is the marginal distribution, while $\mathbf{P}_\vartheta$ denotes the joint distribution of $X_1, \dots, X_n$. In order to describe this situation, we write $\mathbf{P}_\vartheta = \widetilde{\mathbf{P}}_\vartheta^{\otimes n}$. More generally, we denote by $\widehat{\mathbf{P}}^{\otimes n}$ the joint distribution of n independent random elements such that each of them is distributed according to $\widehat{\mathbf{P}}$. For example, $\mathrm{Normal}(0,1)^{\otimes n}$ stands for the joint distribution of n independent standard normal random variables.

Two prominent examples of statistical models are the following:

Example 2.47. The statistical model $(\{0,1\}^n, \mathcal{P}(\{0,1\}^n), (\mathrm{Bernoulli}(p)^{\otimes n})_{p\in[0,1]})$ for some fixed $n \in \mathbb{N}$ describes the situation that a random experiment with unknown success probability p is repeated n times independently. This situation occurs, for example, if n patients get some treatment and one observes whether the treatment is successful.

Example 2.48 (Normal samples)**.** Assume that our sample consists of n realisations of independent normal random variables with identical expectations and means. Depending on whether the variance is known or not, we distinguish the following cases:

(i) *Normal sample with unknown mean and known variance.* The statistical model is $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathrm{Normal}(\mu, \sigma_0^2)^{\otimes n})_{\mu\in\mathbb{R}})$ for some fixed and given $\sigma_0 \in [0, \infty)$.

(ii) *Normal sample with unknown mean and variance.* The statistical model is $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathrm{Normal}(\mu, \sigma^2)^{\otimes n})_{\mu\in\mathbb{R},\sigma\in[0,\infty)})$.

One obtains these statistical models in situations with repeated independent measurements of a deterministic quantity with random measurement errors. Here, μ is the deterministic quantity one aims to determine, but one can only measure μ plus some normal random variable with mean zero. If the variance of the measurement errors is known, one is in the first situation, while in case of unknown variance one is in the second situation, which is harder to deal with but usually more realistic.

Chapter 3

Point estimation

3.1 Point estimators and their properties

Let $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be the underlying statistical model if not specified otherwise. As before we assume that a sample x comes from a distribution $\mathbf{P}_\vartheta$ with an unknown $\vartheta \in \Theta$. The idea of point estimation is to make a guess for the unknown ϑ based on x. Often one does not study ϑ but a transformation of it. For a given map $\tau : \Theta \to \mathbb{R}$, we consider point estimators for $\tau(\vartheta)$ in the following. We start with the formal definition of a point estimator.

Definition 3.1 (Point estimator)**.** A measurable map $T : \mathcal{X} \to \mathbb{R}$ is called a (point) estimator.

In the following we only use the term estimator. For a given sample x, $T(x)$ is a real number. For each $\vartheta \in \Theta$ we can think of T as a random variable on the probability space $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, \mathbf{P}_\vartheta)$. We illustrate Definition 3.1 by several examples.

Example 3.2. For $n \in \mathbb{N}$ and a family of probabilities $(\widetilde{\mathbf{P}}_\vartheta)_{\vartheta\in\Theta}$ on $\mathbb{R}$ we consider the statistical model $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\widetilde{\mathbf{P}}^{\otimes n})_{\vartheta\in\Theta})$. Then, the following maps are estimators:

a) $T^{(1)} : \mathbb{R}^n \ni x \mapsto T^{(1)}(x) = \frac{1}{2}$

b) $T^{(2)} : \mathbb{R}^n \ni x = (x_1, \dots, x_n) \mapsto T^{(2)}(x) = x_1$

c) $T^{(3)} : \mathbb{R}^n \ni x = (x_1, \dots, x_n) \mapsto T^{(3)}(x) = \frac{1}{n}\sum_{i=1}^n x_i =: \overline{x}$

d) $T^{(4)} : \mathbb{R}^n \ni x = (x_1, \dots, x_n) \mapsto T^{(4)}(x) = \frac{1}{n-1}\sum_{i=1}^n (x_i - \overline{x})^2 =: s_x^2$ if $n \geq 2$

The estimator $T^{(3)}$ is denoted as sample mean, while $T^{(4)}$ is called sample variance. Note that the map $x \mapsto \tau(\vartheta)$ is not an estimator since it depends on ϑ.

Although formally an estimator, the estimator $T^{(1)}$ from the previous example does not make sense as it completely ignores the given sample. This leads to the question what a good estimator is. Of course this heavily depends on the particular quantity $\tau(\vartheta)$

we are interested in. The next definition provides several concepts and quantities in this direction. In the following, we denote by $\mathbf{E}_\vartheta$ the expectation with respect to $\mathbf{P}_\vartheta$.

Definition 3.3 (unbiased, bias, mean squared error)**.** Let $T : \mathcal{X} \to \mathbb{R}$ be an estimator such that $\mathbf{E}_\vartheta[|T|] < \infty$ for all $\vartheta \in \Theta$. We say that T is an unbiased estimator for $\tau(\vartheta)$ if

$$\mathbf{E}_\vartheta[T] = \tau(\vartheta) \quad \text{for all} \quad \vartheta \in \Theta.$$

The quantity $B_T(\vartheta) = \mathbf{E}_\vartheta[T] - \tau(\vartheta)$ is called bias of T at ϑ.

If $\mathbf{E}_\vartheta[T^2] < \infty$ for all $\vartheta \in \Theta$,

$$M_T(\vartheta) = \mathbf{E}_\vartheta[(T - \tau(\vartheta))^2]$$

is denoted as mean squared error of T at ϑ.

One can think of an unbiased estimator as an estimator that provides on average the correct value, i.e., we do not have a systematic bias. The mean squared error measures the fluctuations of the estimator around $\tau(\vartheta)$, whence it is desirable to have a small mean squared error.

Remark 3.4. The mean squared error can be rewritten as

$$\begin{aligned} M_T(\vartheta) &= \mathbf{E}_\vartheta[(T - \tau(\vartheta))^2] = \mathbf{E}_\vartheta[(T - \mathbf{E}_\vartheta[T] + \mathbf{E}_\vartheta[T] - \tau(\vartheta))^2] \\ &= \mathbf{E}_\vartheta[(T - \mathbf{E}_\vartheta[T])^2] + 2\mathbf{E}_\vartheta[(T - \mathbf{E}_\vartheta[T])(\mathbf{E}_\vartheta[T] - \tau(\vartheta))] + (\mathbf{E}_\vartheta[T] - \tau(\vartheta))^2 \\ &= \mathbf{Var}_\vartheta(T) + B_T(\vartheta)^2, \end{aligned}$$

where $\mathbf{Var}_\vartheta$ denotes the variance with respect to $\mathbf{P}_\vartheta$. We see that for an unbiased estimator the right-hand side simplifies to its variance. Thus, for unbiased estimators it is desirable to have a small variance.

Example 3.5. We consider the estimators from Example 3.2. For $\vartheta \in \Theta$ let $X_1, \dots, X_n$ be i.i.d. random variables distributed according to $\widetilde{\mathbf{P}}_\vartheta$. Assume that $\mathbf{E}_\vartheta[|X_1|] < \infty$ for all $\vartheta \in \Theta$ and let $\tau(\vartheta) = \mathbf{E}_\vartheta[X_1]$, i.e., our goal is to estimate the expectation. Then, we obtain $\mathbf{E}_\vartheta[T^{(1)}] = \frac{1}{2}$, $\mathbf{E}_\vartheta[T^{(2)}] = \mathbf{E}_\vartheta[X_1]$ and

$$\mathbf{E}_\vartheta[T^{(3)}] = \mathbf{E}_\vartheta\left[\frac{1}{n}\sum_{i=1}^n X_i\right] = \frac{1}{n}\sum_{i=1}^n \mathbf{E}_\vartheta[X_i] = \mathbf{E}_\vartheta[X_1].$$

This means that $T^{(2)}$ and $T^{(3)}$ are unbiased, while $T^{(1)}$ is not unbiased and has the bias $B_{T^{(1)}}(\vartheta) = \frac{1}{2} - \mathbf{E}_\vartheta[X_1]$.

Under the stronger assumption $\mathbf{E}_\vartheta[X_1^2] < \infty$ for all $\vartheta \in \Theta$ we obtain, for $\vartheta \in \Theta$,

$$M_{T^{(1)}}(\vartheta) = \mathbf{E}_\vartheta\left[\left(\frac{1}{2} - \tau(\vartheta)\right)^2\right] = \left(\frac{1}{2} - \mathbf{E}_\vartheta[X_1]\right)^2,$$

$$M_{T^{(2)}}(\vartheta) = \mathbf{E}_\vartheta[(X_1 - \tau(\vartheta))^2] = \mathbf{E}_\vartheta[(X_1 - \mathbf{E}_\vartheta[X_1])^2] = \mathbf{Var}_\vartheta(X_1),$$

and

$$M_{T^{(3)}}(\vartheta) = \mathbf{E}_\vartheta\left[\left(\frac{1}{n}\sum_{i=1}^n X_i - \tau(\vartheta)\right)^2\right] = \mathbf{E}_\vartheta\left[\left(\frac{1}{n}\sum_{i=1}^n X_i - \mathbf{E}_\vartheta[X_1]\right)^2\right] = \mathbf{Var}_\vartheta\left(\frac{1}{n}\sum_{i=1}^n X_i\right)$$
$$= \frac{1}{n^2}\sum_{i=1}^n \mathbf{Var}_\vartheta(X_i) = \frac{1}{n}\mathbf{Var}_\vartheta(X_1).$$

Thus, $T^{(3)}$ has a smaller mean squared error than $T^{(2)}$, whence it might be a better estimator. This is reasonable since $T^{(2)}$ takes only x_1 into account and ignores $x_2, \ldots, x_n$ completely. The estimator $T^{(1)}$ is even worse.

If $\mathbf{E}_\vartheta[X_1^2] < \infty$ for all $\vartheta \in \Theta$, one can show that

$$\mathbf{E}_\vartheta[T^{(4)}] = \mathbf{Var}_\vartheta(X_1)$$

for all $\vartheta \in \Theta$ (see [7, Theorem 6.9]), whence $T^{(4)}$ is an unbiased estimator for $\tau(\vartheta) = \mathbf{Var}_\vartheta(X_1)$.

The next definition concerns the asymptotic behaviour of sequences of estimators for increasing sample size. Here, we consider convergence in probability and almost sure convergence. Since we have a whole family of probabilities $(\mathbf{P}_\vartheta)_{\vartheta\in\Theta}$ given by the statistical model, we write $\xrightarrow{\mathbf{P}_\vartheta}$ and $\xrightarrow{\mathbf{P}_\vartheta - a.s.}$ for these types of convergence.

Definition 3.6 (consistent, asymptotically unbiased)**.** Let $(\mathcal{X}, \mathscr{F}_\mathcal{X}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model such that $\mathbf{P}_\vartheta$ is the joint distribution of random variables $(X_i)_{i\in\mathbb{N}}$ for each $\vartheta \in \Theta$. Let $(T_n)_{n\in\mathbb{N}}$ be a sequence of estimators such that T_n only takes the realisations of $X_1, \ldots, X_n$ into account. We call $(T_n)_{n\in\mathbb{N}}$ weakly consistent for $\tau(\vartheta)$ if

$$T_n \xrightarrow{\mathbf{P}_\vartheta} \tau(\vartheta) \quad \text{as} \quad n \to \infty$$

for all $\vartheta \in \Theta$ and strongly consistent for $\tau(\vartheta)$ if

$$T_n \xrightarrow{\mathbf{P}_\vartheta - a.s} \tau(\vartheta) \quad \text{as} \quad n \to \infty$$

for all $\vartheta \in \Theta$. Assume that $\mathbf{E}_\vartheta[|T_n|] < \infty$ for all $\vartheta \in \Theta$ and $n \in \mathbb{N}$. We say that $(T_n)_{n\in\mathbb{N}}$ are asymptotically unbiased estimators for $\tau(\vartheta)$ if

$$\lim_{n\to\infty} \mathbf{E}_\vartheta[T_n] = \tau(\vartheta) \quad \text{for all} \quad \vartheta \in \Theta.$$

Both weak and strong consistency mean that the estimators converge to the true value as the sample size goes to infinity. Roughly speaking, consistency implies that a large sample size improves the estimation, which is an important paradigm in statistics. When one has a larger sample, more information is provided and it should be possible to draw better conclusions. Note that a sequence of unbiased estimators is obviously asymptotically unbiased.

Example 3.7. Let $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model such that $\mathbf{P}_\vartheta$ is the joint distribution of a sequence of i.i.d. random variables $(X_i)_{i\in\mathbb{N}}$ with $\mathbf{E}_\vartheta[|X_1|] < \infty$ for all $\vartheta \in \Theta$. For $n \in \mathbb{N}$ let $T_n^{(2)}$ and $T_n^{(3)}$ be defined as in Example 3.2 with respect to $X_1, \dots, X_n$.

Let $\vartheta \in \Theta$. For each $n \in \mathbb{N}$, we have $T_n^{(2)} = X_1$, which yields

$$T_n^{(2)} \xrightarrow{\mathbf{P}_\vartheta} X_1 \quad \text{and} \quad T_n^{(2)} \xrightarrow{\mathbf{P}_\vartheta-a.s.} X_1 \quad \text{as} \quad n \to \infty.$$

Hence, we do not have

$$T_n^{(2)} \xrightarrow{\mathbf{P}_\vartheta} \mathbf{E}_\vartheta[X_1] \quad \text{or} \quad T_n^{(2)} \xrightarrow{\mathbf{P}_\vartheta-a.s.} \mathbf{E}_\vartheta[X_1] \quad \text{as} \quad n \to \infty,$$

unless $X_1 = \mathbf{E}_\vartheta[X_1]$ $\mathbf{P}_\vartheta$-a.s. Thus, $(T_n^{(2)})_{n\in\mathbb{N}}$ are neither weakly nor strongly consistent estimators for $\mathbf{E}_\vartheta[X_1]$. Since they are even unbiased estimators for $\mathbf{E}_\vartheta[X_1]$, $(T_n^{(2)})_{n\in\mathbb{N}}$ are asymptotically unbiased estimators for $\mathbf{E}_\vartheta[X_1]$.

For each $\vartheta \in \Theta$ we have by the weak and the strong law of large numbers, Theorem 2.26 and Theorem 2.32,

$$T_n^{(3)} = \frac{1}{n}\sum_{i=1}^n X_i \xrightarrow{\mathbf{P}_\vartheta} \mathbf{E}_\vartheta[X_1] \quad \text{as} \quad n \to \infty$$

and

$$T_n^{(3)} = \frac{1}{n}\sum_{i=1}^n X_i \xrightarrow{\mathbf{P}_\vartheta-a.s.} \mathbf{E}_\vartheta[X_1] \quad \text{as} \quad n \to \infty.$$

Thus, $(T_n^{(3)})_{n\in\mathbb{N}}$ are weakly and strongly consistent estimators for $\mathbf{E}_\vartheta[X_1]$. As they are even unbiased, they are also asymptotically unbiased.

Assume that $\mathbf{E}_\vartheta[X_1^2] < \infty$ for all $\vartheta \in \Theta$ and let

$$\widetilde{T}_n^{(4)} = \frac{1}{n}\sum_{i=1}^n (X_i - \overline{X})^2 \quad \text{with} \quad \overline{X} = \frac{1}{n}\sum_{i=1}^n X_i,$$

i.e., $\widetilde{T}_n^{(4)} = \frac{n-1}{n} T_n^{(4)}$, where $T_n^{(4)}$ is the fourth estimator from Example 3.2. From the unbiasedness of $T_n^{(4)}$ we obtain that

$$\mathbf{E}_\vartheta[\widetilde{T}_n^{(4)}] = \frac{n-1}{n}\mathbf{E}_\vartheta[T_n^{(4)}] = \frac{n-1}{n}\mathbf{Var}_\vartheta(X_1)$$

so that $\widetilde{T}_n^{(4)}$ is not an unbiased estimator for $\mathbf{Var}_\vartheta(X_1)$. But we obtain

$$\lim_{n\to\infty} \mathbf{E}_\vartheta[\widetilde{T}_n^{(4)}] = \lim_{n\to\infty} \frac{n-1}{n}\mathbf{Var}_\vartheta(X_1) = \mathbf{Var}_\vartheta(X_1),$$

whence $(\widetilde{T}_n^{(4)})_{n\in\mathbb{N}}$ are asymptotically unbiased estimators for $\mathbf{Var}_\vartheta(X_1)$.

Remark 3.8. Throughout this section we assumed that we have a map $\tau : \Theta \to \mathbb{R}$, whence we estimated with $\tau(\vartheta)$ a real-valued quantity. The definition of an estimator and the other concepts introduced above can be directly extended to the situation that $\tau : \Theta \to \mathbb{R}^m$, i.e., our goal is to estimate a vector $\tau(\vartheta)$. This situation arises when Θ is a subset of $\mathbb{R}^m$ and our goal is to estimate ϑ itself, i.e., τ is the identity. For example, in case of a normal sample with unknown expectation and variance one is interested in estimating the expectation and the variance simultaneously, i.e., $\vartheta = (\mu, \sigma^2)$.

We can even allow that an estimator takes values in an arbitrary space $\mathcal{Y}$, i.e., we have a measurable map $T : \mathcal{X} \to \mathcal{Y}$.

So far we mostly used T to denote estimators. A common notation in statistics for an estimator for a parameter ϑ is $\hat{\vartheta}$.

3.2 Maximum likelihood estimators

In this section, we introduce maximum likelihood estimators for the parameter ϑ of a statistical model $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$. The idea of maximum likelihood estimation is to choose the parameter $\vartheta \in \Theta$ that maximises the likelihood of the sample $x \in \mathcal{X}$. We have to treat the situations of discrete and absolutely continuous distributions separately.

Definition 3.9. Let $\mathcal{X}$ be an at most countable set and let $(\mathcal{X}, \mathcal{P}(\mathcal{X}), (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model. We call an estimator $T : \mathcal{X} \to \Theta$ a maximum likelihood estimator for ϑ if

$$\mathbf{P}_{T(x)}(\{x\}) = \max_{\vartheta\in\Theta} \mathbf{P}_\vartheta(\{x\})$$

for all $x \in \mathcal{X}$, i.e., $T(x)$ is a maximiser of $L_x : \Theta \ni \vartheta \mapsto \mathbf{P}_\vartheta(\{x\})$, which is called likelihood function.

One can often simplify computations by maximising the logarithm of the likelihood function, which is called log-likelihood function.

Example 3.10. For $n \in \mathbb{N}$ and the statistical model $(\mathbb{N}_0^n, \mathcal{P}(\mathbb{N}_0^n), (\mathrm{Poisson}(\lambda)^{\otimes n})_{\lambda\in[0,\infty)})$ we use the abbreviation $\mathbf{P}_\lambda = \mathrm{Poisson}(\lambda)^{\otimes n}$ for $\lambda \geq 0$. For a sample $x = (x_1, \ldots, x_n) \in \mathbb{N}_0^n$ we obtain

$$\mathbf{P}_\lambda(\{x\}) = \prod_{i=1}^n \frac{\lambda^{x_i}}{x_i!} e^{-\lambda} = \frac{\lambda^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!} e^{-n\lambda} = L_x(\lambda)$$

for $\lambda \geq 0$. Our goal is to maximise the likelihood function $L_x(\lambda)$ with respect to λ. Instead of the likelihood function we maximise the log-likelihood function

$$\log L_x(\lambda) = \log(\lambda) \sum_{i=1}^n x_i - \lambda n - \log\left(\prod_{i=1}^n x_i!\right).$$

We obtain

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} \log L_x(\lambda) = \frac{1}{\lambda} \sum_{i=1}^n x_i - n,$$

whence

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} \log L_x(\lambda) = 0 \quad \Longleftrightarrow \quad \lambda = \frac{1}{n} \sum_{i=1}^{n} x_i.$$

From the form of the log-likelihood function, one can see that this critical point must belong to a maximum. Thus, we have shown that $\widehat{\lambda} = \frac{1}{n} \sum_{i=1}^{n} x_i$ is the maximum likelihood estimator for λ. Note that this estimator is the sample mean.

Next we consider the absolutely continuous case.

Definition 3.11. Let $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathbf{P}_\vartheta)_{\vartheta \in \Theta})$ be a statistical model such that $\mathbf{P}_\vartheta$ has a density $f_\vartheta : \mathbb{R}^n \to [0, \infty)$ for each $\vartheta \in \Theta$. We call an estimator $T : \mathbb{R}^n \to \Theta$ a maximum likelihood estimator for ϑ if

$$f_{T(x)}(x) = \max_{\vartheta \in \Theta} f_\vartheta(x)$$

for all $x \in \mathbb{R}^n$, i.e., $T(x)$ is a maximiser of the function $L_x : \Theta \ni \vartheta \mapsto f_\vartheta(x)$, which is called likelihood function.

In contrast to the discrete case one has in the absolutely continuous case $\mathbf{P}_\vartheta(\{x\}) = 0$ for all $x \in \mathbb{R}^n$ and $\vartheta \in \Theta$, whence the idea to maximise the probability does not work. Instead, we choose the ϑ such that the value of the density for the observation x is maximised. As in the discrete case it can be favourable to study the log-likelihood function.

We begin with an example.

Example 3.12. We consider the statistical model $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathrm{Normal}(\mu, 1)^{\otimes n})_{\mu \in \mathbb{R}})$ for a fixed $n \in \mathbb{N}$, i.e., we have the case of normal samples with unknown expectation and known variance one. For each $\mu \in \mathbb{R}$ we have the density

$$f_\mu(x_1, \ldots, x_n) = \frac{1}{\sqrt{2\pi}^n} e^{-\frac{1}{2} \sum_{i=1}^{n} (x_i - \mu)^2}$$

for $x_1, \ldots, x_n \in \mathbb{R}$. For a sample $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ we obtain the likelihood function

$$L_x(\mu) = \frac{1}{\sqrt{2\pi}^n} e^{-\frac{1}{2} \sum_{i=1}^{n} (x_i - \mu)^2}$$

for $\mu \in \mathbb{R}$ and the log-likelihood function

$$\log L_x(\mu) = -n \log(\sqrt{2\pi}) - \frac{1}{2} \sum_{i=1}^{n} (x_i - \mu)^2$$

for $\mu \in \mathbb{R}$, which we maximise in order to find the maximum likelihood estimator. We have that

$$\frac{\mathrm{d}}{\mathrm{d}\mu} \log L_x(\mu) = \sum_{i=1}^{n} (x_i - \mu)$$

and

$$\frac{\mathrm{d}}{\mathrm{d}\mu}\log L_x(\mu) = 0 \quad \Longleftrightarrow \quad \sum_{i=1}^n (x_i - \mu) = 0 \quad \Longleftrightarrow \quad \mu = \frac{1}{n}\sum_{i=1}^n x_i.$$

From the form of the log-likelihood function we see that this critical point belongs to a maximum, whence $\widehat{\mu} = \frac{1}{n}\sum_{i=1}^n x_i$ is the maximum likelihood estimator for μ. Thus, we have again the situation that the maximum likelihood estimator is the sample mean.

Example 3.13. We consider the statistical model $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathrm{Uniform}([a,b])^{\otimes n})_{a,b\in\mathbb{R},a<b})$ for a fixed $n \in \mathbb{N}$ with $n \geq 2$. For $a, b \in \mathbb{R}$ with $a < b$ we have the density

$$f_{a,b}(x_1,\dots,x_n) = \prod_{i=1}^n \frac{1}{b-a}\mathbf{1}\{x_i \in [a,b]\} = \frac{1}{(b-a)^n}\mathbf{1}\{x_1,\dots,x_n \in [a,b]\}$$

for $x_1,\dots,x_n \in \mathbb{R}$. For a sample $x = (x_1,\dots,x_n) \in \mathbb{R}^n$, we obtain the likelihood function

$$L_x(a,b) = \frac{1}{(b-a)^n}\mathbf{1}\{x_1,\dots,x_n \in [a,b]\}$$

for $a, b \in \mathbb{R}$ such that $a < b$, which we have to maximise in order to find a maximum likelihood estimator for $a, b \in \mathbb{R}$. Note that the factor $1/(b-a)$ becomes large if $b - a$ is small, while the indicator function is zero as soon as one of $x_1,\dots,x_n$ is outside of $[a,b]$. Thus, the likelihood function L_x is maximised for $a = \min\{x_1,\dots,x_n\}$ and $b = \max\{x_1,\dots,x_n\}$. Consequently, $(\widehat{a},\widehat{b}) = (\min\{x_1,\dots,x_n\}, \max\{x_1,\dots,x_n\})$ is the maximum likelihood estimator for (a,b).

3.3 Method of moments

In this section we introduce a further approach for the construction of estimators, the so-called method of moments. For $k \in \mathbb{N}$ and a sample $x = (x_1,\dots,x_n) \in \mathbb{R}^n$ we define the estimator

$$M_k(x) = \frac{1}{n}\sum_{i=1}^n x_i^k,$$

which we denote as k-th sample moment. The sample moments have the following properties.

Theorem 3.14. Let $k \in \mathbb{N}$ be fixed and let $(X_i)_{i\in\mathbb{N}}$ be a sequence of i.i.d. random variables such that $\mathbf{E}[|X_1|^k] < \infty$. Then,

$$\mathbf{E}[M_k(X_1,\dots,X_n)] = \mathbf{E}[X_1^k]$$

for all $n \in \mathbb{N}$ and

$$M_k(X_1,\dots,X_n) \xrightarrow{\mathbf{P}} \mathbf{E}[X_1^k] \quad \text{and} \quad M_k(X_1,\dots,X_n) \xrightarrow{a.s.} \mathbf{E}[X_1^k] \quad \text{as} \quad n \to \infty,$$

i.e., the k-th sample moments are unbiased as well as weakly and strongly consistent estimators for the k-th moment if it exists.

Proof. We have

$$\mathbf{E}[M_k(X_1,\dots,X_n)] = \mathbf{E}\left[\frac{1}{n}\sum_{i=1}^n X_i^k\right] = \frac{1}{n}\sum_{i=1}^n \mathbf{E}[X_i^k] = \mathbf{E}[X_1^k].$$

From the strong law of large numbers, Theorem 2.32, it follows that

$$M_k(X_1,\dots,X_n) = \frac{1}{n}\sum_{i=1}^n X_i^k \xrightarrow{a.s.} \mathbf{E}[X_1^k] \quad \text{as} \quad n\to\infty,$$

which also implies convergence in probability. Alternatively, one could use the weak law of large numbers Theorem 2.26. □

We introduce the method of moments by studying the example of normal samples.

Example 3.15. We consider the statistical model $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\text{Normal}(\mu,\sigma^2)^{\otimes n})_{\mu\in\mathbb{R},\sigma^2\in[0,\infty)})$ for a fixed $n\in\mathbb{N}$. For all $\mu\in\mathbb{R}$ and $\sigma^2\in[0,\infty)$ and $X\sim\text{Normal}(\mu,\sigma^2)$ we have

$$\mathbf{E}_{\mu,\sigma^2}[X] = \mu \quad \text{and} \quad \mathbf{E}_{\mu,\sigma^2}[X^2] = \sigma^2+\mu^2.$$

These equation can be rewritten as

$$\mu = \mathbf{E}_{\mu,\sigma^2}[X] \quad \text{and} \quad \sigma^2 = \mathbf{E}_{\mu,\sigma^2}[X^2] - \mathbf{E}_{\mu,\sigma^2}[X]^2.$$

In the framework of our statistical model the first and the second moment of the underlying distribution are - as the distribution itself - not known, but we can approximate them by the sample moments. By replacing the exact moments by the sample moments in the equations above, we can estimate the unknown parameters. For a sample $x=(x_1,\dots,x_n)\in\mathbb{R}^n$, this leads to the estimators

$$\hat{\mu} = M_1(x) = \frac{1}{n}\sum_{i=1}^n x_i \quad \text{and} \quad \hat{\sigma}^2 = M_2(x) - M_1(x)^2 = \frac{1}{n}\sum_{i=1}^n x_i^2 - \left(\frac{1}{n}\sum_{i=1}^n x_i\right)^2$$

for μ and σ^2. Here, $\hat{\mu}$ is again the sample mean. For $\mu\in\mathbb{R}$ and $\sigma^2\in[0,\infty)$ and $X_1,\dots,X_n\sim\text{Normal}(\mu,\sigma^2)$, we obtain

$$\begin{aligned}
\mathbf{E}_{\mu,\sigma^2}[\hat{\sigma}^2] &= \mathbf{E}_{\mu,\sigma^2}\left[\frac{1}{n}\sum_{i=1}^n X_i^2 - \left(\frac{1}{n}\sum_{i=1}^n X_i\right)^2\right] = \frac{1}{n}\sum_{i=1}^n \mathbf{E}_{\mu,\sigma^2}[X_i^2] - \mathbf{E}_{\mu,\sigma^2}\left[\left(\frac{1}{n}\sum_{i=1}^n X_i\right)^2\right] \\
&= \mathbf{E}_{\mu,\sigma^2}[X_1^2] - \frac{1}{n^2}\sum_{i,j=1}^n \mathbf{E}_{\mu,\sigma^2}[X_iX_j] \\
&= \mathbf{E}_{\mu,\sigma^2}[X_1^2] - \frac{1}{n^2}\sum_{i=1}^n \mathbf{E}_{\mu,\sigma^2}[X_i^2] - \frac{1}{n^2}\sum_{i,j=1,i\neq j}^n \mathbf{E}_{\mu,\sigma^2}[X_i]\mathbf{E}_{\mu,\sigma^2}[X_j] \\
&= \left(1-\frac{1}{n}\right)\mathbf{E}_{\mu,\sigma^2}[X_1^2] - \frac{n(n-1)}{n^2}\mathbf{E}_{\mu,\sigma^2}[X_1]^2 = \frac{n-1}{n}\mathbf{Var}_{\mu,\sigma^2}(X_1) = \frac{n-1}{n}\sigma^2.
\end{aligned}$$

Thus, the derived estimators for σ^2 are asymptotically unbiased.

The approach we used in the previous example to construct estimators is called the method of moments. The general strategy is to express the parameters we aim to estimate in terms of some moments of the distribution. By replacing the exact moments by the sample moments in these representations, we obtain estimators for the parameters.

Chapter 4

Confidence intervals

4.1 Confidence regions

In the previous chapter we studied estimators for a quantity $\tau(\vartheta)$ of an unknown underlying parameter ϑ (often for the situation $\tau(\vartheta) = \vartheta$). The drawback of this procedure is that an estimator gives only a single value and that in general we do not know anything about the error, i.e., the difference between the estimator and $\tau(\vartheta)$. In the following we address this problem by providing a whole set, a so-called confidence region or interval, that contains the true value with some certainty.

Definition 4.1 (Confidence region)**.** Let $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model, let M be some set and let $\tau : \Theta \to M$. We call a map $C : \mathcal{X} \to \mathcal{P}(M)$ a confidence region for $\tau(\vartheta)$ with error level $\alpha \in (0, 1)$ (or confidence level $1 - \alpha$) if

$$\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \tau(\vartheta) \in C(x)\}) \geq 1 - \alpha$$

for all $\vartheta \in \Theta$.

Since $\mathcal{P}(M)$ is the power set of M, i.e., the set of all subsets of M, $C(x)$ is a subset of M for every $x \in \mathcal{X}$. For each $\vartheta \in \Theta$ we can think of the confidence region as a random set on the probability space $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, \mathbf{P}_\vartheta)$ and the inequality in the definition can be rewritten as

$$\mathbf{P}_\vartheta(\tau(\vartheta) \in C(X)) \geq 1 - \alpha$$

with $X \sim \mathbf{P}_\vartheta$. Here, $\tau(\vartheta)$ is a deterministic quantity, while $C(X)$ is a random set, which is the opposite of the common situation that one asks for the probability of a random variable belonging to a deterministic set. We can think of the confidence region as a set, which we expect to contain $\tau(\vartheta)$. Hereby, an error occurs if $\tau(\vartheta) \notin C(X)$. For the probability of this event we have

$$\mathbf{P}_\vartheta(\tau(\vartheta) \notin C(X)) = 1 - \mathbf{P}_\vartheta(\tau(\vartheta) \in C(X)) \leq 1 - (1 - \alpha) = \alpha,$$

which explains the name error level for α.

For Definition 4.1 one needs to assume that $\{x \in \mathcal{X} : \tau(\vartheta) \in C(x)\} \in \mathscr{F}_{\mathcal{X}}$ for all $\vartheta \in \Theta$ since the probabilities are not defined otherwise. Throughout this class we mostly consider the case that $M = \mathbb{R}$ and that the values of C are intervals. In this situation, we call C a confidence interval.

Recall that, for $p \in (0,1)$, $q_p \in \mathbb{R}$ is a p-quantile of a random variable X if

$$\mathbf{P}(X \leq q_p) \geq p \quad \text{and} \quad \mathbf{P}(X \geq q_p) \geq 1 - p.$$

For normal distributions the quantiles are unique and we denote the p-quantile of the standard normal distribution by z_p for $p \in (0,1)$.

Example 4.2 (Confidence interval for the mean of a normal sample with known variance)**.** We consider the statistical model $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathrm{Normal}(\mu, \sigma_0^2)^{\otimes n})_{\mu \in \mathbb{R}})$ for fixed $n \in \mathbb{N}$ and $\sigma_0^2 \in (0, \infty)$. For $\alpha \in (0,1)$,

$$C : \mathbb{R}^n \ni x \mapsto (\overline{x} - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \overline{x} + z_{1-\alpha/2}\sigma_0/\sqrt{n})$$

is a confidence interval with error level α. Recall that $\overline{x}$ stands for the sample mean of $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, i.e., $\overline{x} = \frac{1}{n}\sum_{i=1}^n x_i$. In order to show that C is a confidence interval, let $\mu \in \mathbb{R}$ and let X be a vector of n independent normal random variables with expectation $\mu \in \mathbb{R}$ and variance σ_0^2, whose distribution is denoted by $\mathbf{P}_\mu$. Then, we have $\overline{X} \sim \mathrm{Normal}(\mu, \sigma_0^2/n)$ and obtain

$$\begin{aligned}
\mathbf{P}_\mu(\{x \in \mathbb{R}^n : \mu \in C(x)\}) &= \mathbf{P}_\mu(\mu \in C(X)) \\
&= \mathbf{P}_\mu(\mu \in (\overline{X} - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \overline{X} + z_{1-\alpha/2}\sigma_0/\sqrt{n})) \\
&= \mathbf{P}_\mu(\overline{X} \in (\mu - z_{1-\alpha/2}\sigma_0/\sqrt{n}, \mu + z_{1-\alpha/2}\sigma_0/\sqrt{n})) \\
&= \mathbf{P}_\mu\left(\frac{\overline{X} - \mu}{\sigma_0/\sqrt{n}} \in (-z_{1-\alpha/2}, z_{1-\alpha/2})\right) \\
&= \Phi(z_{1-\alpha/2}) - \Phi(-z_{1-\alpha/2}) = \Phi(z_{1-\alpha/2}) - (1 - \Phi(z_{1-\alpha/2})) \\
&= 2\Phi(z_{1-\alpha/2}) - 1 = 2(1 - \frac{\alpha}{2}) - 1 = 2 - \alpha - 1 = 1 - \alpha.
\end{aligned}$$

When a random experiment with the two possible outcomes success and failure (abbreviated by 1 and 0) and unknown success probability p is repeated n times independently, we obtain the statistical model $(\{0,1\}^n, \mathcal{P}(\{0,1\}^n), (\mathrm{Ber}(p)^{\otimes n})_{p \in [0,1]})$. In this situation it is naturally to count the number of successes $n\overline{x}$, which follows a binomial distribution with known parameter n and unknown parameter $p \in [0,1]$. This leads to the statistical model $(\{0,1,\ldots,n\}, \mathcal{P}(\{0,1,\ldots,n\}), (\mathrm{Binom}(n,p))_{p \in [0,1]})$.

Example 4.3 (Confidence interval for the binomial distribution)**.** Let the statistical model $(\{0,1,\ldots,n\}, \mathcal{P}(\{0,1,\ldots,n\}), (\mathrm{Binom}(n,p))_{p \in [0,1]})$ for a fixed $n \in \mathbb{N}$ be given. Our goal is to determine a confidence interval for p. Motivated by Example 4.2, we let $C : \{0,1,\ldots,n\} \ni k \mapsto \left(\frac{k}{n} - a, \frac{k}{n} + a\right)$ and choose $a \in (0, \infty)$ such that C is a confidence interval for p with error level $\alpha \in (0,1)$. For $p \in [0,1]$ let $\mathbf{P}_p$ be the distribution of

$X \sim \mathrm{Binom}(n,p)$. Then, we obtain

$$\begin{aligned}\mathbf{P}_p(p \notin C(X)) &= \mathbf{P}_p(p \notin (X/n - a, X/n + a)) = \mathbf{P}_p(X/n \notin (p-a, p+a)) \\ &= \mathbf{P}_p(X \notin (np - na, np + na)) = \mathbf{P}_p(|X - np| \geq na).\end{aligned}$$

It follows from the Chebyshev inequality that

$$\mathbf{P}_p(p \notin C(X)) \leq \frac{\mathbf{Var}_p[X]}{(na)^2} = \frac{np(1-p)}{n^2a^2}.$$

The inequality $p(1-p) \leq \frac{1}{4}$ for all $p \in [0,1]$ yields

$$\mathbf{P}_p(p \notin C(X)) \leq \frac{1}{4na^2}$$

for all $p \in [0,1]$. In order to achieve the error level α, the right-hand side needs to be at most α. The equation $\frac{1}{4na^2} = \alpha$ is solved by $a = \frac{1}{\sqrt{4n\alpha}} = \frac{1}{2\sqrt{n\alpha}}$. Thus, $C : \{0,1,\ldots,n\} \ni k \mapsto (k/n - \frac{1}{2\sqrt{n\alpha}}, k/n + \frac{1}{2\sqrt{n\alpha}})$ is a confidence interval for p with error level α.

Since we used several non-sharp inequalities in the derivation of the confidence interval in Example 4.3, the probability of an error is less than α for all $p \in [0,1]$. Such confidence intervals are called conservative.

Now let us provide a general method to construct a confidence region with error level $\alpha \in (0,1)$ for the parameter ϑ of a statistical model $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$. For each $\vartheta \in \Theta$ choose a set $C_\vartheta \subset \mathcal{X}$ such that

$$\mathbf{P}_\vartheta(C_\vartheta) \geq 1 - \alpha.$$

For $x \in \mathcal{X}$ we define

$$C(x) = \{\widetilde{\vartheta} \in \Theta : x \in C_{\widetilde{\vartheta}}\}.$$

Then, we obtain for all $\vartheta \in \Theta$,

$$\begin{aligned}\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \vartheta \in C(x)\}) &= \mathbf{P}_\vartheta(\{x \in \mathcal{X} : \vartheta \in \{\widetilde{\vartheta} \in \Theta : x \in C_{\widetilde{\vartheta}}\}\}) \\ &= \mathbf{P}_\vartheta(\{x \in \mathcal{X} : x \in C_\vartheta\}) = \mathbf{P}_\vartheta(C_\vartheta) \\ &\geq 1 - \alpha,\end{aligned}$$

which shows that C is a confidence region for ϑ with error level α.

We illustrate this strategy by constructing a further confidence interval for the probability p of a binomial distribution.

Example 4.4 (Confidence interval for the binomial distribution)**.** For a fixed $n \in \mathbb{N}$ we consider the statistical model $(\{0,1,\ldots,n\}, \mathcal{P}(\{0,1,\ldots,n\}), (\mathrm{Binom}(n,p))_{p\in[0,1]})$ and let $\mathbf{P}_p \sim \mathrm{Binom}(n,p)$. Our goal is to construct a confidence interval for p with error level $\alpha \in (0,1)$. According to the strategy described above we define for each $p \in [0,1]$ a set $C_p = \{a(p), \ldots, A(p)\}$ such that

$$\mathbf{P}_p(C_p) = \mathbf{P}_p(\{k \in \{0,1,\ldots,n\} : a(p) \leq k \leq A(p)\}) \geq 1 - \alpha.$$

We choose

$$a(p) = \max\left\{\ell \in \{0,1,\dots,n\} : \sum_{j=0}^{\ell-1} \binom{n}{j} p^j (1-p)^{n-j} \le \frac{\alpha}{2}\right\}$$

and

$$A(p) = \min\left\{\ell \in \{0,1,\dots,n\} : \sum_{j=\ell+1}^{n} \binom{n}{j} p^j (1-p)^{n-j} \le \frac{\alpha}{2}\right\}$$

so that

$$\begin{aligned}
&\mathbf{P}_p(\{k \in \{0,1,\dots,n\} : k \notin C_p\}) \\
&= 1 - \mathbf{P}_p(\{k \in \{0,1,\dots,n\} : a(p) \le k \le A(p)\}) \\
&\le \mathbf{P}_p(\{k \in \{0,1,\dots,n\} : k < a(p)\}) + \mathbf{P}_p(\{k \in \{0,1,\dots,n\} : k > A(p)\}) \\
&\le \frac{\alpha}{2} + \frac{\alpha}{2} = \alpha.
\end{aligned}$$

This leads to the confidence region

$$C : \{0,1,\dots,n\} \ni k \mapsto \{p \in [0,1] : a(p) \le k \le A(p)\}$$

for p with error level α. Since, for $\ell \in \{0,1,\dots,n\}$, $[0,1] \ni p \mapsto \sum_{j=0}^{\ell-1} \binom{n}{j} p^j (1-p)^{n-j}$ is monotonously decreasing (this was an exercise in the stochastics class), $a(p)$ is monotonously increasing in p. Moreover, it is right continuous. This implies

$$a(p) \le k \quad \Longleftrightarrow \quad p < \sup\{\tilde{p} \in [0,1] : a(\tilde{p}) = k\} =: L(k).$$

Similarly, $A(p)$ is monotonously increasing in p and left continuous, whence one can establish

$$A(p) \ge k \quad \Longleftrightarrow \quad p > \inf\{\tilde{p} \in [0,1] : A(\tilde{p}) = k\} =: \ell(k).$$

Thus,

$$C : \{0,1,\dots,n\} \ni k \mapsto (\ell(k), L(k))$$

is a confidence interval for p with error level α.

Example 4.5 (Asymptotic confidence intervals for the binomial distribution)**.** For $n \in \mathbb{N}$ we consider the statistical models $(\{0,1,\dots,n\}, \mathcal{P}(\{0,1,\dots,n\}), (\mathrm{Binom}(n,p))_{p\in(0,1)})$ and for $p \in (0,1)$ let $\mathbf{P}_p^{(n)} \sim \mathrm{Binom}(n,p)$ and let $X^{(n)}$ denote a random variable with this distribution. Here we express the dependence on n in our notation since we will let $n \to \infty$ later. Fix $\alpha \in (0,1)$. For $n \in \mathbb{N}$ let

$$C_n : \{0,1,\dots,n\} \ni k \mapsto \left(\frac{k}{n} - \frac{a}{\sqrt{n}}, \frac{k}{n} + \frac{a}{\sqrt{n}}\right),$$

where $a \in (0,\infty)$ will be chosen later. For $p \in (0,1)$ we have

$$\begin{aligned}
&\mathbf{P}_p^{(n)}(\{k \in \{0,1,\dots,n\} : p \in C_n(k)\}) \\
&= \mathbf{P}_p^{(n)}(p \in C_n(X^{(n)})) = \mathbf{P}_p^{(n)}\Big(p \in \Big(\frac{X^{(n)}}{n} - \frac{a}{\sqrt{n}}, \frac{X^{(n)}}{n} + \frac{a}{\sqrt{n}}\Big)\Big) \\
&= \mathbf{P}_p^{(n)}\Big(\sqrt{n}\Big(\frac{X^{(n)}}{n} - p\Big) \in (-a,a)\Big) = \mathbf{P}_p^{(n)}\Big(\frac{X^{(n)} - \mathbf{E}[X^{(n)}]}{\sqrt{n}\sqrt{p(1-p)}} \in \frac{1}{\sqrt{p(1-p)}}(-a,a)\Big).
\end{aligned}$$

Now the central limit theorem yields

$$\begin{aligned}
\lim_{n\to\infty} \mathbf{P}_p^{(n)}(\{k \in \{0,1,\dots,n\} : p \in C_n(k)\}) &= \Phi\Big(\frac{a}{\sqrt{p(1-p)}}\Big) - \Phi\Big(-\frac{a}{\sqrt{p(1-p)}}\Big) \\
&= 2\Phi\Big(\frac{a}{\sqrt{p(1-p)}}\Big) - 1.
\end{aligned}$$

The minimum on the right-hand side is achieved for $p = \frac{1}{2}$ and equals $2\Phi(2a) - 1$. We have that

$$2\Phi(2a) - 1 = 1 - \alpha \quad \Longleftrightarrow \quad \Phi(2a) = 1 - \frac{\alpha}{2} \quad \Longleftrightarrow \quad 2a = z_{1-\frac{\alpha}{2}} \quad \Longleftrightarrow \quad a = \frac{z_{1-\frac{\alpha}{2}}}{2}.$$

For this choice of a we obtain

$$\lim_{n\to\infty} \mathbf{P}_p^{(n)}(\{k \in \{0,1,\dots,n\} : p \in C_n(k)\}) \geq 1 - \alpha$$

for all $p \in (0,1)$. If these inequalities were true without the limit, C_n would be a confidence interval with error level α for each $n \in \mathbb{N}$. We call $(C_n)_{n\in\mathbb{N}}$ asymptotic confidence intervals for p with error level α.

So far we studied confidence intervals that were bounded intervals and, thus, are denoted as two-sided. In contrast, one can also consider intervals that are unbounded in one direction, so-called one-sided confidence intervals.

Example 4.6 (One-sided confidence intervals for normal samples with known variance)**.** Let the statistical model $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathrm{Normal}(\mu,\sigma_0)^{\otimes n})_{\mu\in\mathbb{R}})$ with fixed $\sigma_0^2 \in (0,\infty)$ and $n \in \mathbb{N}$ be given and let $\alpha \in (0,1)$. Then,

$$C_1 : \mathbb{R}^n \ni x \mapsto \Big(-\infty, \overline{x} + \frac{z_{1-\alpha}\sigma_0}{\sqrt{n}}\Big)$$

and

$$C_2 : \mathbb{R}^n \ni x \mapsto \Big(\overline{x} - \frac{z_{1-\alpha}\sigma_0}{\sqrt{n}}, \infty\Big)$$

are one-sided confidence intervals with error level α for μ. Indeed, for $\mu \in \mathbb{R}$ and $X \sim \mathrm{Normal}(\mu,\sigma_0)^{\otimes n} =: \mathbf{P}_\mu$, we have

$$\begin{aligned}
\mathbf{P}_\mu(\mu \in C_1(X)) &= \mathbf{P}_\mu\Big(\mu \in \Big(-\infty, \overline{X} + \frac{z_{1-\alpha}\sigma_0}{\sqrt{n}}\Big)\Big) = \mathbf{P}_\mu\Big(\frac{\overline{X} - \mu}{\sigma_0/\sqrt{n}} > -z_{1-\alpha}\Big) \\
&= 1 - \Phi(-z_{1-\alpha}) = \Phi(z_{1-\alpha}) = 1 - \alpha
\end{aligned}$$

and

$$\mathbf{P}_\mu(\mu \in C_2(X)) = \mathbf{P}_\mu\left(\mu \in \left(\overline{X} - \frac{z_{1-\alpha}\sigma_0}{\sqrt{n}}, \infty\right)\right) = \mathbf{P}_\mu\left(\frac{\overline{X} - \mu}{\sigma_0/\sqrt{n}} < z_{1-\alpha}\right)$$
$$= \Phi(z_{1-\alpha}) = 1 - \alpha.$$

4.2 Chi-squared distribution and Student's t-distribution

In this section we introduce two distributions, which naturally arise in statistics.

Definition 4.7 (Chi-squared distribution)**.** For $k \in \mathbb{N}$ and independent standard normal random variables $N_1, \dots, N_k$ let $Y = \sum_{i=1}^k N_i^2$. The distribution of Y is called chi-squared distribution with k degrees of freedom and we write $Y \sim \chi_k^2$.

Remark 4.8. The characteristic function of $Y \sim \chi_k^2$ with $k \in \mathbb{N}$ is given by

$$\varphi_Y(t) = \mathbf{E}[e^{\mathrm{i}tY}] = \mathbf{E}[e^{\mathrm{i}t\sum_{i=1}^k N_i^2}] = \prod_{i=1}^k \mathbf{E}[e^{\mathrm{i}tN_i^2}] = \mathbf{E}[e^{\mathrm{i}tN_1^2}]^k = \mathbf{E}[e^{\mathrm{i}tN^2}]^k$$

for $t \in \mathbb{R}$, where N is a standard normal random variable.

Definition 4.9 (Student's t-distribution)**.** Let $N \sim \mathrm{Normal}(0,1)$ and $X \sim \chi_k^2$ with $k \in \mathbb{N}$ be independent random variables. The distribution of $Y = \frac{N}{\sqrt{\frac{X}{k}}}$ is called Student's t-distribution with k degrees of freedom and we write $Y \sim t_k$.

In the following, we will denote Student's t-distribution just as t-distribution.

Remark 4.10. For $k \in \mathbb{N}$ and $p \in (0,1)$ we denote by $t_{k,p}$ the p-quantile of a t-distribution with k degrees of freedom. From $N \overset{d}{=} -N$ it follows that $Y \overset{d}{=} -Y$ for $Y \sim t_k$. This means that $\mathbf{P}(Y \le -u) = 1 - \mathbf{P}(Y \le u)$ for $u \in \mathbb{R}$ and, thus, $\mathbf{P}(Y \le -t_{k,p}) = 1 - \mathbf{P}(Y \le t_{k,p})$ for $p \in (0,1)$.

According to the following theorem we observe chi-squared and t-distributed random variables, when studying the sample mean and the sample variance of a normal sample.

Theorem 4.11. For $n \in \mathbb{N}$ with $n \ge 2$, $\mu \in \mathbb{R}$ and $\sigma^2 > 0$ let $X_1, \dots, X_n$ be independent $\mathrm{Normal}(\mu, \sigma^2)$-distributed random variables and let

$$\overline{X} = \frac{1}{n}\sum_{i=1}^n X_i, \quad S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \overline{X})^2 \quad \text{and} \quad S = \sqrt{S^2}.$$

Then, the following statements hold:

a) $\overline{X} \sim \text{Normal}(\mu, \frac{\sigma^2}{n})$

b) The random variables $\overline{X}$ and S^2 are independent.

c) $\frac{(n-1)S^2}{\sigma^2} = \frac{1}{\sigma^2}\sum_{i=1}^n (X_i - \overline{X})^2 \sim \chi^2_{n-1}$

d) $T = \frac{\sqrt{n}(\overline{X}-\mu)}{S} \sim t_{n-1}$

Proof. Since for independent random variables $Y_1 \sim \text{Normal}(\mu_1, \sigma_1^2)$ and $Y_2 \sim \text{Normal}(\mu_2, \sigma_2^2)$ with $\mu_1, \mu_2 \in \mathbb{R}$ and $\sigma_1^2, \sigma_2^2 \in [0, \infty)$, one has $Y_1 + Y_2 \sim \text{Normal}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$, we obtain $\sum_{i=1}^n X_i \sim \text{Normal}(n\mu, n\sigma^2)$ and, thus, $\overline{X} = \frac{1}{n}\sum_{i=1}^n X_i \sim \text{Normal}(\mu, \frac{\sigma^2}{n})$.

Note that the random vector $(X_1, \ldots, X_n)$ follows a multivariate normal distribution. Since $\overline{X}$ and $Y_i = X_i - \overline{X}$ for $i \in \{1, \ldots, n\}$ are linear transformations of $X_1, \ldots, X_n$, the random vector$(\overline{X}, Y_1, \ldots, Y_n)$ also has a multivariate normal distribution. For $i \in \{1, \ldots, n\}$ we get

$$\mathbf{Cov}(\overline{X}, Y_i) = \mathbf{Cov}(\overline{X}, X_i - \overline{X}) = \mathbf{Cov}(\overline{X}, X_i) - \mathbf{Cov}(\overline{X}, \overline{X}).$$

By the multilinearity of the covariance and the independence of $X_1, \ldots, X_n$ we obtain

$$\mathbf{Cov}(\overline{X}, X_i) = \frac{1}{n}\sum_{j=1}^n \mathbf{Cov}(X_j, X_i) = \frac{1}{n}\mathbf{Cov}(X_i, X_i) = \frac{\sigma^2}{n}$$

and

$$\mathbf{Cov}(\overline{X}, \overline{X}) = \mathbf{Var}\left(\frac{1}{n}\sum_{i=1}^n X_i\right) = \frac{1}{n^2}\sum_{i=1}^n \mathbf{Var}(X_i) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n},$$

which leads to

$$\mathbf{Cov}(\overline{X}, Y_i) = \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0.$$

This means that $\overline{X}$ and Y_i are uncorrelated for $i \in \{1, \ldots, n\}$. Thus, $\overline{X}$ and $(Y_1, \ldots, Y_n)$ are independent by similar arguments as used in the proof of Theorem 2.24 since their joint distribution is a multivariate normal distribution and they are uncorrelated. Because of $S^2 = \frac{1}{n-1}\sum_{i=1}^n Y_i^2$, $\overline{X}$ and S^2 are also independent.

We define

$$W = \sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2},$$

which follows a chi-squared distribution with n degrees of freedom since $(X_1 - \mu)/\sigma, \ldots, (X_n - \mu)/\sigma$ are independent standard normal random variables. Moreover, we have

$$\begin{aligned} W &= \sum_{i=1}^n \frac{(X_i - \overline{X} + \overline{X} - \mu)^2}{\sigma^2} \\ &= \sum_{i=1}^n \frac{(X_i - \overline{X})^2}{\sigma^2} + 2\sum_{i=1}^n \frac{(X_i - \overline{X})(\overline{X} - \mu)}{\sigma^2} + \sum_{i=1}^n \frac{(\overline{X} - \mu)^2}{\sigma^2} \\ &= \sum_{i=1}^n \frac{(X_i - \overline{X})^2}{\sigma^2} + 2\frac{\overline{X} - \mu}{\sigma^2}\sum_{i=1}^n (X_i - \overline{X}) + \frac{n(\overline{X} - \mu)^2}{\sigma^2}. \end{aligned}$$

Because of $\sum_{i=1}^n (X_i - \overline{X}) = \sum_{i=1}^n X_i - n\overline{X} = n\overline{X} - n\overline{X} = 0$, this simplifies to

$$W = \frac{(n-1)S^2}{\sigma^2} + \frac{n(\overline{X}-\mu)^2}{\sigma^2}.$$

By the independence of S^2 and $\overline{X}$, we obtain for the characteristic function of W that

$$\varphi_W(t) = \mathbf{E}[e^{\mathbf{i}tW}] = \mathbf{E}[e^{\mathbf{i}t\left(\frac{(n-1)S^2}{\sigma^2} + \frac{n(\overline{X}-\mu)^2}{\sigma^2}\right)}] = \mathbf{E}[e^{\mathbf{i}t\frac{(n-1)S^2}{\sigma^2}}]\mathbf{E}[e^{\mathbf{i}t\frac{n(\overline{X}-\mu)^2}{\sigma^2}}] \tag{4.1}$$

for $t \in \mathbb{R}$. Since $\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma} \sim \text{Normal}(0,1)$, we obtain

$$\mathbf{E}[e^{\mathbf{i}t\frac{n(\overline{X}-\mu)^2}{\sigma^2}}] = \mathbf{E}[e^{\mathbf{i}tN^2}]$$

with a standard normal random variable N, while $W \sim \chi_n^2$ and Remark 4.8 imply

$$\varphi_W(t) = \mathbf{E}[e^{\mathbf{i}tN^2}]^n.$$

Hence, (4.1) yields

$$\mathbf{E}[e^{\mathbf{i}t\frac{(n-1)S^2}{\sigma^2}}] = \mathbf{E}[e^{\mathbf{i}tN^2}]^{n-1}.$$

Since the right-hand side is the characteristic function of a chi-squared distribution with $n-1$ degrees of freedom (see Remark 4.8), we obtain

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2,$$

which proves c).

Since $\sqrt{n}(\overline{X}-\mu)/\sigma \sim \text{Normal}(0,1)$ and $(n-1)S^2/\sigma^2 \sim \chi_{n-1}^2$ are independent, it follows from Definition 4.9 that

$$\frac{\sqrt{n}(\overline{X}-\mu)}{S} = \frac{\frac{\sqrt{n}(\overline{X}-\mu)}{\sigma}}{\sqrt{\frac{(n-1)S^2}{\sigma^2}\frac{1}{n-1}}} \sim t_{n-1},$$

which shows d). □

4.3 Confidence intervals for normal samples with unknown variance

In this section we apply the results from the previous section to derive confidence intervals for normal samples with unknown expectation and variance. Recall that, for $p \in (0,1)$, $t_{k,p}$ denotes the p-quantile of a t-distribution with k degrees of freedom and that, for $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$, the sample mean and the sample variance are given by

$$\overline{x} = \frac{1}{n}\sum_{i=1}^n x_i \quad \text{and,} \quad \text{if } n \geq 2, \quad s_x^2 = \frac{1}{n-1}\sum_{i=1}^n (x_i - \overline{x})^2.$$

Theorem 4.12 (Confidence intervals for μ of normal samples with unknown variance)**.** Let $n \in \mathbb{N}$ with $n \geq 2$, consider the statistical model $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathrm{Normal}(\mu, \sigma^2)^{\otimes n})_{\mu \in \mathbb{R}, \sigma^2 \in (0,\infty)})$ and let $\alpha \in (0,1)$. Then,

$$C : \mathbb{R}^n \ni x \mapsto \left(\overline{x} - \frac{s_x t_{n-1,1-\alpha/2}}{\sqrt{n}}, \overline{x} + \frac{s_x t_{n-1,1-\alpha/2}}{\sqrt{n}}\right)$$

is a confidence interval for μ with error level α.

Proof. For $\mu \in \mathbb{R}$ and $\sigma^2 \in (0,\infty)$ let $X \sim \mathrm{Normal}(\mu, \sigma^2)^{\otimes n}$ and denote its distribution by $\mathbf{P}_{\mu,\sigma^2}$. We obtain

$$\begin{aligned}\mathbf{P}_{\mu,\sigma^2}(\mu \in C(X)) &= \mathbf{P}_{\mu,\sigma^2}\left(\mu \in \left(\overline{X} - \frac{s_X t_{n-1,1-\alpha/2}}{\sqrt{n}}, \overline{X} + \frac{s_X t_{n-1,1-\alpha/2}}{\sqrt{n}}\right)\right) \\ &= \mathbf{P}_{\mu,\sigma^2}\left(\frac{\overline{X} - \mu}{s_X/\sqrt{n}} \in (-t_{n-1,1-\alpha/2}, t_{n-1,1-\alpha/2})\right).\end{aligned}$$

Denoting by Y a random variable with a t-distribution with $n-1$ degrees of freedom, it follows from Theorem 4.11 and the symmetry of the t-distribution (see Remark 4.10) that

$$\begin{aligned}\mathbf{P}_{\mu,\sigma^2}(\mu \in C(X)) &= \mathbf{P}(Y \in (-t_{n-1,1-\alpha/2}, t_{n-1,1-\alpha/2})) \\ &= \mathbf{P}(Y \leq t_{n-1,1-\alpha/2}) - \mathbf{P}(Y \leq -t_{n-1,1-\alpha/2}) \\ &= \mathbf{P}(Y \leq t_{n-1,1-\alpha/2}) - (1 - \mathbf{P}(Y \leq t_{n-1,1-\alpha/2})) \\ &= 2\mathbf{P}(Y \leq t_{n-1,1-\alpha/2})) - 1 = 2(1-\alpha/2) - 1 = 1 - \alpha,\end{aligned}$$

which concludes the proof. □

For $p \in (0,1)$ we denote by $\chi^2_{k,p}$ the p-quantile of a chi-squared distribution with k degrees of freedom.

Theorem 4.13 (Confidence intervals for σ^2 of normal samples)**.** For a fixed $n \in \mathbb{N}$ with $n \geq 2$ let the statistical model $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathrm{Normal}(\mu, \sigma^2)^{\otimes n})_{\mu \in \mathbb{R}, \sigma^2 \in (0,\infty)})$ be given and let $\alpha \in (0,1)$. Then,

$$C : \mathbb{R}^n \ni x \mapsto \left(\frac{(n-1)s_x^2}{\chi^2_{n-1,1-\alpha/2}}, \frac{(n-1)s_x^2}{\chi^2_{n-1,\alpha/2}}\right)$$

is a confidence interval for σ^2 with error level α.

Proof. For $\mu \in \mathbb{R}$ and $\sigma^2 \in (0,\infty)$ let $X \sim \mathrm{Normal}(\mu, \sigma^2)^{\otimes n} =: \mathbf{P}_{\mu,\sigma^2}$. Then, we have

$$\begin{aligned}\mathbf{P}_{\mu,\sigma^2}(\sigma^2 \in C(X)) &= \mathbf{P}_{\mu,\sigma^2}\left(\sigma^2 \in \left(\frac{(n-1)s_X^2}{\chi^2_{n-1,1-\alpha/2}}, \frac{(n-1)s_X^2}{\chi^2_{n-1,\alpha/2}}\right)\right) \\ &= \mathbf{P}_{\mu,\sigma^2}\left(\chi^2_{n-1,\alpha/2} < \frac{(n-1)s_X^2}{\sigma^2} < \chi^2_{n-1,1-\alpha/2}\right).\end{aligned}$$

Since by Theorem 4.11 $\frac{(n-1)s_X^2}{\sigma^2}$ has the same distribution as a random variable $Y \sim \chi^2_{n-1}$, we obtain

$$\begin{aligned}\mathbf{P}_{\mu,\sigma^2}(\sigma^2 \in C(X)) &= \mathbf{P}(\chi^2_{n-1,\alpha/2} < Y < \chi^2_{n-1,1-\alpha/2}) \\ &= \mathbf{P}(Y \leq \chi^2_{n-1,1-\alpha/2}) - \mathbf{P}(Y \leq \chi^2_{n-1,\alpha/2}) = 1 - \frac{\alpha}{2} - \frac{\alpha}{2} = 1 - \alpha,\end{aligned}$$

which concludes the proof. □

Remark 4.14. By the same arguments as used to prove Theorem 4.12 and Theorem 4.13, one can also derive one-sided confidence intervals for μ and σ^2.

Chapter 5

Hypothesis testing

5.1 Statistical tests

Let $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model and let $\Theta_0, \Theta_1 \subset \Theta$ be such that $\Theta_0 \cap \Theta_1 = \emptyset$ and $\Theta_0 \cup \Theta_1 = \Theta$. Assume that we have a sample x that comes from a distribution $\mathbf{P}_\vartheta$ with an unknown $\vartheta \in \Theta$. Our goal is to decide based on x whether

$$H_0 : \vartheta \in \Theta_0 \quad \text{or} \quad H_1 : \vartheta \in \Theta_1.$$

We denote H_0 as null hypothesis and H_1 as alternative. Note that the null hypothesis and the alternative are not interchangeable in the following. When analysing data the alternative describes the situation one would like to verify, while the null hypothesis is the opposite. Thus, the alternative is usually the research hypothesis. In most cases the alternative is that some structure is present in the data, while the null hypothesis is that we do not have this structure. We think of the null hypothesis as the true statement until the data show convincingly the opposite. Then, we have disproved or rejected the null hypothesis and can establish the alternative this way.

A statistical test is now a measurable map $\varphi : \mathcal{X} \to \{0, 1\}$, which tells us for a given sample $x \in \mathcal{X}$ if we reject the null hypothesis (for $\varphi(x) = 1$) or if we cannot reject the null hypothesis (for $\varphi(x) = 0$).

When applying a statistical test φ for $H_0 : \vartheta \in \Theta_0$ against $H_1 : \vartheta \in \Theta_1$, two kinds of errors can occur:

(i) *Type I error (or α-error)*: The null hypothesis is true, but it is rejected, i.e., $\vartheta \in \Theta_0$ but $\varphi(x) = 1$.

(ii) *Type II error (or β-error)*: The null hypothesis is wrong, but it is not rejected, i.e., $\vartheta \in \Theta_1$ but $\varphi(x) = 0$.

	$\vartheta \in \Theta_0$	$\vartheta \in \Theta_1$
$\varphi(x) = 0$ (i.e., H_0 is not rejected)	correct decision	type II error
$\varphi(x) = 1$ (i.e., H_0 is rejected)	type I error	correct decision

For $\vartheta \in \Theta_0$ the probability of a type I error is given by

$$\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\})$$

and a type II error cannot occur. For $\vartheta \in \Theta_1$ the probability of a type II error is

$$\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 0\}) = 1 - \mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\}),$$

while a type I error is not possible.

In order to have small type I and type II errors we need that $\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\})$ is small for $\vartheta \in \Theta_0$ and close to one for $\vartheta \in \Theta_1$. However, usually this cannot be achieved simultaneously. We illustrate this with a toy example.

Example 5.1. We consider the statistical model $(\{0,1\}, \mathcal{P}(\{0,1\}), (\mathbf{P}_p)_{p\in[0,1]})$ with $\mathbf{P}_p =$ Bernoulli(p) for $p \in [0,1]$ and the test problem

$$H_0 : p \leq 0.5 \quad \text{against} \quad H_1 : p > 0.5,$$

i.e., the null hypothesis is that the probability to observe 1 is not larger than that for 0 and the alternative is that the probability of 1 is larger than that of 0. We consider the test

$$\varphi : \{0,1\} \ni x \mapsto \begin{cases} 0, & x = 0, \\ 1, & x = 1, \end{cases}$$

i.e., we reject the null hypothesis if we observe 1. Because of

$$\mathbf{P}_p(\{x \in \{0,1\} : \varphi(x) = 1\}) = \mathbf{P}_p(\{1\}) = p$$

for $p \in [0,1]$, the probability of a type I error is p for $p \in [0, 0.5]$ and the probability of a type II error is $1 - p$ for $p \in (0.5, 1]$.

In the following we focus on the type I error since it is considered as the more serious one. Here, the philosophy is that it is worse to verify something false as true than to fail to discover something true.

Definition 5.2 (Significance level and power of a test)**.** Let $\varphi : \mathcal{X} \to \{0,1\}$ be a test for $H_0 : \vartheta \in \Theta_0$ against $H_1 : \vartheta \in \Theta_1$.

(i) We say that φ is a test of (significance) level $\alpha \in (0,1)$ if

$$\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\}) \leq \alpha$$

for all $\vartheta \in \Theta_0$.

(ii) For $\vartheta \in \Theta_1$ we call

$$\mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\})$$

the power of φ at ϑ.

Note that the test introduced in Example 5.1 has level 0.5 since

$$\mathbf{P}_p(\{x \in \{0,1\} : \varphi(x) = 1\}) = p \leq 0.5$$

for all $p \in [0, 0.5]$. For $p \in (0.5, 1]$ we obtain the power

$$\mathbf{P}_p(\{x \in \{0,1\} : \varphi(x) = 1\}) = p.$$

In the following we always provide the level α of a test. A test with significance level α is also denoted as level-α test and the quantity $1-\alpha$ is sometimes called the confidence level. For a level-α test the probability of a type I error is at most α. Typical choices for α are between 0.1 and 0.01.

The power depends on the choice of $\vartheta \in \Theta_1$ and is the probability that the null hypothesis is rejected, which is in this case the right decision. Thus, it is desirable to have a large power.

Assume we conduct a statistical test with significance level α for a sample $x \in \mathcal{X}$. If we obtain $\varphi(x) = 1$, we reject the null hypothesis at level α. For $\varphi(x) = 0$ we cannot reject the null hypothesis at level α. However, this does not yield that the null hypothesis is true, it only means that there is not sufficient evidence that the null hypothesis is wrong.

Example 5.3 (Two-sided Gauss test)**.** For fixed $n \in \mathbb{N}$ and $\sigma_0^2 \in (0, \infty)$ we consider the statistical model $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathrm{Normal}(\mu, \sigma_0^2)^{\otimes n})_{\mu \in \mathbb{R}})$ and let $\mu_0 \in \mathbb{R}$ and $\alpha \in (0,1)$. For

$$H_0 : \mu = \mu_0 \quad \text{against} \quad H_1 : \mu \neq \mu_0$$

the test

$$\varphi : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \overline{x} \in \left(\mu_0 - \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}, \mu_0 + \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}\right), \\ 1, & \overline{x} \notin \left(\mu_0 - \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}, \mu_0 + \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}\right), \end{cases}$$

has level α. Indeed, for $\mu \in \mathbb{R}$ and $X \sim \mathrm{Normal}(\mu, \sigma_0^2)^{\otimes n} =: \mathbf{P}_\mu$ we have

$$\begin{aligned} \mathbf{P}_\mu(\varphi(X) = 1) &= \mathbf{P}_\mu\left(\overline{X} \notin \left(\mu_0 - \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}, \mu_0 + \frac{z_{1-\frac{\alpha}{2}}\sigma_0}{\sqrt{n}}\right)\right) \\ &= \mathbf{P}_\mu\left(\frac{\overline{X} - \mu}{\sigma_0/\sqrt{n}} \notin \left(\frac{\mu_0 - \mu}{\sigma_0/\sqrt{n}} - z_{1-\frac{\alpha}{2}}, \frac{\mu_0 - \mu}{\sigma_0/\sqrt{n}} + z_{1-\frac{\alpha}{2}}\right)\right), \end{aligned}$$

where $\frac{\overline{X}-\mu}{\sigma_0/\sqrt{n}}$ follows a standard normal distribution. For $\mu = \mu_0$ this simplifies to

$$\mathbf{P}_{\mu_0}(\varphi(X) = 1) = 1 - \Phi(z_{1-\frac{\alpha}{2}}) + \Phi(-z_{1-\frac{\alpha}{2}}) = 2 - 2\Phi(z_{1-\frac{\alpha}{2}}) = 2 - 2\left(1 - \frac{\alpha}{2}\right) = \alpha$$

so that φ is a level-α test. On the other hand, we obtain that the power of φ at $\mu \in \mathbb{R}$ with $\mu \neq \mu_0$ is given by

$$\mathbf{P}_{\mu}(\varphi(X) = 1) = 1 - \mathbf{P}\left(N \in \left(\frac{\mu_0 - \mu}{\sigma_0/\sqrt{n}} - z_{1-\frac{\alpha}{2}}, \frac{\mu_0 - \mu}{\sigma_0/\sqrt{n}} + z_{1-\frac{\alpha}{2}}\right)\right)$$

with a standard normal random variable N.

In Example 5.3, we can observe two typical features of the power. The power is decreasing in the distance from μ to μ_0, whence the probability of rejection is small if μ is too close to μ_0. On the other hand, for $\mu \neq \mu_0$ the power converges to one as $n \to \infty$. This implies that we will reject the null hypothesis with a high probability if n is chosen sufficiently large. In many scientific studies the major part of the work is the collection of the data. Often one has a guess what the difference between the true μ and μ_0 could be. In this case one can use the formula for the power to compute n such that the power equals some given value. This way one can reduce the risk that the sample size is too small to reject the null hypothesis or that one collects too many data.

The formulas in Example 5.3 are very similar to that for two-sided confidence intervals for normal samples. Indeed, there is a general strategy to obtain a test with significance level α from a confidence region with error level α.

Let $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta\in\Theta})$ be a statistical model and let C be a confidence region with error level $\alpha \in (0,1)$ for ϑ. We are interested in the test problem

$$H_0 : \vartheta \in \Theta_0 \quad \text{against} \quad H_1 : \vartheta \notin \Theta_1$$

for $\Theta_0, \Theta_1 \subset \Theta$ with $\Theta_0 \cap \Theta_1 = \emptyset$ and $\Theta_0 \cup \Theta_1 = \Theta$. We define a test $\varphi : \mathcal{X} \to \{0,1\}$,

$$\varphi(x) = \begin{cases} 0, & \Theta_0 \cap C(x) \neq \emptyset, \\ 1, & \Theta_0 \cap C(x) = \emptyset. \end{cases}$$

For $\vartheta \in \Theta_0$ we obtain that

$$\begin{aligned} \mathbf{P}_\vartheta(\{x \in \mathcal{X} : \varphi(x) = 1\}) &= \mathbf{P}_\vartheta(\{x \in \mathcal{X} : \Theta_0 \cap C(x) = \emptyset\}) \\ &\leq \mathbf{P}_\vartheta(\{x \in \mathcal{X} : \{\vartheta\} \cap C(x) = \emptyset\}) \\ &= \mathbf{P}_\vartheta(\{x \in \mathcal{X} : \vartheta \notin C(x)\}) \leq \alpha, \end{aligned}$$

where we used the fact that C is a confidence region with error level α for ϑ. Thus, φ is a test of level α. We illustrate this strategy by several examples.

Example 5.4 (One-sided Gauss test)**.** For a normal sample as in Example 5.3 and $\alpha \in (0,1)$,

$$C_1 : \mathbb{R}^n \ni x \mapsto \Big(-\infty, \overline{x} + \frac{z_{1-\alpha}\sigma_0}{\sqrt{n}} \Big)$$

is a confidence interval for μ with error level α (see Example 4.6). For $\mu_0 \in \mathbb{R}$ we have

$$[\mu_0, \infty) \cap C_1(x) = \emptyset \quad \Longleftrightarrow \quad \overline{x} + \frac{z_{1-\alpha}\sigma_0}{\sqrt{n}} \leq \mu_0 \quad \Longleftrightarrow \quad \overline{x} \leq \mu_0 - \frac{z_{1-\alpha}\sigma_0}{\sqrt{n}}.$$

Thus,

$$\varphi_1 : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \overline{x} > \mu_0 - \frac{z_{1-\alpha}\sigma_0}{\sqrt{n}}, \\ 1, & \overline{x} \leq \mu_0 - \frac{z_{1-\alpha}\sigma_0}{\sqrt{n}}, \end{cases}$$

is a test with significance level α for $H_0 : \mu \geq \mu_0$ against $H_1 : \mu < \mu_0$. Similarly, one obtains that

$$\varphi_2 : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \overline{x} < \mu_0 + \frac{z_{1-\alpha}\sigma_0}{\sqrt{n}}, \\ 1, & \overline{x} \geq \mu_0 + \frac{z_{1-\alpha}\sigma_0}{\sqrt{n}}, \end{cases}$$

is a test with significance level α for $H_0 : \mu \leq \mu_0$ against $H_1 : \mu > \mu_0$.

Example 5.5 (Two-sided t-test)**.** For $n \in \mathbb{N}$ with $n \geq 2$ we consider the statistical model $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), (\mathrm{Normal}(\mu, \sigma^2)^{\otimes n})_{\mu \in \mathbb{R}, \sigma^2 \in (0,\infty)})$ and let $\alpha \in (0,1)$ and $\mu_0 \in \mathbb{R}$. Since

$$C(x) = \Big(\overline{x} - \frac{s_x t_{n-1,1-\frac{\alpha}{2}}}{\sqrt{n}}, \overline{x} + \frac{s_x t_{n-1,1-\frac{\alpha}{2}}}{\sqrt{n}} \Big)$$

for $x \in \mathbb{R}^n$ is a confidence interval for μ with error level α (see Theorem 4.12) and

$$\begin{aligned} C(x) \cap \{\mu_0\} = \emptyset \quad &\Longleftrightarrow \quad \mu_0 \notin \Big(\overline{x} - \frac{s_x t_{n-1,1-\frac{\alpha}{2}}}{\sqrt{n}}, \overline{x} + \frac{s_x t_{n-1,1-\frac{\alpha}{2}}}{\sqrt{n}} \Big) \\ &\Longleftrightarrow \quad \overline{x} \notin \Big(\mu_0 - \frac{s_x t_{n-1,1-\frac{\alpha}{2}}}{\sqrt{n}}, \mu_0 + \frac{s_x t_{n-1,1-\frac{\alpha}{2}}}{\sqrt{n}} \Big), \end{aligned}$$

we see that

$$\varphi : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \overline{x} \in \Big(\mu_0 - \frac{s_x t_{n-1,1-\frac{\alpha}{2}}}{\sqrt{n}}, \mu_0 + \frac{s_x t_{n-1,1-\frac{\alpha}{2}}}{\sqrt{n}} \Big), \\ 1, & \overline{x} \notin \Big(\mu_0 - \frac{s_x t_{n-1,1-\frac{\alpha}{2}}}{\sqrt{n}}, \mu_0 + \frac{s_x t_{n-1,1-\frac{\alpha}{2}}}{\sqrt{n}} \Big), \end{cases}$$

is a test of level α for $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$.

Of course the two-sided t-test from the previous example has one-sided counterparts.

Example 5.6 (One-sided t-test)**.** Assume the statistical model from Example 5.5 and let $\alpha \in (0,1)$. Then

$$\varphi_1 : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \overline{x} > \mu_0 - \frac{s_x t_{n-1,1-\alpha}}{\sqrt{n}}, \\ 1, & \overline{x} \leq \mu_0 - \frac{s_x t_{n-1,1-\alpha}}{\sqrt{n}}, \end{cases}$$

is a level-α test for $H_0 : \mu \geq \mu_0$ against $H_1 : \mu < \mu_0$. For $\mu \in \mathbb{R}$ and $\sigma^2 \in (0, \infty)$ let $X \sim \text{Normal}(\mu, \sigma^2)^{\otimes n} =: \mathbf{P}_{\mu,\sigma^2}$. For $\mu \geq \mu_0$ (i.e., if the null hypothesis is true), we have

$$\begin{aligned}\mathbf{P}_{\mu,\sigma^2}\left(\overline{X} \leq \mu_0 - \frac{S_X t_{n-1,1-\alpha}}{\sqrt{n}}\right) &= \mathbf{P}_{\mu,\sigma^2}\left(\frac{\overline{X} - \mu}{S_X/\sqrt{n}} \leq \frac{\mu_0 - \mu}{S_x/\sqrt{n}} - t_{n-1,1-\alpha}\right) \\ &\leq \mathbf{P}_{\mu,\sigma^2}\left(\frac{\overline{X} - \mu}{S_X/\sqrt{n}} \leq -t_{n-1,1-\alpha}\right).\end{aligned}$$

Since the random variable $\frac{\overline{X}-\mu}{S_X/\sqrt{n}}$ has the same distribution as a random variable $Y \sim t_{n-1}$ (i.e., a t-distribution with $n-1$ degrees of freedom) by Theorem 4.11, we obtain that

$$\mathbf{P}_{\mu,\sigma^2}\left(\overline{X} \leq \mu_0 - \frac{S_X t_{n-1,1-\alpha}}{\sqrt{n}}\right) \leq \mathbf{P}(Y \leq -t_{n-1,1-\alpha}) = 1 - \mathbf{P}(Y \leq t_{n-1,1-\alpha}) = 1 - (1-\alpha) = \alpha.$$

For the test problem $H_0 : \mu \leq \mu_0$ against $H_1 : \mu > \mu_0$ one can show analogously that

$$\varphi_2 : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & \overline{x} < \mu_0 + \frac{s_x t_{n-1,1-\alpha}}{\sqrt{n}}, \\ 1, & \overline{x} \geq \mu_0 + \frac{s_x t_{n-1,1-\alpha}}{\sqrt{n}}, \end{cases}$$

is a level-α test.

Example 5.7 (Two-sided test for the variance of a normal sample)**.** Consider the statistical model from Example 5.5 and the two-sided test problem $H_0 : \sigma^2 = \sigma_0^2$ against $H_1 : \sigma^2 \neq \sigma_0^2$ for some $\sigma_0^2 \in (0, \infty)$. For $\alpha \in (0, 1)$,

$$\varphi : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & s_x^2 \in \left(\frac{\sigma_0^2 \chi^2_{n-1,\alpha/2}}{n-1}, \frac{\sigma_0^2 \chi^2_{n-1,1-\alpha/2}}{n-1}\right), \\ 1, & s_x^2 \notin \left(\frac{\sigma_0^2 \chi^2_{n-1,\alpha/2}}{n-1}, \frac{\sigma_0^2 \chi^2_{n-1,1-\alpha/2}}{n-1}\right), \end{cases}$$

is a test of level α. This follows from Theorem 4.13 and our construction to obtain tests from confidence regions.

The previous two-sided test for the variance has also one-sided counterparts.

Example 5.8 (One-sided tests for the variance of a normal sample)**.** Assume the statistical model from Example 5.5 and let $\alpha \in (0, 1)$ and $\sigma_0^2 \in (0, \infty)$. For $H_0 : \sigma^2 \leq \sigma_0^2$ against $H_1 : \sigma^2 > \sigma_0^2$,

$$\varphi_1 : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & s_x^2 < \frac{\sigma_0^2 \chi^2_{n-1,1-\alpha}}{n-1}, \\ 1, & s_x^2 \geq \frac{\sigma_0^2 \chi^2_{n-1,1-\alpha}}{n-1}, \end{cases}$$

is a level-α test, while

$$\varphi_2 : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & s_x^2 > \frac{\sigma_0^2 \chi^2_{n-1,\alpha}}{n-1}, \\ 1, & s_x^2 \leq \frac{\sigma_0^2 \chi^2_{n-1,\alpha}}{n-1}, \end{cases}$$

is a level-α test for $H_0 : \sigma^2 \geq \sigma_0^2$ against $H_1 : \sigma^2 < \sigma_0^2$. For $\mu \in \mathbb{R}$, $\sigma^2 \in (0, \sigma_0^2]$ and $X \sim \text{Normal}(\mu, \sigma^2)^{\otimes n} =: \mathbf{P}_{\mu,\sigma^2}$, we obtain

$$\begin{aligned}\mathbf{P}_{\mu,\sigma^2}(\varphi_1(X) = 1) &= \mathbf{P}_{\mu,\sigma^2}\left(s_X^2 \geq \frac{\sigma_0^2 \chi^2_{n-1,1-\alpha}}{n-1}\right) = \mathbf{P}_{\mu,\sigma^2}\left(\frac{(n-1)s_X^2}{\sigma^2} \geq \frac{\sigma_0^2}{\sigma^2}\chi^2_{n-1,1-\alpha}\right) \\ &\leq \mathbf{P}_{\mu,\sigma^2}\left(\frac{(n-1)s_X^2}{\sigma^2} \geq \chi^2_{n-1,1-\alpha}\right).\end{aligned}$$

By Theorem 4.11, $\frac{(n-1)s_X^2}{\sigma^2}$ has the same distribution as a random variable $Z \sim \chi^2_{n-1}$, whence

$$\mathbf{P}_{\mu,\sigma^2}(\varphi_1(X) = 1) \leq \mathbf{P}(Z \geq \chi^2_{n-1,1-\alpha}) = 1 - \mathbf{P}(Z \leq \chi^2_{n-1,1-\alpha}) = 1 - (1-\alpha) = \alpha.$$

The computation for φ_2 goes analogously.

In Example 4.5 we studied asymptotic confidence intervals. One can do the same for tests.

Example 5.9 (Asymptotic two-sided test for the binomial distribution)**.** For $n \in \mathbb{N}$ we consider the statistical models $(\{0, 1, \ldots, n\}, \mathcal{P}(\{0, 1, \ldots, n\}), (\text{Binom}(n, p))_{p\in(0,1)})$ and for $p \in (0, 1)$ let $\mathbf{P}_p^{(n)} \sim \text{Binom}(n, p)$ and let $X^{(n)}$ denote a random variable with this distribution. Here we include the dependence on n in our notation since we will let $n \to \infty$ later. Fix $\alpha \in (0, 1)$ and let $p_0 \in (0, 1)$. Our goal is to test $H_0 : p = p_0$ against $H_1 : p \neq p_0$. For $n \in \mathbb{N}$ we define

$$\varphi^{(n)} : \{0, 1, \ldots, n\} \ni k \mapsto \begin{cases} 0, & \frac{k}{n} \in \left(p_0 - \frac{\sqrt{p_0(1-p_0)}z_{1-\alpha/2}}{\sqrt{n}}, p_0 + \frac{\sqrt{p_0(1-p_0)}z_{1-\alpha/2}}{\sqrt{n}}\right), \\ 1, & \frac{k}{n} \notin \left(p_0 - \frac{\sqrt{p_0(1-p_0)}z_{1-\alpha/2}}{\sqrt{n}}, p_0 + \frac{\sqrt{p_0(1-p_0)}z_{1-\alpha/2}}{\sqrt{n}}\right). \end{cases}$$

We have that

$$\begin{aligned}&\mathbf{P}_{p_0}^{(n)}(\varphi^{(n)}(X^{(n)}) = 1) \\ &= \mathbf{P}_{p_0}^{(n)}\left(\frac{X^{(n)}}{n} \notin \left(p_0 - \frac{\sqrt{p_0(1-p_0)}z_{1-\alpha/2}}{\sqrt{n}}, p_0 + \frac{\sqrt{p_0(1-p_0)}z_{1-\alpha/2}}{\sqrt{n}}\right)\right) \\ &= 1 - \mathbf{P}_{p_0}^{(n)}\left(\frac{X^{(n)} - np_0}{\sqrt{n}\sqrt{p_0(1-p_0)}} \in (-z_{1-\alpha/2}, z_{1-\alpha/2})\right).\end{aligned}$$

From the central limit theorem we obtain

$$\begin{aligned}\lim_{n\to\infty} \mathbf{P}_{p_0}^{(n)}(\varphi^{(n)}(X^{(n)}) = 1) &= 1 - (\Phi(z_{1-\alpha/2}) - \Phi(-z_{1-\alpha/2})) = 2 - 2\Phi(z_{1-\alpha/2}) \\ &= 2 - 2\left(1 - \frac{\alpha}{2}\right) = \alpha.\end{aligned}$$

If this equation was satisfied without the limit, $\varphi^{(n)}$ would be a test with level α for each $n \in \mathbb{N}$. However, since this equation holds only in an asymptotic sense, we say that $(\varphi^{(n)})_{n\in\mathbb{N}}$ have asymptotically the level α.

A further asymptotic property of tests is consistency. We say that a test is consistent if for each $\vartheta \in \Theta_1$ the probability of rejection converges to one as the sample size n goes to infinity, which is equivalent to the convergence of the power to one. Note that the Gauss test, the t-test and the test for the variance of a normal sample in their one- and two-sided versions as well as the asymptotic test for the binomial distribution in the previous example are all consistent.

When one uses statistical software for tests, often no level α is chosen since the so-called p-value is computed. The p-value of a given sample is the probability to obtain under the null hypothesis a result that is at least as extreme as the sample. The interpretation of the p-value p^* is as follows. We can reject the null hypothesis at all levels $\alpha \geq p^*$. For a level $\alpha < p^*$ the null hypothesis cannot be rejected.

Example 5.10. We compute for the two-sided Gauss test from Example 5.3 the p-value of a sample $x \in \mathbb{R}^n$. In this test we consider the distance from the sample mean to μ_0. Thus, the p-value p^* is given by

$$p^* = \mathbf{P}_{\mu_0}(|\overline{X} - \mu_0| \geq |\overline{x} - \mu_0|)$$

with $X \sim \text{Normal}(\mu_0, \sigma_0^2)^{\otimes n}$. Since $\frac{\overline{X}-\mu_0}{\sigma_0/\sqrt{n}} \sim \text{Normal}(0,1)$, this yields

$$\begin{aligned} p^* &= \mathbf{P}_{\mu_0}\left(\frac{|\overline{X} - \mu_0|}{\sigma_0/\sqrt{n}} \geq \frac{\sqrt{n}|\overline{x} - \mu_0|}{\sigma_0}\right) = 1 - \Phi\left(\frac{\sqrt{n}|\overline{x} - \mu_0|}{\sigma_0}\right) + \Phi\left(-\frac{\sqrt{n}|\overline{x} - \mu_0|}{\sigma_0}\right) \\ &= 2 - 2\Phi\left(\frac{\sqrt{n}|\overline{x} - \mu_0|}{\sigma_0}\right). \end{aligned}$$

Now let us compute the rejection region for a Gauss test for the null hypothesis $\mu = \mu_0$ with level p^*. We have that

$$\frac{\sigma_0 z_{1-p^*/2}}{\sqrt{n}} = \frac{\sigma_0}{\sqrt{n}} z_{\Phi(\frac{\sqrt{n}|\overline{x}-\mu_0|}{\sigma_0})} = \frac{\sigma_0}{\sqrt{n}} \frac{\sqrt{n}|\overline{x} - \mu_0|}{\sigma_0} = |\overline{x} - \mu_0|,$$

whence we reject if the sample mean is outside of

$$\left(\mu_0 - \frac{\sigma_0 z_{1-p^*/2}}{\sqrt{n}}, \mu_0 + \frac{\sigma_0 z_{1-p^*/2}}{\sqrt{n}}\right) = (\mu_0 - |\overline{x} - \mu_0|, \mu_0 + |\overline{x} - \mu_0|).$$

Thus, we can reject the null hypothesis at level p^* for the sample x.

5.2 Multiple testing

In this section we discuss the situation that one is interested in testing more than one hypothesis. Let the statistical model $(\mathcal{X}, \mathscr{F}_{\mathcal{X}}, (\mathbf{P}_\vartheta)_{\vartheta \in \Theta})$ be given and assume that we are interested in the test problems

$$H_0^{(i)} : \vartheta \in \Theta_0^{(i)} \quad \text{against} \quad H_1^{(i)} : \vartheta \in \Theta_1^{(i)}$$

with $\Theta_0^{(i)} \cup \Theta_1^{(i)} = \Theta$ and $\Theta_0^{(i)} \cap \Theta_1^{(i)} = \emptyset$ for $i \in \{1, \dots, m\}$. Such situations naturally arise if one has a sample of size n whose entries are again vectors and one studies the components of the vectors separately. The next example shows that one must be careful in this case.

Example 5.11. We fix $n \in \mathbb{N}$ and $m \in \mathbb{N}$. Let $X_1, \dots, X_n$ be i.i.d. random vectors such that $X_1 \sim \text{Normal}(0,1)^{\otimes m}$. Our goal is to test

$$H_0^{(i)} : \mathbf{E}[X_1^{(i)}] = 0 \quad \text{against} \quad H_1^{(i)} : \mathbf{E}[X_1^{(i)}] \neq 0$$

for $i \in \{1, \dots, m\}$, i.e., the i-th null hypothesis is that the expectation of the underlying distribution of the i-th components of $X_1, \dots, X_n$ is zero. We apply to each of the components the two-sided Gauss test with level $\alpha \in (0,1)$. Obviously $H_0^{(i)}$ is true for all $i \in \{1, \dots, m\}$. Nevertheless we have that

$$\mathbf{P}(H_0^{(i)} \text{ is rejected}) = \alpha$$

for all $i \in \{1, \dots, m\}$. Since all components are independent, we obtain that the events $\{H_0^{(i)} \text{ is rejected}\}$ for $i \in \{1, \dots, m\}$ are all independent. This yields

$$\mathbf{P}(H_0^{(i)} \text{ is rejected for some } i \in \{1, \dots, m\}) = 1 - (1-\alpha)^m.$$

Consequently, we will reject at least one null hypothesis with a high probability if m is sufficiently large compared with α.

The previous example basically shows that by conducting a large number of tests one might be able to reject some null hypotheses and, thus, find some structure in the data that is not present. The issue with the presented example is that we performed a large number of tests each of them with level α. However, the probabilities of single type I errors accumulate so that the probability that at least one type I error occurs is much higher. Thus, it makes sense to require that the latter probability is bounded by some given $\alpha_{global} \in (0,1)$, i.e.,

$$\mathbf{P}_\vartheta(\exists i \in \{1, \dots, m\} : \vartheta \in \Theta_0^{(i)} \text{and } H_0^{(i)} \text{ is rejected}) \leq \alpha_{global}$$

for all $\vartheta \in \Theta$. In order to achieve such an inequality one adjusts the level α of the tests for $H_0^{(i)} : \vartheta \in \Theta_0^{(i)}$ against $H_1^{(i)} : \vartheta \in \Theta_1^{(i)}$ for $i \in \{1, \dots, m\}$. The simplest way to choose this α is the so-called Bonferroni correction. Here, we conduct each of the m tests at level $\alpha = \frac{\alpha_{global}}{m}$. For $\vartheta \in \Theta$ this leads to

$$\begin{aligned}\mathbf{P}_\vartheta(\exists i \in \{1, \dots, m\} : \vartheta \in \Theta_0^{(i)} \text{and } H_0^{(i)} \text{ is rejected}) &\leq \sum_{i \in \{1,\dots,m\} : \vartheta \in \Theta_0^{(i)}} \mathbf{P}_\vartheta(H_0^{(i)} \text{ is rejected}) \\ &\leq \sum_{i \in \{1,\dots,m\} : \vartheta \in \Theta_0^{(i)}} \frac{\alpha_{global}}{m} \leq \alpha_{global}.\end{aligned}$$

The Bonferroni correction is rather conservative and there exist more sophisticated procedures we will not discuss here (see, for example, the so-called Holm correction). In Example 5.11 we can explicitly compute the required α since the single tests are independent, but often this is not possible.

In general a data sample can be only used to study a limited number of hypotheses. If the number of hypotheses is too large, the required level for each individual test might be too small.

Before one starts to analyse a sample one should have fixed the hypotheses that are considered. If one just checks all possible hypotheses, one is doing multiple testing and runs into the problem described in Example 5.11.

Chapter 6

Two-sample problems

So far we always considered a single sample. A common situation in statistics is that two different samples x and y are given and that the goal is to compare the two samples. We denote the size of x by n, while m stands for the size of y. One interpretation is that two different populations are given and that from each population a sample was drawn. To keep our analysis simple, we will only study the problem for normal samples. We start with the situation that both samples have the same known underlying variance.

Theorem 6.1. For $n, m \in \mathbb{N}$ and $\sigma_0^2 \in (0, \infty)$ consider the statistical model

$$(\mathbb{R}^{n+m}, \mathcal{B}(\mathbb{R}^{n+m}), (\text{Normal}(\mu, \sigma_0^2)^{\otimes n} \otimes \text{Normal}(\nu, \sigma_0^2)^{\otimes m})_{\mu,\nu\in\mathbb{R}})$$

and let $\alpha \in (0, 1)$. Then the following statements hold:

a) The map

$$C : \mathbb{R}^{n+m} \ni (x, y) \mapsto \left(\overline{x} - \overline{y} - \frac{\sqrt{n+m}}{\sqrt{nm}}\sigma_0 z_{1-\frac{\alpha}{2}}, \overline{x} - \overline{y} + \frac{\sqrt{n+m}}{\sqrt{nm}}\sigma_0 z_{1-\frac{\alpha}{2}}\right)$$

is a confidence interval for $\mu - \nu$ with error level α.

b) The test

$$\varphi : \mathbb{R}^{n+m} \ni (x, y) \mapsto \begin{cases} 0, & 0 \in C(x, y), \\ 1, & 0 \notin C(x, y), \end{cases}$$

for

$$H_0 : \mu = \nu \quad \text{against} \quad H_1 : \mu \neq \nu$$

has level α.

Proof. For $\mu, \nu \in \mathbb{R}$ let $X \sim \text{Normal}(\mu, \sigma_0^2)^{\otimes n}$ and $Y \sim \text{Normal}(\nu, \sigma_0^2)^{\otimes m}$ be independent and let $\mathbf{P}_{\mu,\nu}$ denote their joint distribution. Since $\overline{X} \sim \text{Normal}(\mu, \frac{\sigma_0^2}{n})$ and $\overline{Y} \sim \text{Normal}(\nu, \frac{\sigma_0^2}{m})$ and they are independent, we obtain

$$\overline{X} - \overline{Y} \sim \text{Normal}\left(\mu - \nu, \frac{n+m}{nm}\sigma_0^2\right).$$

This implies that, for all $\mu, \nu \in \mathbb{R}$,

$$\begin{aligned}
&\mathbf{P}_{\mu,\nu}\big(\mu - \nu \in C(X,Y)\big) \\
&= \mathbf{P}_{\mu,\nu}\left(\mu - \nu \in \left(\overline{X} - \overline{Y} - \frac{\sqrt{n+m}}{\sqrt{nm}}\sigma_0 z_{1-\frac{\alpha}{2}}, \overline{X} - \overline{Y} + \frac{\sqrt{n+m}}{\sqrt{nm}}\sigma_0 z_{1-\frac{\alpha}{2}}\right)\right) \\
&= \mathbf{P}_{\mu,\nu}\left(\frac{\overline{X} - \overline{Y} - (\mu - \nu)}{\frac{\sqrt{n+m}}{\sqrt{nm}}\sigma_0} \in (-z_{1-\frac{\alpha}{2}}, z_{1-\frac{\alpha}{2}})\right) = \Phi(z_{1-\frac{\alpha}{2}}) - \Phi(-z_{1-\frac{\alpha}{2}}) \\
&= 2\Phi(z_{1-\frac{\alpha}{2}}) - 1 = 2\left(1 - \frac{\alpha}{2}\right) - 1 = 1 - \alpha,
\end{aligned}$$

whence C is a confidence interval for $\mu - \nu$ with error level α. From the previous calculation we obtain for $\mu \in \mathbb{R}$,

$$\begin{aligned}
\mathbf{P}_{\mu,\mu}(\varphi(X,Y) = 1) &= \mathbf{P}_{\mu,\mu}(0 \notin C(X,Y)) = \mathbf{P}_{\mu,\mu}(\mu - \mu \notin C(X,Y)) \\
&= 1 - \mathbf{P}_{\mu,\mu}(\mu - \mu \in C(X,Y)) = 1 - (1 - \alpha) = \alpha
\end{aligned}$$

so that φ is a level-α test. □

Remark 6.2. One can also derive one-sided confidence intervals and tests analogously to those in Theorem 6.1.

Next we study the case that both samples have the same underlying variance, but that this variance is unknown.

Theorem 6.3. For $n, m \in \mathbb{N}$ with $n, m \geq 2$ consider the statistical model

$$(\mathbb{R}^{n+m}, \mathcal{B}(\mathbb{R}^{n+m}), (\mathrm{Normal}(\mu, \sigma^2)^{\otimes n} \otimes \mathrm{Normal}(\nu, \sigma^2)^{\otimes m})_{\mu,\nu \in \mathbb{R}, \sigma^2 \in (0,\infty)})$$

and let $\alpha \in (0,1)$. For $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$ define

$$s_{x,y}^2 = \frac{1}{n+m-2}\big((n-1)s_x^2 + (m-1)s_y^2\big).$$

Then the following statements hold:

a) The map

$$C : \mathbb{R}^{n+m} \ni (x,y) \mapsto \left(\overline{x} - \overline{y} - \frac{\sqrt{n+m}}{\sqrt{nm}} t_{n+m-2,1-\frac{\alpha}{2}} s_{x,y}, \overline{x} - \overline{y} + \frac{\sqrt{n+m}}{\sqrt{nm}} t_{n+m-2,1-\frac{\alpha}{2}} s_{x,y}\right)$$

is a confidence interval for $\mu - \nu$ with error level α.

b) The test

$$\varphi : \mathbb{R}^{n+m} \ni (x,y) \mapsto \begin{cases} 0, & 0 \in C(x,y), \\ 1, & 0 \notin C(x,y), \end{cases}$$

for

$$H_0 : \mu = \nu \quad \text{against} \quad H_1 : \mu \neq \nu$$

has level α.

Proof. For $\mu, \nu \in \mathbb{R}$ and $\sigma^2 \in (0, \infty)$ let $X \sim \text{Normal}(\mu, \sigma^2)^{\otimes n}$ and $Y \sim \text{Normal}(\nu, \sigma^2)^{\otimes m}$ be independent and let $\mathbf{P}_{\mu,\nu,\sigma^2}$ denote their joint distribution. By the independence of X and Y and Theorem 4.11, $\overline{X}$, $\overline{Y}$, s_X^2 and s_Y^2 are independent. This implies the independence of $\overline{X} - \overline{Y}$ and $s_{X,Y}^2$. Moreover, we obtain

$$\overline{X} - \overline{Y} - (\mu - \nu) \sim \text{Normal}\left(0, \frac{\sigma^2}{n} + \frac{\sigma^2}{m}\right) = \text{Normal}\left(0, \frac{(n+m)\sigma^2}{nm}\right).$$

Since, by Theorem 4.11, $\frac{(n-1)s_X^2}{\sigma^2} \sim \chi^2_{n-1}$ and $\frac{(m-1)s_Y^2}{\sigma^2} \sim \chi^2_{m-1}$ are independent, we obtain

$$\frac{(n-1)s_X^2 + (m-1)s_Y^2}{\sigma^2} \sim \chi^2_{n+m-2}.$$

Thus, we see that

$$\frac{\overline{X} - \overline{Y} - (\mu - \nu)}{\sqrt{\frac{n+m}{nm}}\sigma} \sim \text{Normal}(0,1) \quad \text{and} \quad \frac{(n+m-2)s_{X,Y}^2}{\sigma^2} \sim \chi^2_{n+m-2},$$

whence

$$\frac{\overline{X} - \overline{Y} - (\mu - \nu)}{\sqrt{\frac{n+m}{nm}}s_{X,Y}} = \frac{\frac{\overline{X} - \overline{Y} - (\mu - \nu)}{\sqrt{\frac{n+m}{nm}}\sigma}}{\sqrt{\frac{(n+m-2)s_{X,Y}^2}{(n+m-2)\sigma^2}}} \sim t_{n+m-2}.$$

This implies that

$$\begin{aligned}
&\mathbf{P}_{\mu,\nu,\sigma^2}(\mu - \nu \in C(X,Y)) \\
&= \mathbf{P}_{\mu,\nu,\sigma^2}\left(\mu - \nu \in \left(\overline{X} - \overline{Y} - \frac{\sqrt{n+m}}{\sqrt{nm}} t_{n+m-2,1-\frac{\alpha}{2}} s_{X,Y}, \overline{X} - \overline{Y} + \frac{\sqrt{n+m}}{\sqrt{nm}} t_{n+m-2,1-\frac{\alpha}{2}} s_{X,Y}\right)\right) \\
&= \mathbf{P}_{\mu,\nu,\sigma^2}\left(\frac{\overline{X} - \overline{Y} - (\mu - \nu)}{\frac{\sqrt{n+m}}{\sqrt{nm}} s_{X,Y}} \in (-t_{n+m-2,1-\frac{\alpha}{2}}, t_{n+m-2,1-\frac{\alpha}{2}})\right).
\end{aligned}$$

For $Z \sim t_{n+m-2}$ we obtain

$$\begin{aligned}
\mathbf{P}_{\mu,\nu,\sigma^2}(\mu - \nu \in C(X,Y)) &= \mathbf{P}(Z \in (-t_{n+m-2,1-\frac{\alpha}{2}}, t_{n+m-2,1-\frac{\alpha}{2}})) \\
&= \mathbf{P}(Z \le t_{n+m-2,1-\frac{\alpha}{2}}) - \mathbf{P}(Z \le -t_{n+m-2,1-\frac{\alpha}{2}}) \\
&= 2\mathbf{P}(Z \le t_{n+m-2,1-\frac{\alpha}{2}}) - 1 = 2\left(1 - \frac{\alpha}{2}\right) - 1 = 1 - \alpha,
\end{aligned}$$

which implies that C is confidence interval with error level α. Similarly as in the proof of Theorem 6.1 one can show that φ is a level-α test. □

The two-sided confidence intervals and tests of the previous theorem have also one-sided counterparts.

Theorem 6.4. Assume the same framework as in Theorem 6.3. Then, the following holds:

a) The maps

$$C_1 : \mathbb{R}^{n+m} \ni (x, y) \mapsto \left(-\infty, \overline{x} - \overline{y} + \frac{\sqrt{n+m}}{\sqrt{nm}} t_{n+m-2,1-\alpha} s_{x,y} \right)$$

and

$$C_2 : \mathbb{R}^{n+m} \ni (x, y) \mapsto \left(\overline{x} - \overline{y} - \frac{\sqrt{n+m}}{\sqrt{nm}} t_{n+m-2,1-\alpha} s_{x,y}, \infty \right)$$

are confidence intervals for $\mu - \nu$ with error level α.

b) For $H_0 : \mu \geq \nu$ against $H_1 : \mu < \nu$,

$$\varphi : \mathbb{R}^{n+m} \ni (x, y) \mapsto \begin{cases} 0, & 0 \in C_1(x, y), \\ 1, & 0 \notin C_1(x, y), \end{cases}$$

is a level-α test.

c) For $H_0 : \mu \leq \nu$ against $H_1 : \mu > \nu$,

$$\varphi : \mathbb{R}^{n+m} \ni (x, y) \mapsto \begin{cases} 0, & 0 \in C_2(x, y), \\ 1, & 0 \notin C_2(x, y), \end{cases}$$

is a level-α test.

Proof. The proof is analogously to the one of Theorem 6.3. □

Remark 6.5. In the case of so-called paired samples one has n pairs of observations $(x_1, y_1), \ldots, (x_n, y_n)$, where x_i and y_i belong together. Here, one may not apply the above results to the two samples $(x_1, \ldots, x_n)$ and $(y_1, \ldots, y_n)$ since they might not be independent. Instead one can study the single sample $(x_1 - y_1, \ldots, x_n - y_n)$.

Chapter 7

Linear regression

7.1 Linear models

We consider the situation that a quantity y depends on quantities $x_1, \ldots, x_k$ and a random observation error. Our goal is to understand how y is affected by $x_1, \ldots, x_k$. We will always suppose that the dependence of y on $x_1, \ldots, x_k$ is linear. More precisely, we assume that

$$y = \gamma_0 + \gamma_1 x_1 + \ldots + \gamma_k x_k + \varepsilon = \gamma_0 + \sum_{j=1}^{k} \gamma_j x_j + \varepsilon \tag{7.1}$$

with $\gamma_0, \gamma_1, \ldots, \gamma_k \in \mathbb{R}$ and a random variable ε. The idea is that y is a deterministic function of $x_1, \ldots, x_k$ plus random noise. For example, one obtains such a model when one observes a deterministic quantity in an experiment with random measurement errors.

We denote y as dependent variable, while $x_1, \ldots, x_k$ are called independent variables. Here, the words dependent and independent are not to be understood in the probabilistic sense, they only indicate that y is a function of $x_1, \ldots, x_k$.

To the case $k = 1$ (i.e., we have only one independent variable) we refer as simple linear regression in contrast to multiple linear regression for $k > 1$.

It is also possible to incorporate nonlinear dependence into linear regression. For example, if y depends on a single dependent variable x in a quadratic way, one can choose $x_1 = x$ and $x_2 = x^2$. This approach, which can be applied for polynomials of arbitrary order, is called polynomial regression. We can also include the dependence of y on the interaction of two independent variables x and z by adding xz as an independent variable.

In this section, our goal is to determine and investigate $\gamma_0, \gamma_1, \ldots, \gamma_k$, which describe the dependence of y on $x_1, \ldots, x_k$ in the linear model (7.1), from given data. In particular, estimates for $\gamma_0, \gamma_1, \ldots, \gamma_k$ allow to predict y for given $x_1, \ldots, x_k$.

In the following, we assume that n observations $y^{(i)}$ and $x_1^{(i)}, \ldots, x_k^{(i)}$ for $i \in \{1, \ldots, n\}$ are given. More precisely, let

$$y^{(i)} = \gamma_0 + \gamma_1 x_1^{(i)} + \ldots + \gamma_k x_k^{(i)} + \varepsilon^{(i)} = \gamma_0 + \sum_{j=1}^{k} \gamma_j x_j^{(i)} + \varepsilon^{(i)}$$

for $i \in \{1, \dots, n\}$ with $\gamma_0, \gamma_1, \dots, \gamma_k \in \mathbb{R}$ and uncorrelated random variables $\varepsilon^{(1)}, \dots, \varepsilon^{(n)}$ such that $\mathbf{E}[\varepsilon^{(i)}] = 0$ for $i \in \{1, \dots, n\}$. Note that $\varepsilon^{(1)}, \dots, \varepsilon^{(n)}$ do not need to be independent. If they have all the same expectation distinct from zero, we can subtract the expectation and incorporate it into γ_0, whence the assumption of expectation zero is no restriction. We denote γ_0 as intercept.

We can rewrite the situation above as

$$\mathbf{y} = \mathbf{A}\gamma + \boldsymbol{\varepsilon}$$

with $\mathbf{y} \in \mathbb{R}^n$, $\mathbf{A} \in \mathbb{R}^{n \times (k+1)}$, $\gamma \in \mathbb{R}^{k+1}$ and $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ given by

$$\mathbf{y} = \begin{pmatrix} y^{(1)} \\ \vdots \\ y^{(n)} \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 1 & x_1^{(1)} & \dots & x_k^{(1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_1^{(n)} & \dots & x_k^{(n)} \end{pmatrix}, \quad \gamma = \begin{pmatrix} \gamma_0 \\ \gamma_1 \\ \vdots \\ \gamma_k \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon^{(1)} \\ \vdots \\ \varepsilon^{(n)} \end{pmatrix}.$$

The matrix $\mathbf{A}$ is called design matrix.

In the following we estimate γ by its least squares estimator $\widehat{\gamma}$ which is the solution of the optimisation problem

$$\min_{\tilde{\gamma} \in \mathbb{R}^{k+1}} \|\mathbf{y} - \mathbf{A}\tilde{\gamma}\|.$$

This is equivalent to minimising the function

$$\begin{aligned} \tilde{\gamma} \mapsto h(\tilde{\gamma}) := \|\mathbf{y} - \mathbf{A}\tilde{\gamma}\|^2 = \langle \mathbf{y} - \mathbf{A}\tilde{\gamma}, \mathbf{y} - \mathbf{A}\tilde{\gamma} \rangle &= \langle \mathbf{y}, \mathbf{y} \rangle - 2\langle \mathbf{y}, \mathbf{A}\tilde{\gamma} \rangle + \langle \mathbf{A}\tilde{\gamma}, \mathbf{A}\tilde{\gamma} \rangle \\ &= \langle \mathbf{y}, \mathbf{y} \rangle - 2\mathbf{y}^T\mathbf{A}\tilde{\gamma} + \tilde{\gamma}^T\mathbf{A}^T\mathbf{A}\tilde{\gamma}. \end{aligned}$$

Its gradient is given by

$$\nabla h(\tilde{\gamma}) = -2\mathbf{A}^T\mathbf{y} + 2\mathbf{A}^T\mathbf{A}\tilde{\gamma},$$

whence

$$\nabla h(\tilde{\gamma}) = 0 \quad \Longleftrightarrow \quad -2\mathbf{A}^T\mathbf{y} + 2\mathbf{A}^T\mathbf{A}\tilde{\gamma} = 0 \quad \Longleftrightarrow \quad \mathbf{A}^T\mathbf{A}\tilde{\gamma} = \mathbf{A}^T\mathbf{y}.$$

Since h has the positive semi-definite Hessian $2\mathbf{A}^T\mathbf{A}$, the solution $\tilde{\gamma}$ of $\mathbf{A}^T\mathbf{A}\tilde{\gamma} = \mathbf{A}^T\mathbf{y}$ is a minimum of h. Under the assumption that $\mathbf{A}^T\mathbf{A}$ is invertible, we have the solution $\tilde{\gamma} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y}$. Thus, the least squares estimator $\widehat{\gamma}$ is given by

$$\widehat{\gamma} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y}.$$

In general the matrix $\mathbf{A}$ is not quadratic, whence the expression on the right-hand side cannot be further simplified.

We use the notation $\widehat{\mathbf{y}} = \mathbf{A}\widehat{\gamma}$ and denote the components of the vector

$$\widehat{\boldsymbol{\varepsilon}} = \mathbf{y} - \widehat{\mathbf{y}}$$

as residuals. The least squares estimator $\widehat{\gamma}$ has the following properties.

Theorem 7.1. Assume that $\mathbf{A}^T\mathbf{A}$ is invertible.

a) The least squares estimator $\widehat{\gamma} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y}$ is an unbiased estimator for γ.

b) If there exists a $\sigma^2 \in (0,\infty)$ such that $\mathbf{Var}(\varepsilon^{(i)}) = \sigma^2$ for all $i \in \{1,\dots,n\}$,

$$\widehat{\sigma}^2 := \frac{1}{n-k-1}\|\widehat{\boldsymbol{\varepsilon}}\|^2$$

is an unbiased estimator for σ^2.

Proof. The first statement follows from

$$\begin{aligned}\mathbf{E}[\widehat{\gamma}] &= \mathbf{E}[(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y}] = \mathbf{E}[(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T(\mathbf{A}\gamma+\boldsymbol{\varepsilon})]\\ &= (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T(\mathbf{A}\gamma+\mathbf{E}[\boldsymbol{\varepsilon}]) = (\mathbf{A}^T\mathbf{A})^{-1}(\mathbf{A}^T\mathbf{A})\gamma = \gamma.\end{aligned}$$

Let I_n denote the identity matrix in $\mathbb{R}^{n\times n}$. We have

$$\widehat{\boldsymbol{\varepsilon}} = \mathbf{y} - \widehat{\mathbf{y}} = \mathbf{y} - \mathbf{A}\widehat{\gamma} = \mathbf{y} - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y} = (I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{y}.$$

Since the matrix $\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T$ is symmetric, we obtain

$$\begin{aligned}\|\widehat{\boldsymbol{\varepsilon}}\|^2 &= \langle (I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{y}, (I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{y}\rangle\\ &= \mathbf{y}^T(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^T(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{y}\\ &= \mathbf{y}^T(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{y}\\ &= \mathbf{y}^T(I_n - 2\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T + \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{y}\\ &= \mathbf{y}^T(I_n - 2\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T + \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{y} = \mathbf{y}^T(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{y}.\end{aligned}$$

Note that

$$(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{A}\gamma = \mathbf{A}\gamma - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A}\gamma = \mathbf{A}\gamma - \mathbf{A}\gamma = 0,$$

which leads to

$$\begin{aligned}\mathbf{E}[\|\widehat{\boldsymbol{\varepsilon}}\|^2] &= \mathbf{E}[(\mathbf{A}\gamma+\boldsymbol{\varepsilon})^T(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)(\mathbf{A}\gamma+\boldsymbol{\varepsilon})]\\ &= \mathbf{E}[\boldsymbol{\varepsilon}^T(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\boldsymbol{\varepsilon} + (\mathbf{A}\gamma+2\boldsymbol{\varepsilon})^T(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{A}\gamma]\\ &= \mathbf{E}[\boldsymbol{\varepsilon}^T(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\boldsymbol{\varepsilon}].\end{aligned}$$

The assumptions that $\varepsilon^{(1)},\dots,\varepsilon^{(n)}$ are uncorrelated and that $\mathbf{E}[\varepsilon^{(i)}] = 0$ and $\mathbf{Var}(\varepsilon^{(i)}) = \sigma^2$ for all $i \in \{1,\dots,n\}$ imply

$$\mathbf{E}[\varepsilon^{(i_1)}\varepsilon^{(i_2)}] = \begin{cases}\sigma^2, & i_1 = i_2,\\ 0, & i_1 \neq i_2,\end{cases}$$

for $i_1, i_2 \in \{1,\dots,n\}$. This yields

$$\mathbf{E}[\|\widehat{\boldsymbol{\varepsilon}}\|^2] = \sigma^2\,\mathrm{tr}(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T),$$

where the trace $\mathrm{tr}(M)$ of a symmetric matrix M is the sum of its diagonal elements. For two matrices $B = (b_{ij})_{i\in\{1,\dots,p\},j\in\{1,\dots,q\}} \in \mathbb{R}^{p\times q}$ and $C = (c_{ij})_{i\in\{1,\dots,q\},j\in\{1,\dots,p\}} \in \mathbb{R}^{q\times p}$, one has

$$\mathrm{tr}(BC) = \sum_{i=1}^{p}\sum_{j=1}^{q} b_{ij}c_{ji} = \sum_{j=1}^{q}\sum_{i=1}^{p} c_{ji}b_{ij} = \mathrm{tr}(CB).$$

This means that

$$\begin{aligned}\mathrm{tr}(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T) &= \mathrm{tr}(I_n) - \mathrm{tr}(\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\\ &= n - \mathrm{tr}(\mathbf{A}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}) = n - \mathrm{tr}(I_{k+1}) = n-k-1.\end{aligned}$$

Thus, we see that

$$\mathbf{E}[\widehat{\sigma}^2] = \frac{1}{n-k-1}(n-k-1)\sigma^2 = \sigma^2,$$

which completes the proof. □

7.2 The Gaussian linear model

In order to find confidence regions and to test hypotheses, one needs stronger assumptions on the distribution of the error terms $\varepsilon^{(1)},\dots,\varepsilon^{(n)}$. Throughout this section we let

$$\boldsymbol{\varepsilon} \sim \mathrm{Normal}(0, \sigma^2 I_n)$$

for a fixed $\sigma^2 \in (0,\infty)$ and the identity matrix I_n in $\mathbb{R}^{n\times n}$, i.e., $\varepsilon^{(1)},\dots,\varepsilon^{(n)}$ are independent normal random variables with expectation zero and variance σ^2. We assume that σ^2 is unknown and estimate it from the observations as well as $\gamma_0,\gamma_1,\dots,\gamma_k$.

Theorem 7.2. Assume the Gaussian linear model and that $\mathbf{A}^T\mathbf{A}$ is invertible and let $\sigma^2_{ML} = \frac{\|\widehat{\varepsilon}\|^2}{n}$. Then, $(\widehat{\gamma}, \widehat{\sigma}^2_{ML})$ is the maximum likelihood estimator for (γ,σ^2).

Proof. Our assumption is that the observations follow a $\mathrm{Normal}(\mathbf{A}\gamma, \sigma^2 I_n)$-distribution. This leads to the likelihood function

$$L_{\mathbf{y}}(\gamma,\sigma) = \frac{1}{\sqrt{2\pi}^n\sigma^n} e^{-\frac{1}{2\sigma^2}\|\mathbf{A}\gamma-\mathbf{y}\|^2}$$

and the log-likelihood function

$$\log L_{\mathbf{y}}(\gamma,\sigma) = -n\log\sqrt{2\pi} - n\log\sigma - \frac{1}{2\sigma^2}\|\mathbf{A}\gamma-\mathbf{y}\|^2$$

for $\gamma\in\mathbb{R}^{k+1}$ and $\sigma\in(0,\infty)$. We have to maximise the log-likelihood function. For each choice of $\sigma\in(0,\infty)$ the log-likelihood function becomes maximal if γ minimises

$$\|\mathbf{A}\gamma-\mathbf{y}\|^2.$$

From the previous section we know that this expression becomes minimal for $\gamma = \widehat{\gamma}$. Thus, it remains to find $\sigma \in (0, \infty)$ such that

$$\begin{aligned} h(\sigma^2) := -n \log \sqrt{2\pi} - n \log \sigma - \frac{1}{2\sigma^2} \|\mathbf{A}\widehat{\gamma} - \mathbf{y}\|^2 &= -n \log \sqrt{2\pi} - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|\widehat{\mathbf{y}} - \mathbf{y}\|^2 \\ &= -n \log \sqrt{2\pi} - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|\widehat{\boldsymbol{\varepsilon}}\|^2 \end{aligned}$$

is maximised. We think of h as a function of σ^2 instead of σ because this simplifies the calculations. We have

$$h'(\sigma^2) = -\frac{n}{2\sigma^2} + \frac{\|\widehat{\boldsymbol{\varepsilon}}\|^2}{2\sigma^4}$$

and, thus,

$$h'(\sigma^2) = 0 \quad \Longleftrightarrow \quad -\frac{n}{2\sigma^2} + \frac{\|\widehat{\boldsymbol{\varepsilon}}\|^2}{2\sigma^4} = 0 \quad \Longleftrightarrow \quad \sigma^2 = \frac{\|\widehat{\boldsymbol{\varepsilon}}\|^2}{n}.$$

From the form of h' we see that the critical point must be a maximum, whence $(\widehat{\gamma}, \widehat{\sigma}^2_{ML})$ is the maximum likelihood estimator for (γ, σ^2). □

The next result provides the distributions of the least squares estimator $\widehat{\gamma}$ and the estimators for σ^2.

Theorem 7.3. Consider the Gaussian linear model and assume that $\mathbf{A}^T\mathbf{A}$ is invertible. Then,

a) $\widehat{\gamma} \sim \text{Normal}(\gamma, \sigma^2(\mathbf{A}^T\mathbf{A})^{-1})$

b) $\widehat{\gamma}$ and $\widehat{\boldsymbol{\varepsilon}}$ are independent

c) $\frac{n\widehat{\sigma}^2_{ML}}{\sigma^2} = \frac{(n-k-1)\widehat{\sigma}^2}{\sigma^2} = \frac{\|\widehat{\boldsymbol{\varepsilon}}\|^2}{\sigma^2} \sim \chi^2_{n-k-1}$.

Proof. From $\widehat{\gamma} = (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{y}$ and $\mathbf{y} \sim \text{Normal}(\mathbf{A}\gamma, \sigma^2 I_n)$ it follows that

$$\widehat{\gamma} \sim \text{Normal}((\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A}\gamma, (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\sigma^2 I_n((\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^T).$$

Because of $(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A}\gamma = \gamma$ and

$$(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\sigma^2 I_n((\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^T = \sigma^2(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1} = \sigma^2(\mathbf{A}^T\mathbf{A})^{-1},$$

this simplifies to

$$\widehat{\gamma} \sim \text{Normal}(\gamma, \sigma^2(\mathbf{A}^T\mathbf{A})^{-1}).$$

Since

$$\widehat{\boldsymbol{\varepsilon}} = \mathbf{y} - \widehat{\mathbf{y}} = (I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{y},$$

we deduce

$$\begin{pmatrix} \widehat{\gamma} \\ \widehat{\boldsymbol{\varepsilon}} \end{pmatrix} = \begin{pmatrix} (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \\ I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \end{pmatrix} \mathbf{y}.$$

This random vector has the expectation

$$\begin{pmatrix} (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \\ I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \end{pmatrix} \mathbf{A}\gamma = \begin{pmatrix} (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A}\gamma \\ (I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{A}\gamma \end{pmatrix} = \begin{pmatrix} \gamma \\ 0 \end{pmatrix} =: \mu$$

and the covariance matrix

$$\begin{aligned}
&\begin{pmatrix} (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \\ I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \end{pmatrix} \sigma^2 I_n \begin{pmatrix} (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \\ I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \end{pmatrix}^T \\
&= \sigma^2 \begin{pmatrix} (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \\ I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \end{pmatrix} \begin{pmatrix} \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1} & I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \end{pmatrix} \\
&= \sigma^2 \begin{pmatrix} (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1} & (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T) \\ (I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1} & (I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)(I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T) \end{pmatrix} \\
&= \sigma^2 \begin{pmatrix} (\mathbf{A}^T\mathbf{A})^{-1} & (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T - (\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T \\ \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1} - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1} & (I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^2 \end{pmatrix} \\
&= \sigma^2 \begin{pmatrix} (\mathbf{A}^T\mathbf{A})^{-1} & 0 \\ 0 & (I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)^2 \end{pmatrix} =: \Sigma.
\end{aligned}$$

This implies

$$\begin{pmatrix} \widehat{\gamma} \\ \widehat{\boldsymbol{\varepsilon}} \end{pmatrix} \sim \mathrm{Normal}(\mu, \Sigma).$$

Because of the block structure of Σ we see that $\widehat{\gamma}$ and $\widehat{\boldsymbol{\varepsilon}}$ are independent, which is b).

We have

$$\boldsymbol{\varepsilon} = \mathbf{y} - \mathbf{A}\gamma = \mathbf{y} - \mathbf{A}\widehat{\gamma} + \mathbf{A}\widehat{\gamma} - \mathbf{A}\gamma = \mathbf{y} - \widehat{\mathbf{y}} + \mathbf{A}(\widehat{\gamma} - \gamma) = \widehat{\boldsymbol{\varepsilon}} + \mathbf{A}(\widehat{\gamma} - \gamma)$$

so that

$$\|\boldsymbol{\varepsilon}\|^2 = \|\widehat{\boldsymbol{\varepsilon}}\|^2 + 2\widehat{\boldsymbol{\varepsilon}}^T\mathbf{A}(\widehat{\gamma} - \gamma) + \|\mathbf{A}(\widehat{\gamma} - \gamma)\|^2.$$

Due to

$$\widehat{\boldsymbol{\varepsilon}}^T\mathbf{A} = ((I_n - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T)\mathbf{y})^T\mathbf{A} = \mathbf{y}^T(\mathbf{A} - \mathbf{A}(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{A}^T\mathbf{A}) = 0$$

this simplifies to

$$\|\boldsymbol{\varepsilon}\|^2 = \|\widehat{\boldsymbol{\varepsilon}}\|^2 + \|\mathbf{A}(\widehat{\gamma} - \gamma)\|^2$$

and

$$\frac{\|\boldsymbol{\varepsilon}\|^2}{\sigma^2} = \frac{\|\widehat{\boldsymbol{\varepsilon}}\|^2}{\sigma^2} + \frac{\|\mathbf{A}(\widehat{\gamma} - \gamma)\|^2}{\sigma^2}. \tag{7.2}$$

The random variable on the left-hand side of (7.2) follows a chi-squared distribution with n degrees of freedom. Let $(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}$ be the positive definite matrix such that $(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}} = \mathbf{A}^T\mathbf{A}$. Then, we have

$$\|\mathbf{A}(\widehat{\gamma}-\gamma)\|^2 = (\widehat{\gamma}-\gamma)^T\mathbf{A}^T\mathbf{A}(\widehat{\gamma}-\gamma) = (\widehat{\gamma}-\gamma)^T(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}(\widehat{\gamma}-\gamma) = \|(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}(\widehat{\gamma}-\gamma)\|^2.$$

We know from a) that

$$\frac{(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}(\widehat{\gamma}-\gamma)}{\sigma} \sim \text{Normal}((\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}(\gamma-\gamma), (\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}(\mathbf{A}^T\mathbf{A})^{-1}(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}) = \text{Normal}(0, I_{k+1})$$

and, thus, $\frac{\|\mathbf{A}(\widehat{\gamma}-\gamma)\|^2}{\sigma^2} \sim \chi^2_{k+1}$. Since the two expressions on the right-hand side of (7.2) are independent by b), we see that

$$\frac{\|\widehat{\varepsilon}\|^2}{\sigma^2} \sim \chi^2_{n-k-1},$$

which proves c). □

The so-called F-distribution is defined as the ratio of chi-squared distributed random variables.

Definition 7.4 (F-distribution)**.** Let $Z_1 \sim \chi^2_{d_1}$ and $Z_2 \sim \chi^2_{d_2}$ with $d_1, d_2 \in \mathbb{N}$ be independent. The distribution of $\frac{Z_1/d_1}{Z_2/d_2}$ is called F-distribution with d_1 and d_2 degrees of freedom and is denoted by $\mathcal{F}_{d_1,d_2}$.

In the Gaussian linear model, we have a test statistic following an F-distribution.

Corollary 7.5. Let the assumptions of Theorem 7.3 prevail and let $n > k+1$. Then, $\frac{(\widehat{\gamma}-\gamma)^T\mathbf{A}^T\mathbf{A}(\widehat{\gamma}-\gamma)}{(k+1)\widehat{\sigma}^2} \sim \mathcal{F}_{k+1,n-k-1}$.

Proof. Let $(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}$ be the positive definite matrix such that $(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}} = \mathbf{A}^T\mathbf{A}$. It follows from Theorem 7.3 that $(\mathbf{A}^T\mathbf{A})^{\frac{1}{2}}(\widehat{\gamma}-\gamma) \sim \text{Normal}(0, \sigma^2 I_{k+1})$ and $\frac{(n-k-1)\widehat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-k-1}$ are independent. Thus, $\frac{(\widehat{\gamma}-\gamma)^T\mathbf{A}^T\mathbf{A}(\widehat{\gamma}-\gamma)}{\sigma^2} \sim \chi^2_{k+1}$ and we see that

$$\frac{(\widehat{\gamma}-\gamma)^T\mathbf{A}^T\mathbf{A}(\widehat{\gamma}-\gamma)}{(k+1)\widehat{\sigma}^2} \sim \mathcal{F}_{k+1,n-k-1},$$

which concludes the proof. □

Theorem 7.3 and Corollary 7.5 allow us to derive confidence intervals and tests for the Gaussian linear model. For the p-quantile of an F-distribution with d_1 and d_2 degrees of freedom we write $\mathcal{F}_{d_1,d_2,p}$.

Theorem 7.6. Let the assumptions of Theorem 7.3 prevail, let $n > k+1$ and let $\alpha \in (0,1)$.

a) Let $\mathbf{a} \in \mathbb{R}^{k+1}$. The map

$$C : \mathbb{R}^n \ni \mathbf{y} \mapsto \Big(\langle \mathbf{a}, \widehat{\gamma}\rangle - t_{n-k-1,1-\alpha/2}\widehat{\sigma}\sqrt{\mathbf{a}^T(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{a}}, \\ \langle \mathbf{a}, \widehat{\gamma}\rangle + t_{n-k-1,1-\alpha/2}\widehat{\sigma}\sqrt{\mathbf{a}^T(\mathbf{A}^T\mathbf{A})^{-1}\mathbf{a}}\Big)$$

is a confidence interval for $\langle \mathbf{a}, \gamma\rangle$ with error level α. For $m_0 \in \mathbb{R}$,

$$\varphi : \mathbb{R}^n \ni \mathbf{y} \mapsto \begin{cases} 0, & m_0 \in C(\mathbf{y}), \\ 1, & m_0 \notin C(\mathbf{y}), \end{cases}$$

is a level-α test for $H_0 : \langle \mathbf{a}, \gamma\rangle = m_0$ against $H_1 : \langle \mathbf{a}, \gamma\rangle \neq m_0$.

b) The map

$$C : \mathbb{R}^n \ni \mathbf{y} \mapsto \left(\frac{(n-k-1)\widehat{\sigma}^2}{\chi^2_{n-k-1,1-\alpha/2}}, \frac{(n-k-1)\widehat{\sigma}^2}{\chi^2_{n-k-1,\alpha/2}} \right)$$

is a confidence interval with error level α for σ^2. For $\sigma_0^2 \in (0,\infty)$,

$$\varphi : \mathbb{R}^n \ni \mathbf{y} \mapsto \begin{cases} 0, & \sigma_0^2 \in C(\mathbf{y}), \\ 1, & \sigma_0^2 \notin C(\mathbf{y}), \end{cases}$$

is a level-α test for $H_0 : \sigma^2 = \sigma_0^2$ against $H_1 : \sigma^2 \neq \sigma_0^2$.

c) The map

$$C : \mathbb{R}^n \ni \mathbf{y} \mapsto \{z \in \mathbb{R}^{k+1} : (z - \widehat{\gamma})^T \mathbf{A}^T \mathbf{A}(z - \widehat{\gamma}) \leq (k+1)\widehat{\sigma}^2 \mathcal{F}_{k+1,n-k-1,1-\alpha}\}$$

is a confidence region for γ with error level α. For $\overline{\gamma} \in \mathbb{R}^{k+1}$,

$$\varphi : \mathbb{R}^n \ni \mathbf{y} \mapsto \begin{cases} 0, & \overline{\gamma} \in C(\mathbf{y}), \\ 1, & \overline{\gamma} \notin C(\mathbf{y}), \end{cases}$$

is a level-α test for $H_0 : \gamma = \overline{\gamma}$ against $H_1 : \gamma \neq \overline{\gamma}$.

Proof. Throughout the proof we denote by $\mathbf{P}_{\gamma,\sigma^2}$ the distribution of $\mathbf{y}$. From Theorem 7.3 a) it follows that

$$\langle \mathbf{a}, \widehat{\gamma} \rangle \sim \mathrm{Normal}(\langle \mathbf{a}, \gamma \rangle, \sigma^2 \mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a})$$

and, thus,

$$\frac{\langle \mathbf{a}, \widehat{\gamma} \rangle - \langle \mathbf{a}, \gamma \rangle}{\sigma \sqrt{\mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}}} \sim \mathrm{Normal}(0,1).$$

Since, by Theorem 7.3 b) and c), $\widehat{\gamma}$ and $\widehat{\sigma}$ are independent and $\frac{(n-k-1)\widehat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-k-1}$, we obtain

$$\frac{\langle \mathbf{a}, \widehat{\gamma} \rangle - \langle \mathbf{a}, \gamma \rangle}{\widehat{\sigma} \sqrt{\mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}}} = \frac{\frac{\langle \mathbf{a}, \widehat{\gamma} \rangle - \langle \mathbf{a}, \gamma \rangle}{\sigma \sqrt{\mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}}}}{\sqrt{\frac{(n-k-1)\widehat{\sigma}^2}{\sigma^2 (n-k-1)}}} \sim t_{n-k-1}.$$

Let T be a random variable with a t-distribution with $n-k-1$ degrees of freedom. From the previous observation we derive

$$\begin{aligned} \mathbf{P}_{\gamma,\sigma^2}(\langle \mathbf{a}, \gamma \rangle \in C) &= \mathbf{P}_{\gamma,\sigma^2}\left(\frac{\langle \mathbf{a}, \widehat{\gamma} \rangle - \langle \mathbf{a}, \gamma \rangle}{\widehat{\sigma} \sqrt{\mathbf{a}^T (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{a}}} \in (-t_{n-k-1,1-\alpha/2}, t_{n-k-1,1-\alpha/2}) \right) \\ &= \mathbf{P}(T \leq t_{n-k-1,1-\alpha/2}) - \mathbf{P}(T \leq -t_{n-k-1,1-\alpha/2}) \\ &= 2\mathbf{P}(T \leq t_{n-k-1,1-\alpha/2}) - 1 = 2\left(1 - \frac{\alpha}{2}\right) - 1 = 1 - \alpha \end{aligned}$$

for $\gamma \in \mathbb{R}^{k+1}$ and $\sigma^2 \in (0,\infty)$. Thus, C is a confidence interval with error level α for $\langle \mathbf{a}, \gamma \rangle$.

Under the null hypothesis, i.e., $\langle \mathbf{a}, \gamma \rangle = m_0$ we have

$$\mathbf{P}_{\gamma,\sigma^2}(\varphi = 0) = \mathbf{P}_{\gamma,\sigma^2}(\langle \mathbf{a}, \gamma \rangle \in C) = 1 - \alpha$$

by the same computation as above, which shows that φ is a level-α test.

From Theorem 7.3 c) we known that

$$\frac{(n-k-1)\widehat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-k-1}.$$

Denoting by Y a χ^2_{n-k-1}-distributed random variable, we obtain, for $\gamma \in \mathbb{R}^{k+1}$ and $\sigma^2 \in (0, \infty)$,

$$\begin{aligned}
\mathbf{P}_{\gamma,\sigma^2}(\sigma^2 \in C) &= \mathbf{P}_{\gamma,\sigma^2}\left(\sigma^2 \in \left(\frac{(n-k-1)\widehat{\sigma}^2}{\chi^2_{n-k-1,1-\alpha/2}}, \frac{(n-k-1)\widehat{\sigma}^2}{\chi^2_{n-k-1,\alpha/2}}\right)\right) \\
&= \mathbf{P}_{\gamma,\sigma^2}\left(\frac{(n-k-1)\widehat{\sigma}^2}{\sigma^2} < \chi^2_{n-k-1,1-\alpha/2}\right) \\
&\quad - \mathbf{P}_{\gamma,\sigma^2}\left(\frac{(n-k-1)\widehat{\sigma}^2}{\sigma^2} \leq \chi^2_{n-k-1,\alpha/2}\right) \\
&= \mathbf{P}(Y \leq \chi^2_{n-k-1,1-\alpha/2}) - \mathbf{P}(Y \leq \chi^2_{n-k-1,\alpha/2}) = 1 - \frac{\alpha}{2} - \frac{\alpha}{2} = 1 - \alpha.
\end{aligned}$$

This establishes that C is a confidence interval for σ^2. Similarly, we obtain that φ is a test of level α for $H_0 : \sigma^2 = \sigma_0^2$ against $H_1 : \sigma^2 \neq \sigma_0^2$.

Let $Z \sim \mathcal{F}_{k+1,n-k-1}$, $\gamma \in \mathbb{R}^{k+1}$ and $\sigma^2 \in (0, \infty)$. From Corollary 7.5 it follows that

$$\begin{aligned}
\mathbf{P}_{\gamma,\sigma^2}(\gamma \in C) &= \mathbf{P}_{\gamma,\sigma^2}((\gamma - \widehat{\gamma})^T \mathbf{A}^T \mathbf{A}(\gamma - \widehat{\gamma}) \leq (k+1)\widehat{\sigma}^2 \mathcal{F}_{k+1,n-k-1,1-\alpha}) \\
&= \mathbf{P}_{\gamma,\sigma^2}\left(\frac{(\gamma - \widehat{\gamma})^T \mathbf{A}^T \mathbf{A}(\gamma - \widehat{\gamma})}{(k+1)\widehat{\sigma}^2} \leq \mathcal{F}_{k+1,n-k-1,1-\alpha}\right) \\
&= \mathbf{P}(Z \leq \mathcal{F}_{k+1,n-k-1,1-\alpha}) = 1 - \alpha,
\end{aligned}$$

whence C is a confidence region for γ. The same arguments show that φ is a level-α test for $H_0 : \gamma = \overline{\gamma}$ against $H_1 : \gamma \neq \overline{\gamma}$. □

Remark 7.7. In the Gaussian linear model one is often interested in the test problem $H_0 : \gamma_j = 0$ against $H_1 : \gamma_j \neq 0$ for a fixed $j \in \{0, 1, \ldots, k\}$, which is the situation of Theorem 7.6 a) if $\mathbf{a}$ is the $(j+1)$-th unit vector. Here the null hypothesis means that the j-th independent variable does not affect the dependent variable for $j \in \{1, \ldots, k\}$ or that the intercept is zero for $j = 0$. For the situations of Theorem 7.6 a) one has also similar one-sided tests for the test problems $H_0 : \langle \mathbf{a}, \gamma \rangle \leq 0$ against $H_1 : \langle \mathbf{a}, \gamma \rangle > 0$ or $H_0 : \langle \mathbf{a}, \gamma \rangle \geq 0$ against $H_1 : \langle \mathbf{a}, \gamma \rangle < 0$. These can be used when one studies whether γ_j is positive or negative for a fixed $j \in \{0, 1, \ldots, k\}$, i.e., if the effect of the j-th independent variable on the dependent variable is positive or negative for $j \in \{1, \ldots, k\}$ or if the intercept is positive or negative for $j = 0$. Applying a test for each coefficient is multiple testing whence the levels of the tests need to be adjusted.

For $\mathbf{a} = (1, x_1, \ldots, x_k)$, $\langle \mathbf{a}, \widehat{\gamma} \rangle$ is the prediction of the dependent variable y for $x_1, \ldots, x_k$. Thus, Theorem 7.6 a) provides confidence intervals for predictions.

There are also one-sided confidence intervals and tests for the variance similar to the ones in Theorem 7.6 b).

Remark 7.8. Sometimes one is interested in regression models without the intercept γ_0. Then, the results of this and the previous section still hold if one removes the column with ones from the design matrix $\mathbf{A}$ and replaces k by $k-1$ everywhere.

7.3 Analysis of variance

For $m \in \mathbb{N}$ let m samples $\mathbf{y}_1, \ldots, \mathbf{y}_m$ with $\mathbf{y}_i \in \mathbb{R}^{n_i}$ for $i \in \{1, \ldots, m\}$ be given. We assume that $\mathbf{y}_1, \ldots, \mathbf{y}_m$ are realisations of independent random vectors $Y_1 \sim \text{Normal}(\mu_1, \sigma^2)^{\otimes n_1}$, $\ldots, Y_m \sim \text{Normal}(\mu_m, \sigma^2)^{\otimes n_m}$ with unknown $\mu_1, \ldots, \mu_m \in \mathbb{R}$ and $\sigma \in (0, \infty)$. This means that our data are all normally distributed with the same unknown variance and that each sample has its own expectation.

In the following we estimate the parameters $\mu_1, \ldots, \mu_m$ and σ^2 and consider the test problem

$$H_0 : \mu_1 = \ldots = \mu_m \quad \text{against} \quad H_1 : \mu_u \neq \mu_v \quad \text{for some} \quad u, v \in \{1, \ldots, m\}. \tag{7.3}$$

We reformulate the described situation as a regression problem. Let $\mathbf{y}_{ij}$ be the j-th component of $\mathbf{y}_i$ for $j \in \{1, \ldots, n_i\}$ and $i \in \{1, \ldots, m\}$. Then, we have

$$\mathbf{y}_{ij} = \mu_i + \varepsilon_{ij}$$

for $j \in \{1, \ldots, n_i\}$ and $i \in \{1, \ldots, m\}$ with independent random variables $\varepsilon_{ij} \sim \text{Normal}(0, \sigma^2)$ for $j \in \{1, \ldots, n_i\}$ and $i \in \{1, \ldots, m\}$. In matrix notation, this is

$$\mathbf{y} = \mathbf{A}\gamma + \boldsymbol{\varepsilon}$$

with

$$\mathbf{y} = \begin{pmatrix} \mathbf{y}_{11} \\ \vdots \\ \mathbf{y}_{1n_1} \\ \vdots \\ \mathbf{y}_{m1} \\ \vdots \\ \mathbf{y}_{mn_m} \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 1 & & \\ \vdots & & \\ 1 & & \\ & \ddots & \\ & & 1 \\ & & \vdots \\ & & 1 \end{pmatrix}, \quad \gamma = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_m \end{pmatrix} \quad \text{and} \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{11} \\ \vdots \\ \varepsilon_{1n_1} \\ \vdots \\ \varepsilon_{m1} \\ \vdots \\ \varepsilon_{mn_m} \end{pmatrix},$$

i.e., we have m dummy variables such that the i-th dummy variable is one if the observation belongs to the i-th sample and zero, otherwise, for $i \in \{1, \ldots, m\}$.

For $i \in \{1, \dots, m\}$ we define $\overline{\mathbf{y}}_i = \frac{1}{n_i} \sum_{j=1}^{n_i} \mathbf{y}_{ij}$ and let $\overline{\mathbf{y}} = \frac{1}{n} \sum_{i=1}^{m} n_i \overline{\mathbf{y}}_i$ with $n = \sum_{i=1}^{m} n_i$. We have

$$\mathbf{A}^T\mathbf{A} = \begin{pmatrix} n_1 & & \\ & \ddots & \\ & & n_m \end{pmatrix} \quad \text{and} \quad \mathbf{A}^T\mathbf{y} = \begin{pmatrix} n_1\overline{\mathbf{y}}_1 \\ \vdots \\ n_m\overline{\mathbf{y}}_m \end{pmatrix}$$

so that the least squares estimators for $\mu_1, \dots, \mu_m$ are given by $\widehat{\mu}_i = \overline{\mathbf{y}}_i$ for $i \in \{1, \dots, m\}$ and the variance is estimated by

$$\widehat{\sigma}^2 = \frac{1}{n-m} \sum_{i=1}^{m} \sum_{j=1}^{n_i} (\mathbf{y}_{ij} - \overline{\mathbf{y}}_i)^2.$$

Moreover, we have the following test for the null hypothesis that all samples have the same expectation.

Theorem 7.9. Let $\alpha \in (0,1)$ and $n > m$. Then,

$$\varphi = \begin{cases} 0, & T < \mathcal{F}_{m-1,n-m,1-\alpha}, \\ 1, & T \geq \mathcal{F}_{m-1,n-m,1-\alpha}, \end{cases}$$

with

$$T = \frac{\frac{1}{m-1} \sum_{i=1}^{m} n_i (\overline{\mathbf{y}}_i - \overline{\mathbf{y}})^2}{\frac{1}{n-m} \sum_{i=1}^{m} \sum_{j=1}^{n_i} (\mathbf{y}_{ij} - \overline{\mathbf{y}}_i)^2}$$

is a level-α test for the test problem in (7.3).

Proof. In the following, we assume that the null hypothesis is satisfied, i.e., there exists a $\mu \in \mathbb{R}$ such that $\mu_i = \mu$ for all $i \in \{1, \dots, m\}$. It follows from Theorem 7.3 a) that

$$\begin{pmatrix} \overline{\mathbf{y}}_1 \\ \vdots \\ \overline{\mathbf{y}}_m \end{pmatrix} - \begin{pmatrix} \mu \\ \vdots \\ \mu \end{pmatrix} \sim \text{Normal}(0, \Sigma) \quad \text{with} \quad \Sigma = \begin{pmatrix} \frac{\sigma^2}{n_1} & & \\ & \ddots & \\ & & \frac{\sigma^2}{n_m} \end{pmatrix}.$$

Thus, we obtain

$$\begin{pmatrix} \sqrt{n_1}\overline{\mathbf{y}}_1 \\ \vdots \\ \sqrt{n_m}\overline{\mathbf{y}}_m \end{pmatrix} - \begin{pmatrix} \sqrt{n_1}\mu \\ \vdots \\ \sqrt{n_m}\mu \end{pmatrix} \stackrel{d}{=} N$$

with $N \sim \text{Normal}(0, \sigma^2 I_m)$. Let the matrix $\Pi \in \mathbb{R}^{m \times m}$ be given by

$$\Pi_{ij} = \begin{cases} 1 - \frac{\sqrt{n_i}\sqrt{n_j}}{n}, & i = j, \\ -\frac{\sqrt{n_i}\sqrt{n_j}}{n}, & i \neq j, \end{cases}$$

for $i, j \in \{1, \dots, m\}$. A short calculation shows that

$$\Pi \left(\begin{pmatrix} \sqrt{n_1}\overline{\mathbf{y}}_1 \\ \vdots \\ \sqrt{n_m}\overline{\mathbf{y}}_m \end{pmatrix} - \begin{pmatrix} \sqrt{n_1}\mu \\ \vdots \\ \sqrt{n_m}\mu \end{pmatrix} \right) = \begin{pmatrix} \sqrt{n_1}(\overline{\mathbf{y}}_1 - \overline{\mathbf{y}}) \\ \vdots \\ \sqrt{n_m}(\overline{\mathbf{y}}_m - \overline{\mathbf{y}}) \end{pmatrix},$$

whence

$$\begin{pmatrix} \sqrt{n_1}(\overline{\mathbf{y}}_1 - \overline{\mathbf{y}}) \\ \vdots \\ \sqrt{n_m}(\overline{\mathbf{y}}_m - \overline{\mathbf{y}}) \end{pmatrix} \stackrel{d}{=} \Pi N \quad \text{and} \quad \sum_{i=1}^m n_i(\overline{\mathbf{y}}_i - \overline{\mathbf{y}})^2 \stackrel{d}{=} \|\Pi N\|^2.$$

We can also check that Π is a projection onto the hyperplane

$$H = \{(x_1, \dots, x_m) \in \mathbb{R}^m : \sum_{i=1}^m \sqrt{n_i} x_i = 0\}.$$

By the invariance properties of the multivariate standard normal distribution, $\|\Pi N\|^2$ has the same distribution as a projection of N to one of the coordinate planes, whence $\|\Pi N\|^2/\sigma^2$ is χ^2_{m-1}-distributed. This implies

$$\frac{1}{\sigma^2} \sum_{i=1}^m n_i(\overline{\mathbf{y}}_i - \overline{\mathbf{y}})^2 \sim \chi^2_{m-1}.$$

From Theorem 7.3 c) we know that

$$\frac{1}{\sigma^2} \sum_{i=1}^m \sum_{j=1}^{n_i} (\mathbf{y}_{ij} - \overline{\mathbf{y}}_i)^2 \sim \chi^2_{n-m}.$$

Since, by Theorem 7.3 b), the numerator and the denominator of T are independent, we obtain $T \sim \mathcal{F}_{m-1,n-m}$, which concludes the proof. □

Example 7.10. For $m = 2$ we have that

$$\begin{aligned} \sum_{i=1}^2 n_i(\overline{\mathbf{y}}_i - \overline{\mathbf{y}})^2 &= n_1\left(\overline{\mathbf{y}}_1 - \frac{n_1}{n}\overline{\mathbf{y}}_1 - \frac{n_2}{n}\overline{\mathbf{y}}_2\right)^2 + n_2\left(\overline{\mathbf{y}}_2 - \frac{n_1}{n}\overline{\mathbf{y}}_1 - \frac{n_2}{n}\overline{\mathbf{y}}_2\right)^2 \\ &= \frac{n_1 n_2^2}{n^2}(\overline{\mathbf{y}}_1 - \overline{\mathbf{y}}_2)^2 + \frac{n_1^2 n_2}{n^2}(\overline{\mathbf{y}}_1 - \overline{\mathbf{y}}_2)^2 = \frac{n_1 n_2}{n}(\overline{\mathbf{y}}_1 - \overline{\mathbf{y}}_2)^2. \end{aligned}$$

Together with $\mathcal{F}_{1,n-m,1-\alpha} = t^2_{n-m,1-\alpha/2}$, we see that for the case of two samples the test in Theorem 7.9 rejects the null hypothesis of equal means if and only if this null hypothesis is rejected by the two-sample t-test in Theorem 6.3. Thus, for the case of two samples the tests in Theorem 6.3 and in Theorem 7.9 are identical.

Chapter 8

Categorical data

8.1 Fisher's exact test for 2×2 tables

Assume we have two samples $x_1, \dots, x_n$ and $y_1, \dots, y_m$ with $n, m \in \mathbb{N}$ that are realisations of independent random variables $X_1, \dots, X_n \sim \text{Bernoulli}(p_1)$ and $Y_1, \dots, Y_m \sim \text{Bernoulli}(p_2)$ with $p_1, p_2 \in (0,1)$. Such a sample can be described by a table

	Ones	Zeros	Sum
Sample 1	r	$n-r$	n
Sample 2	s	$m-s$	m
Sum	$r+s$	$n+m-r-s$	$n+m$

where r and s are the numbers of ones in both samples. Here, we use one and zero only to encode two different values, which describe, for example, success and failure. Our goal is to compare the underlying distributions of the two samples.

Theorem 8.1 (Fisher's exact test)**.** Let $\alpha \in (0,1)$.

a) For $(x,y) \in \{0,1\}^{n+m}$ let $r = \sum_{i=1}^n x_i$ and $s = \sum_{i=1}^m y_i$, choose $\ell, u \in \{0, 1, \dots, r+s\}$ such that

$$\sum_{j=\ell}^{u} \frac{\binom{n}{j}\binom{m}{r+s-j}}{\binom{n+m}{r+s}} \geq 1 - \alpha$$

and

$$\max_{j \in \{0,1,\dots,r+s\} \setminus \{\ell,\dots,u\}} \binom{n}{j}\binom{m}{r+s-j} \leq \min_{j \in \{\ell,\dots,u\}} \binom{n}{j}\binom{m}{r+s-j}$$

and define

$$\varphi((x,y)) = \begin{cases} 0, & r \in \{\ell, \dots, u\}, \\ 1, & r \notin \{\ell, \dots, u\}. \end{cases}$$

The test φ is a level-α test for $H_0 : p_1 = p_2$ against $H_1 : p_1 \neq p_2$.

b) For $(x,y) \in \{0,1\}^{n+m}$ let $r = \sum_{i=1}^n x_i$ and $s = \sum_{i=1}^m y_i$, choose $\ell, u \in \{0,1,\dots,r+s\}$ such that

$$\sum_{j=\ell}^{r+s} \frac{\binom{n}{j}\binom{m}{r+s-j}}{\binom{n+m}{r+s}} \geq 1-\alpha \quad \text{and} \quad \sum_{j=0}^{u} \frac{\binom{n}{j}\binom{m}{r+s-j}}{\binom{n+m}{r+s}} \geq 1-\alpha.$$

Then,

$$\varphi_1((x,y)) = \begin{cases} 0, & r \geq \ell, \\ 1, & r < \ell, \end{cases}$$

is a level-α test for $H_0 : p_1 \geq p_2$ against $H_1 : p_1 < p_2$ and

$$\varphi_2((x,y)) = \begin{cases} 0, & r \leq u, \\ 1, & r > u, \end{cases}$$

is a level-α test for $H_0 : p_1 \leq p_2$ against $H_1 : p_1 > p_2$.

Proof. For $p_1, p_2 \in (0,1)$ let $X \sim \text{Bernoulli}(p_1)^{\otimes n}$ and $Y \sim \text{Bernoulli}(p_2)^{\otimes m}$ be independent and let $\mathbf{P}_{p_1,p_2}$ denote their joint distribution. Define $R = \sum_{i=1}^n X_i$ and $S = \sum_{i=1}^m Y_i$. For $p_1 \in (0,1)$ we have, by the law of total probability,

$$\mathbf{P}_{p_1,p_1}(\varphi((X,Y)) = 0) = \sum_{k=0}^{n+m} \mathbf{P}_{p_1,p_1}(R+S=k)\mathbf{P}_{p_1,p_1}(\varphi((X,Y)) = 0 | R+S=k).$$

Since $R \sim \text{Binomial}(n,p_1)$, $S \sim \text{Binomial}(m,p_1)$, $R+S \sim \text{Binomial}(n+m,p_1)$ and R and S are independent, for $k \in \{0,1,\dots,n+m\}$ and $j \in \{0,1,\dots,k\}$ we obtain

$$\begin{aligned}
\mathbf{P}_{p_1,p_1}(R=j|R+S=k) &= \frac{\mathbf{P}_{p_1,p_1}(R=j, S=k-j)}{\mathbf{P}_{p_1,p_1}(R+S=k)} = \frac{\mathbf{P}_{p_1,p_1}(R=j)\mathbf{P}_{p_1,p_1}(S=k-j)}{\mathbf{P}_{p_1,p_1}(R+S=k)} \\
&= \frac{\binom{n}{j}p_1^j(1-p_1)^{n-j}\binom{m}{k-j}p_1^{k-j}(1-p_1)^{m-k+j}}{\binom{n+m}{k}p_1^k(1-p_1)^{n+m-k}} \\
&= \frac{\binom{n}{j}\binom{m}{k-j}p_1^k(1-p_1)^{n+m-k}}{\binom{n+m}{k}p_1^k(1-p_1)^{n+m-k}} = \frac{\binom{n}{j}\binom{m}{k-j}}{\binom{n+m}{k}}.
\end{aligned}$$

Since the boundary values ℓ and u of the acceptance region depend on $R+S$, we write $\ell(R+S)$ and $u(R+S)$ in the following. The previous computation and the definitions of $\ell(\cdot)$ and $u(\cdot)$ yield

$$\begin{aligned}
\mathbf{P}_{p_1,p_1}(\varphi((X,Y)) = 0|R+S=k) &= \mathbf{P}_{p_1,p_1}(R \in \{\ell(k),\dots,u(k)\}|R+S=k) \\
&= \sum_{j=\ell(k)}^{u(k)} \frac{\binom{n}{j}\binom{m}{k-j}}{\binom{n+m}{k}} \geq 1-\alpha.
\end{aligned}$$

This implies

$$\mathbf{P}_{p_1,p_1}(\varphi((X,Y)) = 0) \geq \sum_{k=0}^{n+m} \mathbf{P}_{p_1,p_1}(R+S=k)(1-\alpha) = 1-\alpha$$

so that φ is a level-α test.

Next we consider φ_1. For $p_1, p_2 \in (0,1)$ with $p_1 \geq p_2$ we have, by the law of total probability,

$$\mathbf{P}_{p_1,p_2}(\varphi(X,Y)=0) = \sum_{k=0}^{n+m} \mathbf{P}_{p_1,p_2}(R+S=k)\mathbf{P}_{p_1,p_2}(\varphi(X,Y)=0|R+S=k).$$

Let $\ell(R+S)$ be the parameter ℓ from the statement of part b). For $k \in \{0,1,\dots,n+m\}$ we have, by similar arguments as in part a),

$$\begin{aligned}\mathbf{P}_{p_1,p_2}(\varphi_1(X,Y)=0|R+S=k) &= \mathbf{P}_{p_1,p_2}(R \geq \ell(k)|R+S=k)\\ &= \frac{\sum_{i=\ell(k)}^{k}\binom{n}{i}p_1^i(1-p_1)^{n-i}\binom{m}{k-i}p_2^{k-i}(1-p_2)^{n-(k-i)}}{\sum_{i=0}^{k}\binom{n}{i}p_1^i(1-p_1)^{n-i}\binom{m}{k-i}p_2^{k-i}(1-p_2)^{n-(k-i)}}\\ &= \frac{\sum_{i=\ell(k)}^{k}\binom{n}{i}\left(\frac{p_1}{1-p_1}\right)^i\binom{m}{k-i}\left(\frac{1-p_2}{p_2}\right)^i}{\sum_{i=0}^{k}\binom{n}{i}\left(\frac{p_1}{1-p_1}\right)^i\binom{m}{k-i}\left(\frac{1-p_2}{p_2}\right)^i}.\end{aligned}$$

Here, the right-hand side is monotonously increasing in $\frac{1-p_2}{p_2}$ and, thus, monotonously decreasing in p_2. Because of $p_2 \leq p_1$ this yields

$$\begin{aligned}\mathbf{P}_{p_1,p_2}(\varphi_1(X,Y)=0|R+S=k) &\geq \frac{\sum_{i=\ell(k)}^{k}\binom{n}{i}\left(\frac{p_1}{1-p_1}\right)^i\binom{m}{k-i}\left(\frac{1-p_1}{p_1}\right)^i}{\sum_{i=0}^{k}\binom{n}{i}\left(\frac{p_1}{1-p_1}\right)^i\binom{m}{k-i}\left(\frac{1-p_1}{p_1}\right)^i}\\ &= \frac{\sum_{i=\ell(k)}^{k}\binom{n}{i}\binom{m}{k-i}}{\sum_{i=0}^{k}\binom{n}{i}\binom{m}{k-i}} = \frac{\sum_{i=\ell(k)}^{k}\binom{n}{i}\binom{m}{k-i}}{\binom{n+m}{k}}.\end{aligned}$$

Together with the definition of $\ell(k)$ we see that

$$\mathbf{P}_{p_1,p_2}(\varphi_1(X,Y)=0|R+S=k) \geq 1-\alpha,$$

whence

$$\mathbf{P}_{p_1,p_2}(\varphi_1(X,Y)=0) \geq 1-\alpha$$

and φ_1 is a level-α test. The proof for φ_2 goes analogously. □

Remark 8.2. The ratio

$$\frac{\binom{n}{j}\binom{m}{r+s-j}}{\binom{n+m}{r+s}}$$

in the statements of Theorem 8.1 is the probability for j of a hypergeometric distribution with parameters n, m and $r+s$ (i.e., we draw $r+s$ balls without replacement from an urn with n red and m black balls and count the number of drawn red balls). In Theorem 8.1 a) the set $\{\ell,\dots,u\}$ is chosen such that its probability under the hypergeometric distribution is at least $1-\alpha$ and no value outside of $\{\ell,\dots,u\}$ has a higher probability than a value in $\{\ell,\dots,u\}$. In Theorem 8.1 b), one can use quantiles of the hypergeometric distribution for ℓ and u.

Remark 8.3. The rejection regions of the tests in Theorem 8.1 depend on the given data. Such tests are called conditional.

Example 8.4 (Fisher's tea tasting lady)**.** A lady claims that she can taste whether tea or milk was first added to a cup of tea. Therefore, an experiment is conducted where the lady has to guess what was poured first. The results are given in the following table, where the rows show what was added first and the columns are the guesses:

	Tea guessed	Milk guessed
Tea first	3	1
Milk first	1	3

Our goal is to decide whether the lady can taste if tea or milk was added first to the cup. Let p_1 be the probability that she guesses tea if tea is the correct answer and let p_2 be the probability that she guesses tea if milk is the correct answer.

In the following we consider the test problem

$$H_0 : p_1 \leq p_2 \quad \text{against} \quad H_1 : p_1 > p_2.$$

When we reject H_0, we can establish that the lady can taste what was poured first. Alternatively, one can study the two-sided problem

$$H_0' : p_1 = p_2 \quad \text{against} \quad H_1' : p_1 \neq p_2,$$

where the null hypothesis is also rejected if the guesses are systematically wrong in the sense that milk is guessed if tea is correct and vice versa.

In the following we apply Fisher's exact test φ_2 from Theorem 8.1 b). We have $n = m = 4$, $r = 3$ and $s = 1$. Instead of providing a level α, we compute here the p-value, i.e., we determine the smallest α such that we can reject H_0. Since we have $r = 3$ and reject if $r > u$, we need $u = 2$ for rejection. We have

$$\alpha \geq \sum_{j=3}^{4} \frac{\binom{4}{j}\binom{4}{4-j}}{\binom{8}{4}} = \frac{\binom{4}{3}\binom{4}{1} + \binom{4}{4}\binom{4}{0}}{\binom{8}{4}} = \frac{16+1}{70} = \frac{17}{70} \approx 0.2429$$

so that the p-value is 0.2429. Thus, we cannot reject the null hypothesis at an usual level such as 0.01, 0.05 or 0.1.

8.2 The chi-square test of independence

In this section we consider $k \times m$ tables instead of 2×2 tables, i.e., we have k different samples of sizes $n_{1+}, \ldots, n_{k+}$ and m categories. This can be described by a table of the following form:

	1	2	...	m	Sum
Sample 1	n_{11}	n_{12}	...	n_{1m}	n_{1+}
Sample 2	n_{21}	n_{22}	...	n_{2m}	n_{2+}
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
Sample k	n_{k1}	n_{k2}	...	n_{km}	n_{k+}
Sum	n_{+1}	n_{+2}	...	n_{+m}	n

For $i \in \{1, \dots, k\}$ and $j \in \{1, \dots, m\}$ let p_{ij} be the probability of category j for sample i. We define $p^{(i)} = (p_{i1}, \dots, p_{im})$ for $i \in \{1, \dots, k\}$. We are interested to check if the probabilities are the same for all samples. Thus, we consider the test problem

$$H_0 : p^{(1)} = \ldots = p^{(k)} \quad \text{against} \quad H_1 : p^{(u)} \neq p^{(v)} \quad \text{for some} \quad u, v \in \{1, \dots, k\}. \tag{8.1}$$

Under H_0, the distribution of the categories is the same for all samples, which can be interpreted in the way that the samples and the categories are independent, i.e., the information which sample we have does not affect the probabilities of the categories. This explains the name of the following test.

Theorem 8.5 (Chi-square test of independence)**.** Let $\alpha \in (0, 1)$. For the test problem (8.1) we study the test

$$\varphi = \begin{cases} 0, & T < \chi^2_{(k-1)(m-1),1-\alpha}, \\ 1, & T \geq \chi^2_{(k-1)(m-1),1-\alpha}, \end{cases}$$

with the test statistic

$$T = \sum_{i=1}^{k} \sum_{j=1}^{m} \frac{\left(n_{ij} - \frac{n_{i+}n_{+j}}{n}\right)^2}{\frac{n_{i+}n_{+j}}{n}}.$$

Under the null hypothesis, one has

$$\lim_{n\to\infty} \mathbf{P}(\varphi = 1) = \alpha,$$

i.e., φ is an asymptotic test of level α for (8.1).

We refer to [4, Theorem 11.18] for a detailed proof of Theorem 8.5, which relies on similar arguments as we will employ in the proof of Theorem 9.10.

Remark 8.6. In this chapter, we always assumed that the sizes of the different samples are deterministic and given. A slightly different set up is that only the total number over all samples is fixed and that in a first step the observations are randomly allocated to the samples. This situation arises if we consider two different categorical quantities and are interested if they are independent. In this case Fisher's exact test and the chi-square test of independence can both still be applied.

Chapter 9

Nonparametric statistics

So far, we usually studied statistical models with an underlying parametric family of distributions. In nonparametric statistics, one considers larger classes of distributions without fixed parameters (such as all continuous distributions).

9.1 Confidence intervals and tests for quantiles

In nonparametric statistics, quantiles are often used to describe an unknown distribution since one does not have underlying parameters. An important special case of quantiles are medians, which are 0.5-quantiles. Every distribution possesses a median but not necessarily an expectation.

Let X be a random variable with a continuous distribution function. For $p \in (0,1)$ we call $q_p \in \mathbb{R}$ a p-quantile of the distribution of X if

$$\mathbf{P}(X \leq q_p) = p \quad \text{and} \quad \mathbf{P}(X \geq q_p) = 1 - p.$$

In general, p and $1-p$ are only lower bounds for the probabilities, but the continuity of the distribution function of X implies even equality. For $n \in \mathbb{N}$ let $X_1, \ldots, X_n$ be i.i.d. copies of X. By ordering $X_1, \ldots, X_n$ according to their size starting with the smallest value, we obtain the order statistics $X_{(1)} < \ldots < X_{(n)}$, i.e., $X_{(k)}$ is the k-th smallest value of $X_1, \ldots, X_n$. Because of the assumption that the distribution of X is continuous, we have $\mathbf{P}(X_1 = X_2) = 0$ and do not need to deal with possible ties.

For quantiles we have the following confidence intervals.

Theorem 9.1. For $p \in (0,1)$ let q_p be a p-quantile of X and let $\alpha \in (0,1)$.

a) For $\ell, u \in \{1, \ldots, n\}$ such that $\ell < u$ and

$$\sum_{j=0}^{\ell-1} \binom{n}{j} p^j (1-p)^{n-j} + \sum_{j=u}^{n} \binom{n}{j} p^j (1-p)^{n-j} \leq \alpha,$$

one has

$$\mathbf{P}(q_p \in [X_{(\ell)}, X_{(u)}]) \geq 1 - \alpha,$$

i.e., $[X_{(\ell)}, X_{(u)}]$ is a confidence interval for q_p with error level α.

b) For $\ell, u \in \{1, \dots, n\}$ such that

$$\sum_{j=0}^{\ell-1} \binom{n}{j} p^j (1-p)^{n-j} \le \alpha \quad \text{and} \quad \sum_{j=u}^{n} \binom{n}{j} p^j (1-p)^{n-j} \le \alpha,$$

one has

$$\mathbf{P}(q_p \in [X_{(\ell)}, \infty)) \ge 1 - \alpha \quad \text{and} \quad \mathbf{P}(q_p \in (-\infty, X_{(u)}]) \ge 1 - \alpha,$$

i.e., $[X_{(\ell)}, \infty)$ and $(-\infty, X_{(u)}]$ are confidence intervals for q_p with error level α.

Proof. By our assumptions the indicator random variables $\mathbf{1}\{X_i \le q_p\}$ for $i \in \{1, \dots, n\}$ are independent and are one with probability $\mathbf{P}(X \le q_p) = p$, whence

$$\sum_{i=1}^{n} \mathbf{1}\{X_i \le q_p\}$$

follows a binomial distribution with parameters n and p. For $m \in \{1, \dots, n\}$ we obtain

$$\mathbf{P}(X_{(m)} > q_p) = \mathbf{P}\Big(\sum_{i=1}^{n} \mathbf{1}\{X_i \le q_p\} \le m-1 \Big) = \sum_{j=0}^{m-1} \binom{n}{j} p^j (1-p)^{n-j}$$

and

$$\begin{aligned} \mathbf{P}(X_{(m)} < q_p) &= \mathbf{P}\Big(\sum_{i=1}^{n} \mathbf{1}\{X_i < q_p\} \ge m \Big) = \mathbf{P}\Big(\sum_{i=1}^{n} \mathbf{1}\{X_i \le q_p\} \ge m \Big) \\ &= \sum_{j=m}^{n} \binom{n}{j} p^j (1-p)^{n-j}. \end{aligned}$$

Together with the assumptions on ℓ and u, this implies

$$\mathbf{P}(q_p \notin [X_{(\ell)}, X_{(u)}]) = \mathbf{P}(X_{(\ell)} > q_p) + \mathbf{P}(X_{(u)} < q_p) \le \alpha.$$

The proofs for the one-sided confidence intervals are analogously. □

The same approach as in Theorem 9.1 leads to the following tests for quantiles.

Theorem 9.2. Let $p \in (0,1)$, $\alpha \in (0,1)$ and $z \in \mathbb{R}$.

a) For ℓ and u as in Theorem 9.1 a),

$$\varphi = \begin{cases} 0, & z \in [X_{(\ell)}, X_{(u)}], \\ 1, & z \notin [X_{(\ell)}, X_{(u)}], \end{cases}$$

is a level-α test for $H_0 : q_p = z$ against $H_1 : q_p \ne z$.

b) For ℓ and u as in Theorem 9.1 b),

$$\varphi = \begin{cases} 0, & X_{(\ell)} \leq z, \\ 1, & X_{(\ell)} > z, \end{cases}$$

is a level-α test for $H_0 : q_p \leq z$ against $H_1 : q_p > z$ and

$$\varphi = \begin{cases} 0, & X_{(u)} \geq z, \\ 1, & X_{(u)} < z, \end{cases}$$

is a level-α test for $H_0 : q_p \geq z$ against $H_1 : q_p < z$.

Proof. The statements follow from Theorem 9.1 and our general strategy to obtain tests from confidence intervals. □

Remark 9.3. For $p \in (0,1)$ the p-quantile of a distribution does not need to be unique. If the random variable X considered in this section has p-quantiles $q_p^{(1)}, q_p^{(2)}$ with $q_p^{(1)} < q_p^{(2)}$, we obtain

$$\mathbf{P}(X \in [q_p^{(1)}, q_p^{(2)}]) = \mathbf{P}(X \leq q_p^{(2)}) - \mathbf{P}(X \leq q_p^{(1)}) = p - p = 0.$$

Thus, $X_1, \ldots, X_n$ almost surely do not lie between different p-quantiles so that the confidence intervals in Theorem 9.1 almost surely contain all p-quantiles or no p-quantile.

If the p-quantile is not unique, the null hypotheses in Theorem 9.2 need to be understood in the sense that z is a p-quantile or that there exists a p-quantile that is at most or at least z.

Remark 9.4. By using the normal approximation of the binomial distribution, one can construct asymptotic confidence intervals and tests for quantiles. They can be useful for large values of n, where the computation of the quantities ℓ and u in the previous results can be tedious.

9.2 The Wilcoxon rank sum test

Let F and G be continuous distribution functions and let $n, m \in \mathbb{N}$. Let the samples $x \in \mathbb{R}^n$ and $y \in \mathbb{R}^m$ be realisations of independent random variables $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ distributed according to F and G, respectively. Our goal is to compare the distribution functions F and G. This is a two sample problem as studied in Chapter 6, but here we consider a much larger family of distributions than only normal distributions.

We combine $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_m)$ to a joint sample and order them according to their size. The rank of x_i with $i \in \{1, \ldots, n\}$ is given by

$$r(x_i) = \sum_{k=1}^{n} \mathbf{1}\{x_k \leq x_i\} + \sum_{k=1}^{m} \mathbf{1}\{y_k \leq x_i\},$$

i.e., x_i has the rank k if it is the k-th smallest value of $x_1, \dots, x_n, y_1, \dots, y_m$. Since the samples come from continuous distributions, we do not need to take care of ties. As test statistic we consider the rank sum

$$\sum_{i=1}^{n} r(x_i),$$

i.e., the sum of the ranks of $x_1, \dots, x_n$. In order to construct tests, we study the behaviour of the test statistic under the null hypothesis. Recall that we denote the cardinality of a set A by $|A|$.

Theorem 9.5. Assume that $F = G$ and let $W = \sum_{i=1}^{n} r(X_i)$. For $w \in \mathbb{N}$,

$$\mathbf{P}(W = w) = \frac{|\{T \subset \{1, \dots, n+m\} : |T| = n \text{ and } \sum_{t \in T} t = w\}|}{\binom{n+m}{n}}.$$

Proof. From the $n + m$ possible ranks n belong to components of x. For the ranks of x, whose set we denote by T, there are $\binom{n+m}{n}$ possible choices. Since $F = G$, each of these choices has the same probability, i.e., we have a Laplace distribution on the set of subsets of $\{1, \dots, n+m\}$ with n elements. □

Note that the distribution of W in the previous theorem does not depend on F and G.

Example 9.6. For $n = 2$ and $m = 3$, the possible ranks for the two observations from x are

$$\{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{1,5\}, \{2,4\}, \{2,5\}, \{3,4\}, \{3,5\}, \{4,5\}.$$

Thus, W has the distribution

$$\mathbf{P}(W = i) = \frac{1}{10} \quad \text{for} \quad i \in \{3,4,8,9\} \quad \text{and} \quad \mathbf{P}(W = i) = \frac{1}{5} \quad \text{for} \quad i \in \{5,6,7\}.$$

For $p \in (0,1)$ we denote by $w_{n,m,p}$ a p-quantile of the distribution of W in Theorem 9.5.

Theorem 9.7 (Wilcoxon rank sum test)**.** Let $\alpha \in (0,1)$.

a) For $H_0 : F = G$ against $H_1 : F \neq G$,

$$\varphi : \mathbb{R}^{n+m} \ni (x,y) \mapsto \begin{cases} 0, & \sum_{i=1}^{n} r(x_i) \in [w_{n,m,\alpha/2}, w_{n,m,1-\alpha/2}], \\ 1, & \sum_{i=1}^{n} r(x_i) \notin [w_{n,m,\alpha/2}, w_{n,m,1-\alpha/2}], \end{cases}$$

is a level-α test.

b) For $H_0 : F = G$ against $H_1 : F \geq G$ and $F(x) > G(x)$ for some $x \in \mathbb{R}$,

$$\varphi : \mathbb{R}^{n+m} \ni (x,y) \mapsto \begin{cases} 0, & \sum_{i=1}^{n} r(x_i) \geq w_{n,m,\alpha}, \\ 1, & \sum_{i=1}^{n} r(x_i) < w_{n,m,\alpha}, \end{cases}$$

is a level-α test. For $H_0 : F = G$ against $H_1 : F \leq G$ and $F(x) < G(x)$ for some $x \in \mathbb{R}$,

$$\varphi : \mathbb{R}^{n+m} \ni (x, y) \mapsto \begin{cases} 0, & \sum_{i=1}^n r(x_i) \leq w_{n,m,1-\alpha}, \\ 1, & \sum_{i=1}^n r(x_i) > w_{n,m,1-\alpha}, \end{cases}$$

is a level-α test.

Proof. The statements follow directly from Theorem 9.5. □

Note that $F \geq G$ implies $\mathbf{P}(X_1 \leq t) = F(t) \geq G(t) = \mathbf{P}(Y_1 \leq t)$ and, thus, $\mathbf{P}(X_1 > t) \leq \mathbf{P}(Y_1 > t)$ for $t \in \mathbb{R}$, i.e., X_1 tends to take smaller values than Y_1. Thus, the observations from the sample x have the smaller ranks and the rank sum becomes smaller. This explains the rejection region for the first test in Theorem 9.7 b). For the second test in Theorem 9.7 b) one can argue analogously.

Remark 9.8. A similar test to the Wilcoxon rank sum test is the so-called Man-Whitney U test, where one studies the test statistic

$$U = \sum_{i=1}^n r(x_i) - \frac{n(n+1)}{2}.$$

Note that $\frac{n(n+1)}{2}$ is the smallest possible value of the rank sum $\sum_{i=1}^n r(x_i)$.

9.3 Goodness of fit tests

Assume that a sample x is the realisation of n independent random variables distributed according to a distribution function F, which is unknown. Our goal is to check whether F is a given distribution function F_0. More formally, we consider the test problem

$$H_0 : F = F_0 \quad \text{against} \quad H_1 : F \neq F_0.$$

In other words, we study whether the sample fits to the distribution function F_0, which explains the name goodness of fit test.

We start with the situation of discrete probability distributions with finitely many values. We prepare this with the following limit theorem for the multinomial distribution, where we denote by $e_1, \ldots, e_k$ the standard unit vectors in $\mathbb{R}^k$.

Theorem 9.9. Let $(X_i)_{i \in \mathbb{N}}$ be i.i.d. random vectors in $\mathbb{R}^k$ such that

$$\mathbf{P}(X_1 = e_j) = p_j$$

for $j \in \{1, \ldots, k\}$ and $\sum_{j=1}^k p_j = 1$. Then,

$$\left(\frac{\sum_{i=1}^n \mathbf{1}\{X_i = e_j\} - np_j}{\sqrt{np_j}} \right)_{j \in \{1,\ldots,k\}} \xrightarrow{d} N \quad \text{as} \quad n \to \infty \tag{9.1}$$

with $N \sim \mathrm{Normal}(0, \Sigma)$ and $\Sigma = (\Sigma_{ij})_{i,j\in\{1,\dots,k\}}$ given by

$$\Sigma_{ij} = \begin{cases} 1 - p_i, & i = j, \\ -\sqrt{p_i}\sqrt{p_j}, & i \neq j, \end{cases}$$

for $i, j \in \{1, \dots, k\}$. Moreover, one has $\Sigma = OE_{k-1}O^T$ with an orthogonal matrix $O \in \mathbb{R}^{k\times k}$ and the diagonal matrix E_{k-1} in $\mathbb{R}^{k\times k}$ whose first $k-1$ diagonal elements are ones and whose last diagonal element is zero.

Proof. For $i, j \in \{1, \dots, k\}$ one has $\mathbf{E}[\mathbf{1}\{X_1 = e_i\}] = p_i$ and

$$\widetilde{\Sigma}_{i,j} := \mathbf{Cov}(\mathbf{1}\{X_1 = e_i\}, \mathbf{1}\{X_1 = e_j\}) = \begin{cases} p_i(1 - p_i), & i = j, \\ -p_i p_j, & i \neq j. \end{cases}$$

Define $\widetilde{\Sigma} = (\widetilde{\Sigma}_{i,j})_{i,j\in\{1,\dots,k\}}$. From the multivariate central limit theorem it follows that

$$\left(\frac{\sum_{i=1}^n \mathbf{1}\{X_i = e_j\} - np_j}{\sqrt{n}} \right)_{j\in\{1,\dots,k\}} \xrightarrow{d} \widetilde{N} \quad \text{as} \quad n \to \infty$$

with $\widetilde{N} \sim \mathrm{Normal}(0, \widetilde{\Sigma})$, which implies (9.1).

We define

$$H = \left\{ x \in \mathbb{R}^k : \sum_{j=1}^k \sqrt{p_j} x_j = 0 \right\}$$

and $e_H = (\sqrt{p_1}, \dots, \sqrt{p_k}) \in \mathbb{R}^k$. Note that, for $\ell \in \{1, \dots, k\}$,

$$\big(\Sigma e_H\big)_\ell = (1 - p_\ell)\sqrt{p_\ell} - \sum_{i\in\{1,\dots,k\}\setminus\{\ell\}} \sqrt{p_\ell} p_i = (1 - p_\ell)\sqrt{p_\ell} - (1 - p_\ell)\sqrt{p_\ell} = 0,$$

whence $\Sigma e_H = 0$. For $x = (x_1, \dots, x_k) \in H$ and $\ell \in \{1, \dots, k\}$ we obtain

$$\big(\Sigma x\big)_\ell = (1 - p_\ell)x_\ell - \sum_{i\in\{1,\dots,k\}\setminus\{\ell\}} \sqrt{p_\ell}\sqrt{p_i} x_i = (1 - p_\ell)x_\ell - \sqrt{p_\ell}(-\sqrt{p_\ell} x_\ell) = x_\ell$$

so that $\Sigma x = x$. Thus, Σ is the matrix of the projection onto H. This allows to write Σ as the desired product of matrices. □

Assume that the underlying distribution of the sample x takes only the distinct values $z_1, \dots, z_k$ for some $k \in \mathbb{N}$ and that the probability of z_j is p_j for $j \in \{1, \dots, k\}$. Moreover, suppose that a second distribution on $z_1, \dots, z_k$ is given such that z_j has probability $\pi_j \neq 0$ for $j \in \{1, \dots, k\}$. Our goal is to test

$$H_0 : p_j = \pi_j \quad \text{for} \quad j \in \{1, \dots, k\} \quad \text{against} \quad H_1 : p_j \neq \pi_j \quad \text{for some} \quad j \in \{1, \dots, k\}. \tag{9.2}$$

For $j \in \{1, \dots, k\}$ define $n_j = \sum_{i=1}^n \mathbf{1}\{x_i = z_j\}$ and let

$$D = \sum_{j=1}^k \frac{(n_j - n\pi_j)^2}{n\pi_j}.$$

Theorem 9.10 (Chi-square test of goodness of fit)**.** For $\alpha \in (0,1)$ let

$$\varphi : \mathbb{R}^n \ni x \mapsto \begin{cases} 0, & D \in [0, \chi^2_{k-1,1-\alpha}), \\ 1, & D \notin [0, \chi^2_{k-1,1-\alpha}). \end{cases}$$

For i.i.d. random variables $(X_i)_{i\in\mathbb{N}}$ such that $\mathbf{P}(X_1 = z_j) = \pi_j$ for $j \in \{1, \dots, k\}$, one has

$$\lim_{n\to\infty} \mathbf{P}(\varphi((X_1, \dots, X_n)) = 1) = \alpha,$$

i.e., φ is an asymptotic test for the test problem in (9.2) with level α.

Proof. Let n_j, $j \in \{1, \dots, k\}$, and D be the quantities introduced before with respect to $X_1, \dots, X_n$. It follows from Theorem 9.9 that

$$\left(\frac{n_j - n\pi_j}{\sqrt{n\pi_j}}\right)_{j\in\{1,\dots,k\}} \xrightarrow{d} \tilde{N} \quad \text{as} \quad n \to \infty,$$

where $\tilde{N}$ is centred normal random vector whose covariance matrix Σ can be written as $\Sigma = OE_{k-1}O^T$ with an orthogonal matrix O and a diagonal matrix E_{k-1} such that $k-1$ diagonal elements are one and one is zero. The continuous mapping theorem leads to

$$D \xrightarrow{d} \|\tilde{N}\|^2 \quad \text{as} \quad n \to \infty.$$

Since $\tilde{N}$ has the same distribution as ON with $N \sim \text{Normal}(0, E_{k-1})$, we have $\|\tilde{N}\|^2 \stackrel{d}{=} \|ON\|^2 \stackrel{d}{=} \|N\|^2 \sim \chi^2_{k-1}$. This shows the convergence in distribution of D to a random variable $Z \sim \chi^2_{k-1}$ and, thus,

$$\begin{aligned} \lim_{n\to\infty} \mathbf{P}(\varphi((X_1, \dots, X_n)) = 1) &= \lim_{n\to\infty} \mathbf{P}(D \notin [0, \chi^2_{k-1,1-\alpha})) \\ &= \mathbf{P}(Z \notin [0, \chi^2_{k-1,1-\alpha})) = 1 - \mathbf{P}(Z \le \chi^2_{k-1,1-\alpha}) \\ &= 1 - (1-\alpha) = \alpha, \end{aligned}$$

which completes the proof. □

Remark 9.11. If we replace the quantile of the chi-squared distribution by the corresponding quantile of the distribution of the test statistic D for sample size n under the null hypothesis, the test has even level-α for sample size n. Unfortunately, the quantiles of the distribution of D under the null hypothesis are not known. However, one can estimate them by simulations, i.e., one computes the test statistic for n random numbers distributed according to F_0 many times and takes the quantiles of the empirical distribution function. This procedure can be often used to obtain an exact test from an asymptotic test.

Next we consider the goodness of fit problem for continuous distribution functions. We prepare this by the following result. Recall that the empirical distribution function F_n with respect to random variables $X_1, \dots, X_n$ is given by

$$F_n(t) = \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{X_i \le t\}$$

for $t \in \mathbb{R}$.

Theorem 9.12. Let $(X_i)_{i\in\mathbb{N}}$ be i.i.d. random variables with a continuous distribution function F_0 and let F_n be the empirical distribution function with respect to $X_1,\dots,X_n$ and

$$K_n = \sup_{t\in\mathbb{R}} |F_n(t) - F_0(t)|$$

for $n \in \mathbb{N}$. Then,

a) $K_n = \max_{j\in\{1,\dots,n\}} \max\{\frac{j}{n} - F_0(X_{(j)}), F_0(X_{(j)}) - \frac{j-1}{n}\}$, where $X_{(1)} < \ldots < X_{(n)}$ are the order statistics of $X_1,\dots,X_n$

b) the distribution of K_n does not depend on F_0

c) $\sqrt{n}K_n \xrightarrow{d} Z$ as $n \to \infty$, where the random variable Z is Kolmogorov distributed, i.e., it has the distribution function

$$G(x) = \begin{cases} 1 - 2\sum_{k=1}^{\infty}(-1)^{k-1}e^{-2k^2x^2}, & x \geq 0, \\ 0, & x < 0. \end{cases} \tag{9.3}$$

Proof. Note that $F_n(X_{(j)}) = \frac{j}{n}$, while the left-sided limit of F_n in $X_{(j)}$ is $\frac{j-1}{n}$. Now part a) follows from the facts that F_n is piecewise constant and that F_0 is non-decreasing.

For a random variable X with distribution function F_0 and $u \in [0,1]$, one has

$$\mathbf{P}(F_0(X) \leq u) = \mathbf{P}(X \leq \max\{x \in \mathbb{R} : F_0(x) \leq u\}) = F_0(\max\{x \in \mathbb{R} : F_0(x) = u\}) = u,$$

whence $F_0(X)$ is uniformly distributed on $[0,1]$. Thus, $(F_0(X_{(1)}),\dots,F_0(X_{(n)}))$ has the same distribution as $(U_{(1)},\dots,U_{(n)})$, where $U_{(1)} < \ldots < U_{(n)}$ are the order statistics of independent $U_1,\dots,U_n \sim \text{Uniform}([0,1])$. This implies that the distribution of K_n is independent of F_0.

For the proof of part c) we refer to the original work by Kolmogorov (see Kolmogorov, A., Sulla determinazione empirica di una legge di distributione, Giornale dell' Istituto Italiano degli Attuari 4 (1933), 83–91). □

Assume that the sample x comes from n independent random variables with distribution function F. Let F_0 be a continuous distribution function. We aim to test

$$H_0 : F = F_0 \quad \text{against} \quad H_1 : F \neq F_0. \tag{9.4}$$

The idea of the following test is that the empirical distribution function should be close to F_0 under H_0.

Theorem 9.13 (Kolmogorov-Smirnov test)**.** Let $\alpha \in (0,1)$ and let $q_{Kol,1-\alpha}$ be the $(1-\alpha)$-quantile of the Kolmogorov distribution (see (9.3)). Define the test

$$\varphi : \mathbb{R}^n \ni x = (x_1,\dots,x_n) \mapsto \begin{cases} 0, & K \leq q_{Kol,1-\alpha}, \\ 1, & K > q_{Kol,1-\alpha}, \end{cases}$$

with

$$K = \sqrt{n} \sup_{t \in \mathbb{R}} \left| \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}\{x_i \leq t\} - F_0(t) \right|.$$

For i.i.d. random variables $(X_i)_{i \in \mathbb{N}}$ distributed according to F_0,

$$\lim_{n \to \infty} \mathbf{P}(\varphi((X_1, \ldots, X_n)) = 1) = \alpha,$$

i.e., φ is a test for the test problem (9.4) with asymptotic level α.

Proof. From Theorem 9.12 it follows that

$$K \xrightarrow{d} Z \quad \text{as} \quad n \to \infty,$$

where Z is a random variable following the Kolmogorov distribution. This implies

$$\begin{aligned} \lim_{n \to \infty} \mathbf{P}(\varphi((X_1, \ldots, X_n)) = 1) &= \lim_{n \to \infty} \mathbf{P}(K > q_{Kol,1-\alpha}) = \mathbf{P}(Z > q_{Kol,1-\alpha}) \\ &= 1 - \mathbf{P}(Z \leq q_{Kol,1-\alpha}) = 1 - (1 - \alpha) = \alpha, \end{aligned}$$

which completes the proof. □

Remark 9.14. Before applying parametric methods, one should check whether the data follow the assumed distributions. Apart from goodness of fit tests, one can use so-called qq-plots (or quantile-quantile plots). Let $x = (x_1, \ldots, x_n)$ be a sample and let $x_{(1)} < \ldots < x_{(n)}$ be the ordered values. If the sample x comes from independent copies of a random variable X with unique quantiles and n is large, one can expect that $x_{(k)}$ is approximately the $\frac{k}{n}$-quantile $q_{\frac{k}{n}}$ of the distribution of X for $k \in \{1, \ldots, n\}$. Thus, we plot the pairs $(x_{(k)}, q_{\frac{k}{n}})$ for $k \in \{1, \ldots, n\}$. Here, the points should be close to the diagonal. Otherwise, the underlying distribution of the sample might not be the distribution of X. The name qq-plot, which is an abbreviation for quantile-quantile plot, comes from the fact that empirical and theoretical quantiles are plotted.

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